Nutri-Scan Launch Pack

Business, customer, technical and operational planning documents

Prepared from the saved Nutri-Scan workspace documentation on 15 October 2026.

Document status: planning and review drafts. This PDF is not commercial, legal, scientific, regulatory or production approval.

Generated locally from Markdown and CSV source files. Refer to the source files for ongoing edits and the claims matrix before external use.

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README

Nutri-Scan launch pack — start here

Prepared: 2026-10-15 · **Owner:** [FOUNDER NAME] · **Entity/contact:** [LEGAL ENTITY] / [CONTACT]

Status: planning documents saved; commercial, legal, scientific and production approval not recorded. This pack prepares a restricted, paid **workflow pilot**, not a laboratory replacement or a feed-safety product launch. Drafts, example prices and schedules are not offers, certifications, forecasts, customer evidence or professional advice. Do not send a template externally until its placeholders and reviews are complete.

Starting as a student? Read the student-founder start plan for a low-cost first 14 days: one mentor, two teammates, three farmer interviews and one agreed small pilot. Real-data use still requires the relevant readiness gates.

Recommended launch decision

Lead with **silage/feed batch records, observation-only checks, evidence and follow-up**. Find a farm with a named operator, a repeated record-retrieval problem and an authorized buyer. Propose a six-week, one-farm pilot with explicit caps and measured workflow outcomes. Keep scientific validation and any hardware/laboratory service separately scoped and funded. Cooperatives/advisers can introduce or sponsor farms; a cooperative multi-farm administration console and in-app billing are not implemented.

The existing complete-screening flow uses ten supplied inputs and an **uncalibrated prototype score**. Photos do not diagnose feed; the ESP32 default sketch/emulator uses synthetic values. No toxin detection, lab accuracy, certified calibration or feeding clearance is established. Read the claims matrix before a pitch, demo, quotation or website publication.

Start with these seven steps

1. Read the one-pager, business plan

and go-to-market plan; confirm buyer and narrow scope.

1. Run customer discovery and record permissioned,

pseudonymous evidence in the discovery log.

1. Rehearse the demo script on the actual browser/device.

Keep synthetic data separate; disclose unsupported serial/manual fallback honestly.

1. Review unit economics and the

cash-flow plan; replace assumptions before quoting.

1. Complete the proposal and

quotation; have counsel/accountant review terms, data handling, taxes and payment prerequisites before signature or collection.

1. Resolve both the business checklist and

technical readiness checklist.
Missing privacy, access, restore or safety evidence blocks real-data use.

1. Record the permitted scope and actual go/no-go decision in the

production/pilot launch checklist.
Use the 90-day action tracker, not assumed approvals.

Document library

Section	Saved materials
Student-founder start plan	Practical first steps for a mechanical engineering student at Sinhgad, Solapur
Business and sales index	Strategy, market approach, pricing, pitch, demo, FAQ, interviews, outreach, pilot proposal/agreement and laboratory MoU outline
Customer privacy notice draft	Customer-facing data categories, purposes, optional services, retention and request process
Customer service terms outline	Counsel-reviewed continuation/service terms beyond a one-off pilot
Data and publicity permissions	Separate, optional research, case-study, logo and marketing decisions
Website and launch copy	Bounded landing-page, announcement and short introduction drafts
Business setup and IP checklist	Entity, authority, finance, IP, regulatory applicability and insurance review
Cash flow and funding	Assumptions, formulas, funding-gap illustration and spending gates
Quote and order template	Scope, itemized charges, exclusions and commercial review; not an invoice or ready-to-sign agreement
Technical and operations index	Readiness, deployment/restore, hardware, QA, calibration, privacy, incidents, risks, field SOP and go/no-go

Business CSVs: pricing scenarios,
12-month cash flow,
90-day actions,
discovery,
pilot KPIs,
partner pipeline.

Technical CSVs: calibration,
device inventory,
QA results,
incidents,
sample custody.

PDF exports

These generated PDFs are convenience copies of the saved Markdown/CSV sources;
edit the source files, then regenerate the PDFs when content changes.

PDF	Contents
Complete launch pack	All 35 launch Markdown documents and 11 CSV templates
Business pack	Business, sales, customer/commercial documents and business CSVs
Technical pack	Technical/operations documents and technical CSVs
Student start plan	Short student-focused first-14-day plan

The PDF copies are planning/review drafts, not legal agreements or launch approval. Do not add private customer forms, credentials or raw field data to repository PDFs.

Working-copy and privacy rules

- Keep these repository files as reusable, redacted templates. Put completed customer

forms, contracts, contact mappings, actual financials, field evidence and release approvals in access-controlled storage outside Git. Do not replace placeholders here with private information, credentials, tokens or raw customer data.

- Use the canonical placeholder register.

Technical `PLACEHOLDER_*` values and `.example.invalid` domains are also unresolved. A published/signed customer copy must have its own completed, reviewed version.

- CSVs are ordinary UTF-8 tables, not application import files. Illustrative numbers

are not facts; blank or not-recorded values do not mean zero, pass or calibrated. Treat untrusted spreadsheet-like text as text rather than executable formulas.

- All checklists and action statuses begin unverified/unstarted. Documentation

completion does not complete the work they describe. A scheduled day is not a release gate or evidence that someone performed an action.

Inputs needed before a customer-ready offer

Decision to supply/verify	Where it is used
Founder/entity, authority, contact and operating jurisdiction	Business setup, quotation, reviewed legal/data documents
First customer farm, operator, sponsor/payer and repeated task	Discovery, proposal, acceptance baseline
Actual pilot fee, included effort, exclusions, support and payment/tax treatment	Pricing, cash flow, quote and agreement
Actual deployment, providers, privacy roles, retention and closeout operators	Privacy notice, governance and technical gates
Demo versus real sensor scope, actual modules and qualified methods	Hardware records, SOP and any separate validation program
Named approvals, funded budget, dated QA/restore/access evidence	Scoped go/no-go; not supplied by this pack

No real outreach, payment collection, company registration, trademark application, partner execution, deployment or public website publication is performed by saving these documents. Account recovery/disablement, full erasure and broad production administration gaps must be resolved for the approved scope; unsupported guarantees cannot be fixed merely by adding contract language.

Evidence and review

The document validation report records checks performed for this pack. Existing application verification and calibration workflow are historical references, not new production checks. Before a release, rerun the real application checks on the identified candidate, test actual devices/environment, and record current results under the QA plan.

Immediate next action: assign owners and review the first-week discovery and cash-budget actions. Choose **proceed / revise / pause** only after the relevant business and technical gates have evidence; no launch approval is prefilled here.

README

Nutri-Scan business launch pack

> **DRAFT — legal/accounting review required.** Prepared against the repository state reviewed on 15 October 2026. Commercial terms, budgets, targets and schedules below are proposals, not offers, forecasts or achieved results.

Owner: [FOUNDER NAME], [LEGAL ENTITY] · **Contact:** [CONTACT] · **Next review:** [REVIEW DATE]

Start with the one-pager, rehearse the demo, then use discovery to qualify a bounded workflow pilot. Complete the proposal and obtain reviewed terms before collecting customer data or charging. This pack provides no certification or legal opinion.

Index

Artifact	Use
Student-founder start plan	Low-cost mentor/team/farmer discovery and a proposed 14-day target
Executive one-pager	First customer/investor introduction
Business plan	Wedge, scope, differentiation, milestones and risks
Go-to-market plan	ICP, funnel, partners and conversion
Pricing and unit economics	Illustrative INR model and break-even
Investor pitch outline	Ten-slide evidence-led narrative
Sales demo script	Current app and ESP32-compatible emulator walkthrough
Customer FAQ	Approved practical answers
Pilot proposal	Customer-specific scope and success measures
Pilot agreement	Non-legal discussion template for counsel
Lab partner MoU	Non-legal collaboration outline
Discovery interview	Problem and willingness-to-pay evidence
Partner outreach	Editable introduction/follow-up emails
Launch checklist	Business readiness and accountable next actions
Privacy notice template	Customer-facing data notice; review before collection
Customer service terms outline	Counsel drafting decisions for continuation/service
Separate data/publicity permissions	Granular optional reuse and branding decisions
Website and announcement copy	Claims-bounded drafts, not a deployed website
Business setup and IP	Entity, rights, applicability, finance and insurance questions
Cash-flow and funding plan	Worked scenario and spending gates, not a forecast

Artifact	Use
Quote and order schedule	Customer scope, charges and review record

CSV working templates

Template	Instructions
Pricing scenarios	All four rows are illustrative calculations; replace assumptions before quoting.
Discovery log	Placeholder row only; use consented, pseudonymous notes.
Pilot KPI log	Placeholder row only; record denominators, evidence and missing data.
Partner pipeline	Placeholder row only; no verified partners or contacts are supplied.
12-month cash flow	Static illustrative arithmetic; see the cash-flow plan before editing.
90-day action plan	Proposed relative days; all actions unstarted and approvals unset.

CSV use: preserve headers and remove the placeholder row before logging real work. Quote fields containing commas/newlines. Keep evidence links access-restricted and use explicit `not_measured` / `not_applicable` rather than zero for missing numeric data. The KPI log holds one metric per period; define its unit, numerator and denominator before measuring. Set `record_type` to `real` or `demo` and exclude demo rows from adoption/acceptance summaries. The partner contact column is a reference to an approved restricted contact record, not permission to publish personal details. `review_status` records review state; its initial value is not approval.

The cash-flow CSV is a separate worked scenario, not an actuals ledger; its zeros and excluded costs follow the cash-flow assumptions.
The action plan's day 1 is the agreed planning start, not a calendar commitment.
Move dates when gates are blocked; schedule pressure never overrides approval.
Replace owner tokens, track actual completion/evidence privately, and adapt the plan to the real pilot start. The example assumes pilot days 36–77 (six weeks).

Product truth and source priority

- Today's demo: actual same-day app/serial demonstration.
- Fieldwork contract: current persistence, permissions, local testing requests and offline boundaries.
- Calibration workflow: synthetic scaffold, real collection protocol and validation gates.
- Fieldwork verification: recorded software checks and real-device gaps; these are existing results, not checks performed for this pack.
- Farmer usability: guided entry, evidence, explicit draft sync and all-ten-readings requirement.
- Serial firmware guide, input model, prototype engine, guided form: implementation references.

Older architecture/marketing text includes future capabilities. For business copy, use the current contract and implementation. In particular, the later fieldwork contract adds an installable **public-help** PWA shell beyond the earlier usability guide; it does not add offline private-record access. Recheck these sources before each external demo or quote.

Claims matrix

“Can say now” means supported by the reviewed prototype; rehearse on the actual demo device. A future gate is necessary evidence, not permission to promise a capability today. Any future claim requires a separately approved, bounded statement.

Topic	Can say now	Must not say	Validation gate for any expanded claim
Core value	Record feed/silage batches, observations and follow-up; demonstrate a reviewed screening workflow.	Proven savings, higher milk yield, disease prevention or improved feed safety.	Consented baseline and prospective customer study; measured outcomes, attribution limits and expert review.
Scoring / AI	Transparent, deterministic, uncalibrated prototype score and heuristic confidence from supplied inputs.	Lab accuracy; trained field model; confidence is diagnostic accuracy or a validated probability; LOW means safe to feed.	Representative paired references; train-only fitting; locked held-out farm/batch/time evaluation, uncertainty and expert release approval for a defined use.
ESP32 / serial	User-selected serial capture and confirmation; emulator/default sketch emits illustrative moisture, pH, temperature and humidity values.	Live field measurements from the emulator, plug-and-play meter compatibility, certified calibration, autonomous submission.	Validated sensor hooks, traceable calibration/maintenance records, repeatability and target-browser/device field tests. Device-reported status alone proves none of these.
Nutrients / NIR	Accepts reviewed protein, dry matter and crude-fiber numerical inputs.	Measures/predicts protein, fiber or energy from a phone/photo or this ESP32 demo; deployed calibrated NIR.	Suitable hardware, defined units/basis and methods, real paired lab-labelled dataset and target-specific held-out validation.
Photos / contamination	Private evidence photos and operator-supplied visual indicators; observations preserve Unknown.	AI photo diagnosis, toxin detection, aflatoxin/mycotoxin detection, certified adulteration detection, absence of contamination.	Separate analyte-specific method/hardware and qualified reference testing, limits and false-negative evaluation; expert/legal review. A generic score cannot clear this gate.
Advice / safety	Prototype guidance prompts qualified review and confirmatory testing.	Diagnosis, guaranteed safety, feeding clearance or replacement for a laboratory/adviser.	Qualified domain review of guidance and translations; any regulated intended use needs its own applicable review. No guarantee is offered.
Testing help	Farm-local request/status tracking, manual contact details and private report download.	Automatic lab booking, sending samples/requests, lab directory, OCR or automatic interpretation; named/accredited partner without evidence.	Executed partner scope, verified methods/contact details, explicit farmer authorization and an implemented/tested integration before claiming automation.
Offline / reminders	Public-help PWA shell; explicit local complete-reading drafts/queue in an already-loaded signed-in app; in-app reminders.	Fully offline private records, background auto-sync, retained instrument provenance in drafts, SMS/background notifications.	Implemented/tested scope and device tests before changing copy. Direct online submission currently preserves selected instrument provenance.
Accounts / reports	Farm-scoped owner/member/viewer access, saved history, HTML reports and browser Print / Save PDF.	Certified compliance, audited security, public QR access, guaranteed production uptime or native PDF generation.	Deployment/privacy review, access and restore tests, supported-browser checks and any specifically commissioned audit.
Calibration demo	Offline synthetic cleaning/splitting/metrics/fitting scaffold; coefficients are not integrated into the app.	Synthetic metrics demonstrate accuracy or certified calibration.	Follow the real-data protocol in the calibration workflow and obtain a separate release decision.

Placeholder register

Keep the exact tokens below across this pack. Square-bracket values are unresolved fields, never real identities, quotes or commitments. Retain `ILLUSTRATIVE` / `PLACEHOLDER` labels in working CSVs; replace tokens and remove example rows in a reviewed customer copy. Use YYYY-MM-DD for dates and timezone-aware timestamps for captures. Do not paste credentials or identifiable farm notes into pipeline sheets.

Group	Canonical tokens and meaning
Identity / review	[LEGAL ENTITY] = Nutri-Scan contracting entity; [FOUNDER NAME] = accountable founder; [CONTACT] = verified Nutri-Scan contact; [REVIEW DATE] = next review date.

Group	Canonical tokens and meaning
Customer / pilot	[CUSTOMER LEGAL NAME], [CUSTOMER CONTACT], [PILOT ID], [PILOT START DATE], [PILOT END DATE].
Commercial	[PILOT PRICE] = total pilot fee in INR; [MONTHLY PRICE] = recurring fee per farm/month in INR; [PAYMENT TERMS] = schedule/due dates/refund treatment; [TAX TREATMENT] = accountant-reviewed invoicing/taxes.
Delivery	[SUPPORT HOURS], [RESPONSE TARGET], [SAMPLE COUNT], [ACCEPTANCE TARGET], [NOTICE PERIOD]. Sample count means paired physical samples only when a separately agreed reference study exists.
Data / counsel	[DATA USE PERMISSION], [RETENTION PERIOD], [EXPORT WINDOW], [LIABILITY CAP], [GOVERNING LAW], [DISPUTE PROCESS].
Partner / funding	[LAB PARTNER], [LAB CONTACT], [LAB METHOD], [LAB FEES], [FUNDING ASK], [RUNWAY MONTHS]. None implies an existing relationship or funding.
Tracking	[OWNER], [DUE DATE], [NEXT ACTION], [STATUS], [EVIDENCE LINK], [SEGMENT], [CONSENT STATUS], [INTERVIEW ID], [DATE], [SUMMARY], [PARTNER ID], [ROLE], [KPI NAME], [KPI VALUE], [BASELINE MINUTES], [CURRENT MINUTES].

Release rule: [FOUNDER NAME] checks the claims matrix and evidence; counsel reviews customer/data/partner terms; an accountant reviews prices, tax treatment and costs. Record the review date and decision in the launch checklist. Unresolved placeholders block a customer-ready copy.

Business Plan

Nutri-Scan business plan

> **DRAFT — legal/accounting review required.** Strategy and milestones are hypotheses; dates run from an agreed pilot start, not a promise of readiness.

Accountable owner: [FOUNDER NAME], [LEGAL ENTITY] · [CONTACT] · [REVIEW DATE]

Problem and customer segments

Test whether fragmented batch notes, incomplete readings and disconnected testing reports create repeated work or missed follow-up. Quantify recent examples before proposing software.

Segment	Job / likely buying condition	Initial priority
Dairy farm with several silage batches	Retrieve batch evidence, assign checks and coordinate adviser review; named operator and budget owner.	Primary ICP
Cooperative or producer organization	Help member farms adopt a consistent workflow.	Introductions/pilot sponsor; cross-farm management console is not present.
Independent nutrition adviser	Review farmer-shared records and identify where qualified testing is needed.	Referral/design partner, not an automated adviser replacement.
Feed/silage supplier	Preserve lot evidence and handle queries.	Later discovery; no lot certification or dispute guarantee.
Qualified laboratory	Receive traceable samples under an agreed protocol and return reference reports.	Validation/service partner, not presumed software buyer.

Wedge: silage/feed screening workflow

Lead with one silage storage/batch routine: create identity → record observations → set follow-up → retain testing report. A complete screening is optional and only possible with all required inputs from appropriate sources. This avoids making nutrient instrumentation a prerequisite for basic recordkeeping. Feed is supported, but keep the first pilot's sample type and operator routine narrow.

Product scope

Current prototype: authenticated farm roles; batch/observation/reminder/test-request records; evidence photos; guided reviewed screening; serial protocol capture with device-reported calibration; uncalibrated deterministic scoring; history/trends; HTML/browser-print reports; multilingual guidance; explicit local drafts; public-help PWA shell.

Contract boundaries: testing requests are local and manually acted upon; no lab booking, OCR or report interpretation. Photos do not drive AI analysis. All ten scoring inputs remain required. Draft save/load drops instrument provenance; direct online submission preserves selected capture metadata. Private fieldwork writes need connectivity. See the current contract.

Future research, separately gated: real sensor module validation, calibrated NIR/nutrient targets and lab-labelled evaluation. The synthetic calibration scaffold is not an app model or evidence of accuracy. No future research item belongs in a pilot acceptance promise without its own funded protocol.

Business model

1. Fixed-fee, time-boxed workflow pilot with capped delivery effort and explicit exclusions.
2. Optional per-farm recurring software/support service after measured value and a signed continuation; no automatic renewal assumption.
3. Separately quoted hardware, visits or reference testing where feasible. Show laboratory costs transparently; do not assume referral margins or bundling rights.

Use pricing hypotheses; verify willingness to pay and delivery load before fixing packages. The current farm-scoped app is not a cooperative-wide tenant-management product.

Competition and positioning

Alternative	Customer advantage to respect	Nutri-Scan positioning to test
Notebook / spreadsheet / messaging	Familiar, inexpensive and flexible.	Easier batch retrieval and structured follow-up, measured in the customer's routine.
Laboratory testing	Qualified methods and reference results for specific tests.	Organize sampling/report follow-up; complement testing, never replace it.
Standalone meter / NIR system	Direct measurements under its validated operating conditions.	Reviewed record capture where protocol-compatible; no equivalence claim.
Farm-management software	Established broader farm workflows.	Narrow silage/feed evidence loop; test whether switching/integration burden outweighs value.
Adviser-led service	Context and professional judgment.	Shared evidence and history supporting human review.

No competitor-specific superiority, price comparison or logo is asserted. Discovery should record the actual incumbent and reason to switch.

Potential moat and evidence needed

- **Workflow fit:** repeat usage with low training/support burden; evidence = cohort retention and task completion.
- **Operational reliability:** traceability, retrieval and repeatable onboarding; evidence = readback, permissions and recovery checks on pilot setup.
- **Partner trust:** defined methods, handling and responsibilities; evidence = reviewed written arrangements, not outreach responses alone.
- **Data learning:** only explicitly authorized, representative and traceable reference data could support later validation. Ownership/permission, label quality and held-out evaluation are prerequisites. No exclusive data or accuracy moat is claimed today.

Proposed milestones and go/no-go gates

Window	Deliverable	Gate / evidence
Weeks 1–2	Proposed 12 discovery interviews across farm operators, managers and advisers.	Document recent problem instances, existing spend and buyer/payer; revise ICP if no repeated need.

Window	Deliverable	Gate / evidence
Weeks 3–4	Proposed three qualified pilot discussions.	Named sponsor/operator, lawful data scope, baseline task and reviewed price/terms.
Six-week pilot per farm	Bounded one-farm workflow and weekly KPI review.	Customer-agreed [ACCEPTANCE TARGET]; logs distinguish real use from demo/test entries.
Within 14 days of pilot end	Outcome review and written continuation decision.	Signed paid scope plus first payment; otherwise document objection/closeout.
Separate validation program	Protocol, lab methods, representative paired samples and split manifest.	Predefined target thresholds, locked tests, uncertainty/subgroups and expert release approval; no calendar-based accuracy promise.

Principal risks and response

Risk	Early signal	Owner / response
Users interpret score as safety clearance	Interview/demo comprehension failure.	[FOUNDER NAME]: restate boundary, change sales emphasis; qualified review remains necessary.
Missing nutrient/visual inputs	Screening cannot be completed reliably.	Pilot lead: use observation-only workflow; never invent readings.
Weak willingness to pay	Interest without sponsor, signature or payment.	Sales owner: test smaller scope/price; stop free open-ended delivery.
Support/hardware costs overwhelm margin	Hours/visits exceed agreed cap.	Delivery owner: log costs weekly and re-scope; no unpriced hardware promise.
Connectivity, browser or device failure	Serial unavailable or private saves fail.	Delivery owner: rehearse manual fallback; distinguish unsaved data and offline public help.
Data/confidentiality concerns	Unclear permissions, sharing or retention.	[LEGAL ENTITY]: agree purpose/access/export and separate training permission; review deployment.
Biased/insufficient validation data	Correlated farms/batches or unit/basis mismatches.	Methods owner with [LAB PARTNER]: freeze protocol, quarantine failed QC and preserve independent test groups.

Operational next steps: go-to-market plan, proposal, launch checklist. All public wording follows the claims matrix.

Business Setup And Ip Checklist

Business setup, IP and applicability checklist

> **DRAFT — professional review required.** This is a question list, not advice to incorporate a particular entity or an exhaustive registration/licensing list. INR examples suggest an Indian pilot planning context only; confirm actual jurisdiction, activities and rules in force with qualified advisers.

Accountable founder: [FOUNDER NAME] · **Review:** [REVIEW DATE]
Every box starts unverified. Keep identity documents, tax records, banking details, private contracts and credentials in protected business storage—not this repository.

Entity, authority and finance

- ☐ Decide the legal operating/contracting entity with counsel/accountant; verify founders, ownership, authority and any institutional/employment restrictions.
- ☐ Confirm whether the project name can be used; investigate relevant company, domain and trademark conflicts before public expenditure. No trademark is claimed.
- ☐ Verify required entity/tax/local registrations for actual activities and territory. Do not assume startup recognition, incentives or tax treatment applies.
- ☐ Establish authorized invoicing, payment collection, expense approval, bookkeeping, refunds and reconciliation; protect account/payment credentials.
- ☐ Document founder contributions, equity/vesting, decision rights, departures, related-party arrangements and signing/spending authority appropriately.
- ☐ Review budgets and cash timing; treat grants, investments and partner support as zero committed funding until execution and actual availability are verified.

Ownership and third-party rights

- ☐ Inventory source code, presentations, artwork, firmware, datasets, methods and commissioned work; identify authors and provenance without assuming assignment.
- ☐ Review SIH/institution/employer/sponsor rules, submission terms and contributor agreements; resolve ownership and commercialization authority in writing.
- ☐ Review open-source licences/notices and third-party media/data terms; keep a dependency/licence inventory for the delivered release. No licence audit is implied.
- ☐ Obtain appropriate written rights for contractors, joint research, lab reports,

training/reference data and generated material. Farm authorization may not cover a lab's or an individual's rights.

- ☐ Separate customer-owned records from supplier pre-existing software and later

research. No exclusive dataset, patent, accuracy moat or proprietary model is claimed without evidence and rights review.

- ☐ Use the permission record for any

optional research, testimonial, customer logo or public case study.

Product activity, safety and insurance

- ☐ Describe actual intended use to a qualified adviser: workflow software versus

measurement hardware, laboratory services, advisory or another regulated activity. Determine applicable obligations before supply; a disclaimer is not an exemption.

- ☐ If supplying/importing/assembling hardware, review applicable electrical,

radio, battery, environmental, warranty and distribution requirements for the exact product and territory. Do not imply certification from a board's branding.

- ☐ If handling physical samples or providing tests, confirm provider qualification,

methods, handling/transport, chain of custody, reporting and relevant permissions.

- ☐ Review professional/product/public liability, cyber and equipment insurance

needs with a qualified broker; record coverage and exclusions, not assumed cover.

- ☐ Align contracts, demonstrations, translations and marketing with the current

claims matrix; expert review is separate from sales approval.

Data, contracts and operations

- ☐ Approve the privacy notice, organization roles,

retention, provider terms, access withdrawal, deletion and incident processes.

- ☐ Obtain reviewed pilot/continuation terms, proposal, quotation, support scope,

tax treatment, refund/termination and dispute procedures before charging.

- ☐ Complete technical readiness,

restore/privacy tests and scoped release approval before real customer data.

- ☐ Assign accountable support, finance, domain, hardware and data owners; one

founder may hold several roles but each responsibility still needs coverage.

Review record: [OWNER] / [DATE] / [STATUS] / [EVIDENCE LINK]
Open matters and next action: [SUMMARY] / [NEXT ACTION] / [DUE DATE]

Cash Flow And Funding Plan

Cash-flow planning and funding gates

> **ILLUSTRATIVE SCENARIO — NOT A FORECAST, VALUATION OR FUNDING COMMITMENT.**
> Accountant review required. The 12-month CSV
> is a worked arithmetic example, not evidence of customers, cash, financing or demand.

Owner: [FOUNDER NAME] · **Review:** [REVIEW DATE]

Assumptions and scope

Use the recurring **base** case from unit economics:
₹1,499/farm/month, ₹579.98 direct cost/farm/month, ₹120,000 fixed monthly cost and
₹3,000 incremental acquisition spend per new paying farm. Fixed spend and acquisition
cost must be reconciled to avoid counting the same outreach/team cost twice.

- The example starts with **₹600,000 hypothetical available cash**, not confirmed

founder/investor funds. Set actual unrestricted cash from a verified private record.

- Active farms are arbitrary scenario inputs: 0, 0, 1, 3, 5, 8, 12, 18, 25, 35, 50, 70.

They are not funnel targets or expected sales. No churn is assumed solely to simplify
the example; test delayed sales, nonpayment, churn and higher support in real copies.

- Each active farm pays a whole month immediately; receipts equal recurring fees only

under this deliberately simplified same-month collection assumption. Proration,
receivables, refunds and revenue recognition need a real accounting model.

- All modeled direct and fixed costs are paid that month. No new funding is received.
- `other_reviewed_cash_costs_inr = 0` means **excluded/not budgeted in this example**,

not that hardware, labs, calibration, capex, insurance, taxes, shipping or debt are free.
The recurring unit model also excludes these costs. Add every applicable cash outflow
and prevent double counting before using the sheet for a spending decision.

- One-time pilot cash/costs are omitted, not counted as MRR. Model pilot timing separately

if offered. Hardware/validation funds and grants are not assumed.

Row formulas

```
active_farms = previous_active_farms + new_farms - churned_farms
customer_receipts_inr = active_farms * monthly_price_inr
direct_cash_cost_inr = active_farms * direct_cost_per_farm_inr
acquisition_spend_inr = new_farms * cac_inr
net_cash_flow_inr = customer_receipts_inr + funding_received_inr
                  - direct_cash_cost_inr - fixed_cash_cost_inr
                  - acquisition_spend_inr - other_reviewed_cash_costs_inr
closing_cash_before_gap_inr = opening_cash_inr + net_cash_flow_inr
next_opening_cash_inr = closing_cash_before_gap_inr
cumulative_funding_gap_inr = max(0, -closing_cash_before_gap_inr)
```

The CSV contains static numbers, **not spreadsheet formulas**. Recalculate all derived cells after changing assumptions. Negative cash is an **unfunded scenario deficit**, not permission to overdraw or a real bank balance. The cumulative gap column is not a monthly bill: do not sum it. If finance is actually received, insert its cash timing and recalculate from that month; do not also treat the same gap as a receipt.

What this illustration shows

Illustration	Arithmetic result, not an operating promise
First negative closing position	Month 5: −₹6,728.82
Month 12 recurring receipts	₹104,930.00 at 70 hypothetical paying farms
Month 12 closing deficit before any added funding	−₹841,382.46
Additional cash needed to stay nonnegative through this path	At least ₹841,382.46 beyond the assumed initial ₹600,000; excluded costs and a prudent reserve would increase this

Positive per-farm contribution does not establish company cash break-even. Sales timing, delivery load and fixed spend matter. Do not advertise a runway number based on an average burn when month-by-month cash changes materially.

Stage-gated funding/spending plan

1. **Discovery:** cap founder-approved spending; establish a repeated problem and a

reachable buyer before commitments to hardware, leases or broad campaigns.

1. **Scoped pilot:** approve the delivery budget, cash/payment schedule, privacy and

readiness gates; reserve effort for support, closeout and refunds if applicable.

1. **Continuation:** add spend only with documented collections, support capacity,

retention evidence and a reviewed cash plan; a signed interest letter is not cash.

1. **Scientific validation:** separately cost sampling, qualified reference methods,

hardware, handling, dataset rights, held-out evaluation and expert review. No model accuracy release is triggered by a fundraising or sales deadline.

1. **Fundraising:** use pitch outline and verified use of

funds; replace [FUNDING ASK] / [RUNWAY MONTHS] with an evidence-backed plan. Confirm eligibility and current terms directly before applying for any grant/program.

Monthly finance review: compare actual collections/costs with plan, reconcile bank and invoices privately, refresh scenarios and log the named spending decision. Keep account numbers, payment credentials, payroll and investor identity records outside Git.

Customer Discovery Interview

Customer discovery interview

> **DRAFT — legal/accounting review required.** Research guide, not evidence that interviews or customer commitments exist. Use consented, minimal notes; approved data handling is required before outreach.

Interviewer: [OWNER] · **Interview:** [INTERVIEW ID] · **Date:** [DATE] · **Segment/role:** [SEGMENT] / [ROLE]

Preparation and permission — 2 minutes

Introduce [FOUNDER NAME] from [LEGAL ENTITY], contact [CONTACT]. Say: “We are learning how dairy teams manage silage/feed batch records and testing follow-up. May I take notes for product/business research? You may skip questions or stop. Please avoid identifying details or confidential reports.”

Record [CONSENT STATUS]; no audio/photo/report capture by default. Discuss separate permission before recording or receiving documents. Use an interview ID, not a personal name, in the shared discovery log. Keep any verified contact details in an appropriately restricted system.

Recent behavior — 8 minutes

1. “Tell me about the **last** silage/feed batch you checked. What happened from receiving/opening it to the next action?”
2. “Who wrote down observations? Where could someone else find them later? Can you walk through the last handoff without sharing private data?”
3. “When did a missing reading, report or follow-up last cause extra work? What did you do, and roughly how much time did it take?” Do not suggest it caused animal harm or loss.
4. “What did you do when something looked concerning or you were unsure? Who decided whether to test?”
5. “How was the last laboratory test arranged, paid for and matched back to its batch? What went wrong or worked well?”
6. “Which measurements can you actually obtain, from which instruments/reports and with what units/basis? Which are unavailable?”
7. “How often does this workflow occur? How many batches/operators are involved, and when is it busiest?”

Ask for recent instances, not agreement with a sales claim. Record exact quotes only with permission, and record “no problem” responses as carefully as problems.

Alternatives, purchasing and constraints — 5 minutes

- “What notebook, spreadsheet, software, adviser or service do you use now? What do you like about it?”
- “What did you spend on the workflow or testing recently? Which part was software, laboratory work, labor or travel?” Accept “unknown”; do not estimate for them.
- “Who would use a new tool, approve it and pay its invoice? What approval steps and timing apply?”
- “What would prevent adoption: phone/browser, language, connectivity, effort, trust, privacy or something else?” Ask open-ended first, then probe.

- “What data could be shared with a provider, and what must stay private? Who can authorize that?”

Concept check and pilot offer — 5 minutes

Only after the behavior questions, explain: “Nutri-Scan currently organizes batch observations, reminders and farm-local testing/report records. It also demonstrates reviewed, uncalibrated screening; photos are evidence only and testing requests are not booked automatically.”

1. “Which, if any, step would be useful in your current routine? What would you remove?”
2. “What do you understand the score, photo and testing-request features **cannot** establish?” Correct any inference of lab accuracy, toxin detection, certified calibration, diagnosis or guaranteed safety using the claims matrix.
3. “If you lacked the required measurements, would observation-only recording still help? Why or why not?” Do not propose guessed readings.
4. “Would you consider a six-week, one-farm workflow pilot at [PILOT PRICE], excluding hardware/lab work? What task and result would justify payment? Who would need to approve it?” Only use a completed, reviewed price in a live offer.
5. “What evidence would justify a separate [MONTHLY PRICE] continuation? May we schedule a decision with the buyer?” Distinguish stated interest from a signed scope/payment.

Optional baseline task

With permission, time a non-sensitive task such as finding the latest report and next action for one batch. Record [BASELINE MINUTES], task/device, attempts and obstacles; repeat a comparable task later for [CURRENT MINUTES]. No observed task result means “not measured,” not zero. Do not record screen contents without permission.

Debrief and next action

Record	Required distinction
Problem / incumbent	Recent observed/reported event versus interviewer inference.
Buyer / user / payer	Roles separately confirmed; no unidentified “customer” shorthand.
Measurement availability	Complete-input screening feasible, observation-only suitable or neither.
Price evidence	Current spend, reaction to reviewed offer, approval/signature/payment; do not equate them.
Qualification	Named owner, recurring task, feasible data boundary, dates and budget path. Record rejection reasons too.
Next action	[NEXT ACTION] by [OWNER] on [DUE DATE], with [EVIDENCE LINK] in a restricted location.

Complete [SUMMARY] and [STATUS] in the CSV. Do not log passwords, actual farm reports, unsupported revenue or invented testimonials. Review results by segment weekly before changing the GTM plan.

Customer Faq

Customer FAQ

> **DRAFT — legal/accounting review required.** Use with the claims matrix; customer-specific fees, support and data terms need review.

Contact: [CONTACT] · [LEGAL ENTITY] · **Review:** [REVIEW DATE]

What can Nutri-Scan help us do today?

Record feed/silage batches, private evidence photos, observations, in-app follow-up tasks and local testing-request/report records. The prototype also supports reviewed complete-input screening, saved history, trends and HTML/browser-print reports. Start with the task your team needs to retrieve or follow up.

Is a score a laboratory result or permission to feed?

No. Scoring is a transparent, uncalibrated prototype. Quality/risk, nutrient-related scores, fermentation/spoilage labels and heuristic confidence are not validated measurements, probabilities, diagnoses or safety guarantees. Even a LOW result does not clear feed for use. Seek a qualified adviser and appropriate confirmatory testing for concerns.

Does it detect toxins, adulterants or disease?

No toxin detection, aflatoxin/mycotoxin detection, certified adulteration detection or diagnosis is offered. Operator observations and numerical visual indicators do not establish presence or absence of contamination. A photo is evidence, not an AI diagnosis.

Does my phone or the ESP32 measure protein, fiber or energy?

Not in this demo. The app accepts numerical nutrient inputs; it does not produce them from a photograph or the default ESP32 sketch. Real nutrient entries must come from a suitable report/instrument with the right method, units and basis. Do not substitute NDF/ADF for crude fiber. Energy prediction is not a validated current capability.

What if we do not have all readings?

All ten inputs are required for scoring: moisture, pH, temperature, humidity, protein, dry matter, crude fiber, mould indicator, colour anomaly and surface damage. Do not guess or replace missing values with zero. Use **Quick observations** instead: Yes/No/Unknown records are explicitly observation-only and have no screening score. Real visual scoring requires a defined, reviewed method and trained operator.

Is the device calibrated? Which meters work?

Calibration status is reported by the device, not verified or certified by Nutri-Scan. The emulator/default firmware uses synthetic values and `unknown` calibration. Supported capture requires the documented newline JSON protocol, units and 115200 baud on a compatible Chromium browser at HTTPS/localhost. A particular real meter/sensor needs separate compatibility and method checks; manual entry remains available.

Will a lab receive my testing request automatically?

No. **Testing help** tracks a farm-local request, collection date and report. You or an agreed human coordinator must contact the chosen provider separately. There is no automatic lab booking, directory, OCR or report interpretation. Qualified providers set sample requirements, fees and turnaround; none is assumed here.

What works offline?

The production PWA caches public shell/help, including public English/Hindi/Marathi help, not private API data, photos, reports or sign-in details. In an already-loaded signed-in app, deliberately saved complete-reading drafts can be queued locally; reconnect and press **Sync now**. Reconnection alone does not upload. Instrument provenance is not retained in draft save/load; use direct online submission to preserve reviewed capture metadata. Private fieldwork writes need connectivity.

Who sees our farm data? What happens when we share?

The current app uses farm-scoped roles: owners/members can write, viewers are read-only. QR links still require sign-in. Review report contents and sharing previews before sending outside the farm; shared files/text can be retained by recipients. Photos are evidence only. These prototype controls do not imply a security audit or certification. Deployment, access, backup and recovery responsibilities must be agreed for each pilot.

Who owns our records, and may they train a model?

Proposed terms retain customer rights in its supplied farm records, subject to third-party rights; [LEGAL ENTITY] retains its software. Record [DATA USE PERMISSION] explicitly. Training, research, onward sharing and publicity require separately agreed permission and applicable review; they are not presumed from app use. Agree [RETENTION PERIOD] and [EXPORT WINDOW]. Current HTML reports/individual downloads are not a promise of self-service bulk export or deletion.

Can the app remind us by SMS or give professional advice?

Reminders are in-app tasks, not SMS/background notifications. Guidance and English/Hindi/Marathi translations are prototype material requiring qualified review. Listening and voice entry depend on browser/device support; voice may use network speech services, proposes a reviewed value and never submits a scan. No nutrition prescription, diagnosis or professional service is included by default.

What does a pilot cost and what happens afterward?

Customer quote: [PILOT PRICE] for the written scope; optional continuation: [MONTHLY PRICE] per farm/month. [PAYMENT TERMS] and [TAX TREATMENT] require review. The financial model contains illustrative hypotheses only. Hardware, laboratory work and extra visits are separately scoped. Continuation requires a separate written decision; no silent renewal.

What support and results can we expect?

Agree [SUPPORT HOURS], [RESPONSE TARGET], support contact [CONTACT] and measurable [ACCEPTANCE TARGET] before kickoff. Evaluate adoption, retrieval and follow-up, not feed-safety outcomes. No production uptime, regulatory certification, lab accuracy or savings guarantee is offered. See the proposal and non-legal agreement discussion template.

Customer Terms Outline

Customer service terms — counsel drafting outline

> **DRAFT — NOT A READY-TO-SIGN CONTRACT.** Qualified counsel must draft appropriate terms for the chosen jurisdiction and intended use. This outline adds continuation decisions to the pilot agreement outline; it does not create enforceable terms, a warranty exclusion or a liability limit.

Supplier/customer: [LEGAL ENTITY] / [CUSTOMER LEGAL NAME]
Contacts: [CONTACT] / [CUSTOMER CONTACT] · **Review:** [REVIEW DATE]

Service schedule to attach

Decision	Required agreement
Authorized scope	Named farm, users, supported browsers, workflow, deployment, attachment/storage limits and explicitly included support. Cross-farm administration and billing automation are not current product promises.
Duration and renewal	Start/end dates, renewal decision and notice procedure. No silent transfer from a trial/pilot to paid recurring service.
Fees	[MONTHLY PRICE] INR per farm/month or a separately defined unit; [PAYMENT TERMS], approved charges and [TAX TREATMENT]. Example model prices are not offers.
Support	[SUPPORT HOURS], [RESPONSE TARGET], escalation and included effort; distinguish first response from resolution and uptime.
Availability/change	Maintenance notification, supported versions, material-change process and any measured service commitment. No service level is established here.
Exit	[NOTICE PERIOD], [EXPORT WINDOW], [RETENTION PERIOD], access termination, outstanding fees and named closeout operators.

Drafting instructions

1. **Parties and authority:** verify entity, addresses, signatory/payer authority,

order of precedence, effective version and a reliable acceptance record.

1. **Use and reliance:** state uncalibrated prototype limits prominently. Do not

promise lab accuracy, toxin detection, diagnosis, feeding clearance or guaranteed safety. Full screening requires ten suitable inputs; observation-only use has no score. Qualified advice/reference testing remains separate.

1. **Feature boundaries:** photos are evidence, not AI diagnosis; laboratory requests

are local records, reminders are in-app, offline reload is public help only and draft save/load drops instrument provenance. No unsupported device compatibility.

1. **Customer duties:** authorize supplied material, assign individual users, maintain

suitable devices/connectivity, review shared reports, protect credentials and report suspected exposure. Do not transfer all supplier responsibilities to users.

1. **Data/confidentiality:** agree rights and organization roles, permitted delivery use [DATA USE PERMISSION], actual providers, security duties, incidents and the privacy notice. Optional research/publicity requires separate permission; customer data does not become supplier-owned by hosting it.
1. **Administrative gaps:** account disabling/recovery, bulk export, farm erasure, backup expiry and audit operations are not complete self-service MVP features. Confirm a reliable, authorized manual process before promising deadlines.
1. **IP and third parties:** distinguish pre-existing software, customer material, generated reports, laboratory rights and commissioned work. Review dependency licences, third-party service terms and any hardware loan/supply separately.
1. **Fees and remedies:** agree invoicing, taxes, collection timing, disputed charges, change approval, cancellation/refunds and suspension/cure. No unquoted pass-throughs.
1. **Risk allocation:** counsel to draft appropriate commitments, remedies, mandatory rights, insurance, third-party claims and any lawful [LIABILITY CAP]. No blanket waiver or prototype disclaimer substitutes for responsible operation.
1. **Termination/disputes:** set notice, cure, export/access/retention handling, confidentiality survival, [GOVERNING LAW] and [DISPUTE PROCESS].

Execution gate

- ☐ Completed scope and price; accountant and counsel reviews recorded.
- ☐ Deployment, security, retention, access withdrawal and restore responsibilities verified.
- ☐ All customer-facing claims match the claims matrix.
- ☐ Final privacy notice and permissions supplied in an understandable language.
- ☐ Authorized execution copy/version and acceptance evidence retained privately.

Review owner/decision: [OWNER] / [STATUS] / [EVIDENCE LINK]. All boxes start unverified.

Data Permission And Publicity Form

Data reuse and publicity — separate permission record

> **DRAFT — legal review required.** A blank box is **no permission**, never implied consent. This record is not itself a lawful basis for every processing purpose. Counsel must confirm authority, notices and the appropriate permission process. Keep completed forms privately; repository copies contain placeholders only.

Customer/pilot: [CUSTOMER LEGAL NAME] / [PILOT ID]
Responsible organization/contact: [LEGAL ENTITY] / [CONTACT]
Authorized representative/contact: [OWNER] / [CUSTOMER CONTACT]
Date/review: [DATE] / [REVIEW DATE]

Core delivery is a separate decision

Core pilot processing is limited to the reviewed agreement/proposal and privacy notice: [DATA USE PERMISSION]. Record the purpose, scope, organization roles and authority there. Refusing any optional use below must not quietly change access to the agreed core pilot or its price.

Optional choices — each requires its own explicit decision

No choice is preselected. For each option, record **permit / decline / not asked**, the exact approved scope and evidence of an informed authorized decision. Do not interpret a signature on a pilot proposal as selecting these options.

Optional use	Initial decision	Required scope before permission can be used
Research/calibration analysis	not asked	Specific dataset categories and sample IDs; methods/targets; recipients; duration; re-identification risk; withdrawal handling.
Model training/evaluation	not asked	Specific model/program and permitted stages; retention; lab/third-party rights; whether derived artifacts can be withdrawn and any limits. Synthetic demo metrics prove no field accuracy.
Named testimonial/quotation	not asked	Exact approved text, name/role, placement, channels, review/expiry and withdrawal route.
Farm/customer name and logo	not asked	Approved assets, purpose, channels, brand rules and publication period; signatory must hold relevant rights.
Photo/video or public case study	not asked	Exact approved media/draft, identifiable people, locations, results/context, channels and expiry. Separate individual permissions where needed.
Marketing contact	not asked	Recipient, channel, message purpose/frequency, duration and an effective opt-out process.

Per-option decision record — duplicate for each selected purpose

Field	Complete in restricted working copy
Purpose / decision	[SUMMARY] / [CONSENT STATUS]
Exact approved records, text or asset version	[SUMMARY] / [EVIDENCE LINK]
Permitted recipients, channels and organization roles	[SUMMARY]
Start, expiry and retention	[DATE] / [DUE DATE] / [RETENTION PERIOD]
Withdrawal contact and verified procedure	[CONTACT] / [SUMMARY]
Rights/authority for customer, people and lab material	[SUMMARY] / [EVIDENCE LINK]
Decision-maker and record of informed authorization	[OWNER] / [DATE] / [EVIDENCE LINK]
Responsible reviewer and next review	[OWNER] / [REVIEW DATE]

Explain plain-language risks: de-identification may not prevent re-identification; published/downloaded copies may not be recallable; deleting source rows does not necessarily remove influence from trained artifacts. Resolve a realistic withdrawal process **before** reuse rather than promise erasure that cannot be delivered.

Withdrawal and publication gate

1. Verify requester authority and identify affected permissions/assets/data.
2. Stop new optional use in scope; record the request in restricted evidence.
3. Assess removal from controlled channels, recipients, datasets and artifacts with

the responsible reviewers; explain actual limits, holds and approved timelines.

1. Record action, remaining copies/limits and the response; update permissions so

future imports, publications or restores do not silently reinstate the use.

No approved optional choices are supplied by this pack. Permission to discuss a pilot is not permission to publish its name, outcomes, photos, logo or contacts.

Executive One Pager

Nutri-Scan — executive one-pager

> **DRAFT — legal/accounting review required.** External-use draft; no validated performance, revenue or customer traction is asserted.

Sponsor: [FOUNDER NAME], [LEGAL ENTITY] · **Contact:** [CONTACT] · **Review:** [REVIEW DATE]

The problem to test

Dairy teams may keep silage/feed observations, readings and reports in separate notebooks and messages, making batch follow-up difficult. Discovery must establish how often this occurs, who owns the task and what it costs; no market-size or savings claim is assumed.

Initial customer and wedge

Start with a dairy farm managing multiple silage batches and a named operator, manager and qualified adviser/testing route. Offer **silage/feed screening workflow and traceable follow-up**, beginning with batch cards and observation-only checks when complete measurements are unavailable. Cooperatives/advisers are potential introduction channels, not confirmed customers.

What exists today

- Farm accounts and roles; feed/silage batch records, private photos, Yes/No/Unknown observations and in-app reminders.
- Farm-local testing-request tracking and report storage; staff contact their chosen provider separately.
- Four-step screening with all ten inputs required, reviewed optional USB-serial entry, uncalibrated scoring, saved history/trends and HTML/print reports.
- English/Hindi/Marathi interface guidance, explicit local reading drafts and a public-help PWA shell with limited offline scope.

The current demo uses synthetic serial values and a separate synthetic calibration scaffold. Photos are evidence only; nutrient values are supplied inputs, not measurements produced by this demo. See the demo script.

Proposed offer and economics

A bounded six-week workflow pilot: one farm, two operators and up to ten managed batches, with agreed onboarding and review. Quote **[PILOT PRICE]**; any continuation requires a separate **[MONTHLY PRICE]** offer. Hardware, laboratory work and extra visits require separate scope/pricing.

For internal hypothesis testing only, a ₹9,000 pilot produces ₹5,020 contribution before acquisition and fixed overhead; a ₹1,499/farm/month plan produces ₹919.02 monthly contribution under the illustrative cost assumptions. These are not approved prices or evidence of demand.

Why this could win

A practical batch-to-follow-up workflow combines record retrieval, evidence and explicit human review. The potential defensibility is consented workflow learning, reliable operations and eventually properly governed reference data. Neither an exclusive dataset nor a validated model moat exists today.

Next decision

Seek a discovery conversation, a scoped pilot sponsor or a qualified laboratory methods discussion. Agree a workflow baseline, ownership and [ACCEPTANCE TARGET] before starting; measure retrieval time, weekly usage and follow-up completion, then seek a signed paid continuation. Investment ask, if relevant: [FUNDING ASK] for [RUNWAY MONTHS] months, subject to a reviewed budget.

Claim boundary: No lab accuracy, toxin detection, certified calibration, diagnosis, guaranteed safety or automatic lab booking is offered. Use the claims matrix, pilot proposal and launch checklist before sharing an offer.

Go To Market Plan

Go-to-market plan

> **DRAFT — legal/accounting review required.** Funnel counts, conversion targets and timing are planning experiments, not traction.

Owner: [FOUNDER NAME] · [CONTACT] · **Review:** [REVIEW DATE]

ICP and buying roles

First ICP: a dairy farm actively storing silage in several identifiable batches, with recurring checks, a smartphone/browser, an operator and a manager willing to compare the current process with a six-week pilot. Prioritize a reachable cluster to control travel cost. Geography and farm-size thresholds must come from discovery.

Role	Candidate	Qualification question
User	Feed-store operator / farmer	Who records observations, retrieves reports and completes reminders?
Buyer	Farm owner / manager	Who approves a new workflow and evaluates its value?
Payer	Farm entity; possibly a sponsoring cooperative under separate terms	Who signs, receives invoices and controls the budget?
Influencer	Nutrition adviser / veterinarian / testing provider	Who reviews concerns and determines appropriate confirmatory testing?

A sponsor paying for several farms does not imply a cross-farm dashboard or shared data rights. Define each farm's account and permissions explicitly.

Message and channels

Opening: “Keep silage/feed batch observations, follow-up and reports together. Nutri-Scan also demonstrates reviewed, uncalibrated screening; it does not certify feed safety.”

Channel	First experiment	Record / stop condition
Founder-led farm introductions	Ask for consented 20-minute discovery calls.	Problem evidence and qualified meetings; stop repeatedly unqualified outreach.
Adviser referrals	Demo the batch/report handoff to a qualified adviser.	Accepted introductions and role clarity; no endorsement inferred.
Cooperative/producer organization	Propose a small opt-in member-farm cohort.	Named sponsor and independent farm permissions; no bulk data access assumption.
Laboratory methods outreach	Discuss sampling and report identity mapping.	Methods/fees/handling feasibility, not booking volume promises.
Small field demonstration	Use synthetic records and the demo script.	Comprehension and next meetings; no public customer data.

Use the outreach drafts and pipeline CSV. No grants, contacts, endorsements or customer logos are supplied.

Measurable sales funnel

Illustrative first eight-week outreach experiment: **40 consent-respecting contacts → 12 discovery interviews → 6 qualified opportunities → 3 pilot proposals → 2 signed/paid pilot starts**. This is a proposed activity funnel; pilots started late in this window finish later. It is not a revenue forecast. Keep unique farms/deals as denominators, not email counts.

Stage	Exit criterion	Action / evidence
Contacted	Relevant person agrees to a conversation.	Log source, role, permission and next date.
Discovered	Recent problem instance and incumbent process understood.	Interview guide; record objections and no-problem responses.
Qualified	Named buyer/user/payer, task frequency, data boundary, pilot dates and plausible budget.	Disqualify safety-certification/automatic-booking expectations and missing operational owner.
Demo completed	Prospect correctly explains limits and identifies a useful task.	Rehearse app/serial demo; capture comprehension and required inputs.
Proposal	Baseline, support cap, acceptance and commercial fields agreed in principle.	Customer-specific pilot proposal.
Pilot started	Reviewed terms signed, required payment received and setup/readback checked.	Agreement discussion template, invoice evidence and kickoff.
Paid continuation	Separate recurring scope signed and first continuation payment received.	Record offer/decision/payment dates and monthly fee; a verbal “yes” is not converted revenue.

Metrics: discovery qualification = qualified farms / interviewed farms; proposal win = paid starts / proposals; sales cycle = paid-start date − first qualified date. Track cash collected separately from signed value. Use a defined mature cohort for each rate.

Pilot-to-paid conversion playbook

1. **Before kickoff:** select one retrieval/follow-up task, baseline minutes, named operators and an agreed [ACCEPTANCE TARGET]. Quote [PILOT PRICE], support limits and separately priced exclusions.
2. **Week 1:** onboard up to two operators, create agreed batches, verify persisted readback and observation-only fallback. Label any demonstration scans synthetic; exclude them from adoption counts.
3. **Weeks 2–4:** review weekly active farm usage, timely follow-ups, missing readings, support minutes and failure reports. Document value evidence and adverse feedback, not just usage totals.
4. **Week 5:** buyer reviews baseline/current task evidence and a written [MONTHLY PRICE] offer. Identify approval and invoice steps; no silent conversion or automatic renewal.
5. **Week 6:** acceptance review and issue list. Separate workflow acceptance from any scientific accuracy discussion.
6. **Within 14 days after end:** log signed/paid continuation, decline or pending decision. Export/retention closeout follows reviewed terms if declining.

Conversion definition: farms with a signed continuation **and first payment** within 14 days / all pilot farms whose end dates are at least 14 days old. Include terminated pilots in that denominator and report reasons separately; do not hide failed pilots or count unfinished pilots as losses. A planning threshold of 50% after at least four mature pilots is a hypothesis, not statistical validation.

Partnership terms to settle

- **Adviser:** referral permission, role in interpretation, any fee/conflict disclosure and explicit customer sharing consent.
- **Cooperative:** sponsor/payer, per-farm access, training responsibilities and no implied member-data ownership.
- **Laboratory:** sample protocol, immutable IDs, methods/basis, fees, reporting, failed-QC rules and separate research permissions; use the MoU outline.
- **Device supplier:** protocol compatibility and maintenance evidence before naming a supported device; the default ESP32 sketch remains a demonstration.

Weekly review: [OWNER] updates stage, [NEXT ACTION], [DUE DATE], cost and evidence; [FOUNDER NAME] decides continue/revise/stop using the launch checklist and claims matrix.

Investor Pitch Outline

Investor pitch outline — 10 slides

> **DRAFT — legal/accounting review required.** Fundraising discussion outline only. No traction, market size, valuation, grant or investment commitment is asserted.

Presenter: [FOUNDER NAME], [LEGAL ENTITY] · [CONTACT] · **Review:** [REVIEW DATE]

Slide	Headline and content	Evidence / preparation
1. Mission and wedge	Make silage/feed batch evidence and follow-up easier for dairy teams. Lead with workflow value, not an accuracy claim.	Current one-pager; describe the product as a prototype.
2. Problem	Show a recent batch-to-report workflow and the handoff/retrieval friction to investigate.	Consented interview evidence and actual baseline task time; until collected, mark as a hypothesis.
3. Customer and beachhead	Farm operator = user; manager/owner = buyer; farm or explicit sponsor = payer. Several active silage batches and a named owner are qualification signals.	ICP and funnel; no unsupported farmer counts or TAM.
4. Product today	Batch → observation → reminder → local testing tracker → saved report; separately, reviewed complete-input screening with serial capture.	Rehearsed demo; synthetic demo records labelled on every screenshot.
5. Technical truth and validation	Deterministic uncalibrated scoring; device-reported calibration; synthetic offline metrics scaffold with no app integration.	Calibration workflow; separate future reference-study protocol and budget. Never show synthetic metrics as measured accuracy.
6. Positioning and defensibility	Complement labs, meters and advisers; compete with fragmented records for one focused task. Potential moat is workflow reliability and permitted learning, not existing exclusive data.	Competition/moat discussion; evidence still to collect. No customer/partner logos without authorization.
7. Business model and economics	Paid bounded pilot → separately signed per-farm monthly service; lab/hardware costs separately scoped.	Illustrative ₹9,000 pilot and ₹1,499/month base case only; unit economics shows ₹919.02 monthly contribution and 131-farm modeled operating break-even.
8. GTM and traction evidence	Founder discovery, adviser/cooperative introductions and a small cohort. Show actual stage counts with dates and definitions when available.	Proposed 40 contacts → 12 interviews → 6 qualified → 3 proposals → 2 paid starts is a hypothesis. Separate interest, signed pilots, cash collected and paid continuations.
9. Team, milestones and risks	[FOUNDER NAME] owns commercial delivery; identify actual methods, engineering and field-support gaps. Discovery → bounded pilot → paid decision; validation is a separate gate.	Use only verified biographies. State data, support-cost, input availability and claim-misinterpretation risks with accountable actions.
10. Ask and use of funds	Seek [FUNDING ASK] for [RUNWAY MONTHS] months, subject to budget and agreed financing terms. Request a defined next diligence meeting.	Month-by-month allocation for product/deployment, pilot delivery, reference validation, sales and working capital; amounts must reconcile to ask and runway after review. No grants or commitments assumed.

Presenter preparation

- Attach an evidence date to every number. Keep proposed KPIs and observed results visibly distinct.
- Build market sizing bottom-up only after discovering reachable farms, qualification rates and an evidenced price: reachable qualified farms × tested annual fee. Do not present the product as serving every dairy farmer today.
- Carry a limitations slide/appendix using the claims matrix. No lab accuracy, toxin detection, certified calibration, diagnosis, guaranteed safety or automatic lab booking claim.
- Have counsel review the fundraising communication and any customer/data references; have an accountant reconcile the financial model. Use placeholders until identity, budget and terms are verified.

Lab Partner Mou Outline

Lab partner MoU — non-legal discussion outline

> **DRAFT — legal/accounting review required. NON-LEGAL, NON-BINDING DISCUSSION OUTLINE.** Counsel must determine which obligations, if any, belong in a separate executed agreement. No laboratory appointment, accreditation, capacity or turnaround is asserted.

Proposed parties: [LEGAL ENTITY] / [LAB PARTNER] · **Contacts:** [CONTACT] / [LAB CONTACT] · **Review:** [REVIEW DATE]

1. Purpose and workstream selection

Discuss one or both **separate** workstreams: (a) customer-arranged testing/report follow-up, and (b) a prospective reference-data study. A service discussion does not authorize research/model training. The app only tracks farm-local requests; contacting a provider, arranging samples and fees remains a human process. No automatic lab booking, directory, guaranteed referral volume or exclusive relationship is implied.

2. Scope and division of responsibilities

Topic	Nutri-Scan proposal	Laboratory discussion point
Customer testing	Record farmer-selected contact, request/collection/report status; store customer-authorized reports.	Confirm accepted sample types, ordering steps, methods, fees, receipt and reporting process.
Sampling	Coordinate an agreed chain-of-custody form; no claim the current UI captures a complete scientific sample record.	Approve collection quantity, container, mixing/homogenization, storage/time limits, transport and rejection rules.
Reference study	Capture raw device output and immutable IDs under separate consent/protocol.	Provide appropriate qualified methods, QC metadata and traceable reference reports.
Interpretation	Present reports for download, without OCR or automated interpretation.	Define any professional interpretation service, responsibility, limits and separate fee.
Logistics / incidents	Name human coordinator and escalation contacts.	Define lost/damaged/insufficient/failed-QC sample handling, recollection and costs.

3. Method and identity contract

Follow the real collection protocol before reference collection:

- Specify target(s), sample types, [LAB METHOD], units, wet/dry/as-fed basis, ranges and any applicable detection/quantification limits. Verify actual method suitability and qualification evidence; do not insert invented accreditation.
- Agree [SAMPLE COUNT] based on expert study design, intended variation and independent groups; no statistically adequate sample size is supplied here.
- Assign immutable sample IDs linked to farm, batch, sample type, collection/capture time, operator, device/version, conditions and handling events. Preserve raw outputs/reports and correction history. Never join results by row order

or informal description.

- Record whether device and lab aliquots came from the same homogenized material. Include receipt, storage, preparation, analysis time, QC status and replicates; agree rejection/exclusion rules **before** looking at performance.
- Keep failed-QC/contaminated samples traceable and quarantined pending expert review. Any intended contamination target requires its own method/design; this outline does not create toxin detection capability.

4. Study design and release gate

Pre-register target definitions, acceptance tolerances and farm/batch/time grouping. Fit preprocessing/models on train only; tune on validation; freeze before opening the locked test set. Duplicate aliquots, repeats and lab replicates must not cross split boundaries.

Report count/coverage, MAE, RMSE, signed bias, R^2, confidence intervals where feasible, subgroup error, repeatability and reference uncertainty. Evaluate sensitivity/specificity and false negatives when threshold decisions are intended. Seek laboratory/domain and product-safety review of any proposed release. The synthetic scaffold is not validation, its coefficients are not app-integrated, and a high R^2 alone authorizes no safety claim.

5. Data, IP, confidentiality and publication

Agree customer, laboratory and supplier rights separately: source records, reports, raw measurements, derived datasets, model artifacts and any joint work. Document [DATA USE PERMISSION], approved access, onward sharing, research/training permission, [RETENTION PERIOD] and [EXPORT WINDOW]. No ownership transfer, exclusive dataset or training right is presumed.

Set de-identification, correction, secure transfer and incident procedures appropriate to the agreed project. Publications, names, logos and case studies need separate written approval and claim review. Commercial referral relationships and any fee/conflict disclosure require review.

6. Fees, service expectations and exit

- [LAB FEES]: specify per-test/method charges, collection/transport, repeat/rejected sample fees, invoicing payer, [PAYMENT TERMS] and [TAX TREATMENT]. Do not assume referral margin or pass-through authority.
- Discuss turnaround from accepted receipt, capacity, holiday handling, report corrections and escalation. Record only verified commitments; none is supplied by this outline.
- Define duration, [NOTICE PERIOD], termination, outstanding samples/fees, data closeout and continuing confidentiality.
- Counsel to resolve any binding status, responsibility allocation, [LIABILITY CAP], [GOVERNING LAW] and [DISPUTE PROCESS]. No liability term is established here.

7. Next meeting agenda

Confirm methods/handling feasibility → obtain reviewed fee schedule → select service versus reference-study scope → draft protocol and permissions → agree counsel-reviewed terms → authorize collection. Owner [OWNER], action [NEXT ACTION], due [DUE DATE]. Apply the claims matrix to every partner description and output.

Launch Checklist

Business launch checklist

> **DRAFT — legal/accounting review required.** Business readiness checklist for a bounded prototype workflow pilot. No item is pre-completed; a checked item needs dated evidence. Scientific/commercial targets are not achieved results.

Accountable founder: [FOUNDER NAME] · **Entity/contact:** [LEGAL ENTITY] / [CONTACT] · **Review:** [REVIEW DATE]

For each item record [OWNER], [DUE DATE], [STATUS] and [EVIDENCE LINK] in the pilot/launch working record. Use statuses `not_started`, `in_progress`, `blocked`, `done`; `done` requires evidence and reviewer, not intention. Do not interpret this checklist as regulatory certification.

Positioning and customer fit

- ☐ Agree the first ICP, buyer/user/payer and silage/feed workflow wedge; document discovery evidence and objections, not only positive feedback. GTM plan
- ☐ Review the one-pager, business plan, pitch and FAQ against the claims matrix.
- ☐ Remove any lab-accuracy, toxin-detection, certified-calibration, diagnosis, guaranteed-safety or automatic-lab-booking promise from customer-ready copy. Do not equate a LOW score or heuristic confidence with validated safety/accuracy.
- ☐ Verify identities and permission before using contacts, testimonials, grants or customer/partner logos; none is supplied by this pack.

Demo and pilot operational fit

- ☐ Rehearse the actual app/emulator script on the target browser/device; label synthetic data, unknown calibration and manual fallbacks honestly.
- ☐ Show observation-only checks for missing measurements; confirm all ten suitable readings are required for full screening. No invented inputs or photo-derived diagnosis.
- ☐ Confirm local-only testing requests, in-app reminders and report-download limits with the buyer. Name the human responsible for contacting a chosen lab/adviser.
- ☐ Explain public-help PWA versus private data, explicit draft sync and loss of instrument provenance in draft save/load. Agree connectivity and manual fallback expectations.
- ☐ Review the existing verification evidence and device gaps. Check deployment, farm-role access, saved readback and backup/restore responsibilities on the chosen pilot setup before real data; record actual results separately from historical tests.
- ☐ Agree language/measurement guidance review by a qualified domain reviewer and a route for concerns requiring professional review.

Commercial, accounting and legal readiness

- ☐ Complete the proposal: dates, caps, deliverables, baseline, [ACCEPTANCE TARGET], exclusions and named operator/buyer/payer.

- ☐ Replace illustrative pricing with [PILOT PRICE] / [MONTHLY PRICE] supported by delivery assumptions. Review support hours, processing, travel, hardware/lab exclusions, fixed-cost allocations and cash timing. Unit economics
- ☐ Obtain accountant review of [TAX TREATMENT], invoice/payment handling, withholding if relevant, refunds and gross/net pass-through treatment; record [PAYMENT TERMS]. No tax status/rate is assumed.
- ☐ Have counsel turn the agreement discussion template into appropriate execution terms: authority, scope, acceptance, data rights, support, termination, liability, disputes and notices. Resolve [LIABILITY CAP], [GOVERNING LAW] and [DISPUTE PROCESS].
- ☐ Agree [DATA USE PERMISSION], farm-role access, external sharing, [RETENTION PERIOD] and [EXPORT WINDOW]. Training/research and branding need separate permission; manually delivered export/deletion commitments need technical confirmation.
- ☐ Set [SUPPORT HOURS], [RESPONSE TARGET], escalation contact and change-control process; distinguish response from resolution/uptime. Complete signature/payment prerequisites under reviewed terms.

Partners and scientific boundary

- ☐ Record verified partner status in the pipeline; outreach/interest is not an executed relationship or accreditation.
- ☐ If laboratory collaboration is needed, settle methods, sampling/chain of custody, reports, fees, responsibility and permissions using the MoU outline.
- ☐ Keep any reference study separate from workflow acceptance and pricing. Before collection, freeze intended targets, representative sampling, units/basis, immutable IDs, QC rules, grouped/time splits and acceptance thresholds with qualified owners. Calibration protocol
- ☐ Keep synthetic scaffold reports and coefficients out of accuracy claims. Future expanded claims need target-specific held-out evaluation, uncertainty and expert release approval; no sales deadline substitutes for validation.

Measurement, conversion and closeout

- ☐ Complete consented discovery logs; distinguish observed behavior, stated intent, signed commitments and cash received.
- ☐ Agree baseline/current task method and KPI log denominators; exclude demo entries from adoption counts and show missing data explicitly.
- ☐ Review support load, collections, missing-input failures, task retrieval and follow-up weekly; re-scope if costs or confusion exceed the agreed boundary.
- ☐ Hold the end-of-pilot acceptance review and document issues/remedies. Seek a separate signed/paid continuation; measure conversion using mature cohorts, including terminated pilots. Conversion definition
- ☐ If ending, deliver the agreed manual export/access/retention closeout; document costs, reasons and lessons. No silent renewal or presumed research reuse.

Release decision record

Decision field	Complete with evidence
Pilot / customer	[PILOT ID] / [CUSTOMER LEGAL NAME]
Business review	[FOUNDER NAME] — [DATE] — [STATUS] — [EVIDENCE LINK]
Counsel review	[OWNER] — [DATE] — [STATUS] — [EVIDENCE LINK]
Accounting review	[OWNER] — [DATE] — [STATUS] — [EVIDENCE LINK]
Domain review where required	[OWNER] — [DATE] — [STATUS] — [EVIDENCE LINK]
Open blockers / next action	[SUMMARY] / [NEXT ACTION] / [DUE DATE]

Decision field	Complete with evidence
Decision	[SUMMARY] — explicitly record proceed, revise or pause, the agreed scope and decision-maker.

Before releasing an external copy, resolve its placeholders against the canonical register, check links, label examples, and retain a dated approved version in the agreed working record. This checklist itself is not approval or proof that a pilot is ready.

Partner Outreach Emails

Partner outreach emails

> **DRAFT — legal/accounting review required.** Replace placeholders, verify recipient/contact permission and review any proposed fee/data terms before sending. These are drafts; no message has been sent and no relationship is implied.

Use the partner pipeline. Send from a verified [CONTACT]; no contacts, accreditations, grants, endorsements or logos are supplied. Stop on decline and follow agreed contact preferences.

1. Dairy farm discovery introduction

To: [CUSTOMER CONTACT]

Subject: A short conversation about silage/feed batch follow-up

Hello,

I am [FOUNDER NAME] from [LEGAL ENTITY]. We are exploring how dairy teams keep silage/feed observations, follow-ups and testing reports linked to the right batch.

Could we ask about your most recent batch-check and report-retrieval process in a 20-minute conversation? We are looking for practical problems, including cases where your current process works well; no private reports are needed.

Nutri-Scan is a prototype workflow tool with optional reviewed, uncalibrated screening. It is not a laboratory replacement or feed-safety certification. If there is a fit, we can discuss a separately scoped pilot rather than assume a commitment.

Would [DATE] suit, or should we close this enquiry?

[FOUNDER NAME] · [LEGAL ENTITY] · [CONTACT]

2. Adviser or cooperative introduction

To: [CUSTOMER CONTACT]

Subject: Explore a small, opt-in silage recordkeeping pilot

Hello,

I am [FOUNDER NAME] from [LEGAL ENTITY]. Nutri-Scan currently supports farm-scoped batch records, observation-only checks, in-app follow-up and local testing/report tracking. We would value your view on whether this could help an operator retrieve evidence for qualified adviser review.

Could we show a 12-minute synthetic-data demo and discuss introductions to suitable, willing farms? There is no assumed endorsement, referral fee, multi-farm dashboard or access to member-farm data. Each farm's scope, payer and sharing permission would be agreed separately.

Scores are uncalibrated prototype indicators; photos are evidence only, and testing requests are not sent/booked automatically. We offer no lab accuracy, toxin detection, diagnosis or safety guarantee.

If useful, our next step would be a discovery meeting on [DATE], followed by a reviewed proposal with [PILOT PRICE] and explicit exclusions.

[FOUNDER NAME] · [LEGAL ENTITY] · [CONTACT]

3. Qualified laboratory methods discussion

To: [LAB CONTACT] at [LAB PARTNER]

Subject: Feed/silage sampling and reference-method feasibility discussion

Hello,

I am [FOUNDER NAME] from [LEGAL ENTITY]. We are developing a prototype for reviewed feed/silage record capture and follow-up. We would like to understand your accepted sample types, sampling/transport requirements, suitable methods, units/basis, fees and reporting process.

Separately, we may explore a consented reference-data study under an expert-designed protocol. The current calibration demonstration uses synthetic data and is not evidence of accuracy. We have no validated toxin-detection or lab-equivalent claim.

Would you be open to a methods meeting on [DATE]? Please advise what verified method/qualification information and fee schedule can be shared. No volume, turnaround, accreditation, referral or data-use right is assumed. The app does not book tests automatically.

We can use our non-legal MoU outline as a discussion agenda, with any service/research commitments subject to separate legal/accounting review.

[FOUNDER NAME] · [LEGAL ENTITY] · [CONTACT]

4. Device supplier compatibility enquiry

To: [CUSTOMER CONTACT]

Subject: Reviewed USB-serial feed-reading workflow — compatibility enquiry

Hello,

We are checking whether a device you supply could support our documented 115200-baud newline-JSON capture workflow, with explicit user selection and confirmation. We would need actual interface details, units, timestamps, supported environments and calibration/maintenance evidence before naming any compatible device.

Our default ESP32 sketch and emulator use synthetic readings with unknown calibration; we do not claim a calibrated meter integration, nutrient measurement or plug-and-play compatibility. Could we arrange a technical discussion on [DATE]? Any hardware tests, warranty responsibilities and commercial terms would be separately agreed.

[FOUNDER NAME] · [LEGAL ENTITY] · [CONTACT]

5. Follow-up after a real meeting

To: [CUSTOMER CONTACT] or [LAB CONTACT] — select the actual recipient before sending

Subject: Agreed next step from our Nutri-Scan discussion

Hello,

Thank you for the discussion on [DATE]. Our understanding is: [SUMMARY]. Please correct anything inaccurate.

Proposed next step: [NEXT ACTION], owned by [OWNER], by [DUE DATE]. Any pilot fee [PILOT PRICE], laboratory fees [LAB FEES], data use and delivery responsibilities remain subject to a completed, reviewed scope. This message creates no booking or automatic renewal.

If this is no longer useful, please let us know and we will close the enquiry.

[FOUNDER NAME] · [LEGAL ENTITY] · [CONTACT]

Before sending

Use the claims matrix, attach only an appropriate reviewed one-pager or proposal, and remove unresolved fields in that external copy. Log only verified contact details and permission in a restricted location; a reply is interest, not a partner agreement or customer logo authorization.

Pilot Agreement Template

Pilot agreement — non-legal discussion template

> **DRAFT — legal/accounting review required. NOT A LEGAL AGREEMENT OR READY-TO-SIGN CONTRACT.** This is a discussion checklist for qualified counsel/accountants to turn into appropriate terms. No clause is represented as sufficient or enforceable in any jurisdiction.

1. Parties, authority and term

- Proposed supplier: [LEGAL ENTITY], represented by [FOUNDER NAME], notice contact [CONTACT].
- Proposed customer: [CUSTOMER LEGAL NAME], authorized representative/notice contact [CUSTOMER CONTACT]. Verify signatory and payer authority in the reviewed contract.
- Pilot [PILOT ID], from [PILOT START DATE] to [PILOT END DATE]; counsel to specify effective date, addresses and notice method.
- Review owner/date: [OWNER] / [REVIEW DATE]. No automatic continuation or renewal; later paid service needs separately agreed terms.

2. Scope and deliverables

Attach a completed pilot proposal with farm/operator/batch caps, included hours, devices/browser, deployment responsibilities, dates, deliverables and explicit exclusions. Counsel must resolve order of precedence between proposal and agreement.

Current scope may include batch records, observation-only checks, reminders, farm-local testing requests, private evidence/reports and reviewed uncalibrated screening. All ten suitable inputs remain required for a full screening. Photos are evidence only. Device calibration status is a report, not verification. The prototype is not a laboratory replacement.

Exclude hardware supply/calibration, laboratory work, sample logistics, professional advice and research validation unless separately agreed. There is no automatic lab booking, toxin detection, certified calibration, diagnosis, guaranteed safety or lab accuracy commitment. No customer acceptance term may imply such a claim.

3. Acceptance and changes

- Attach predefined workflow tests, baseline and [ACCEPTANCE TARGET], evidence sources, review date and named reviewers. Separate observation-only use from full-screening demonstration.
- Customer records accepted deliverables, specific defects and agreed remedies in a written review. Discuss a remediation/retest period and dispute route; do not assume silence means acceptance.
- Define how partial acceptance affects payment/refunds in [PAYMENT TERMS]. Accuracy, feeding clearance or safety are not acceptance criteria.
- Both parties approve any added scope, hours, price, dates or data purpose in writing before work begins; prototype features are not an open-ended development obligation.

4. Data ownership, permissions and confidentiality

Proposed allocation for counsel review: customer retains its rights in customer-supplied farm records, photos and reports, subject to third-party rights. [LEGAL ENTITY] retains pre-existing software/methods and does not acquire farm-data ownership merely by hosting records. Resolve rights to generated reports, joint work and lab-provided data explicitly; no transfer of a laboratory's rights is implied.

Record [DATA USE PERMISSION] covering the minimum processing needed to deliver the pilot. Specify data categories, permitted purposes, controller/processor or other applicable roles, authorized staff, hosting/subprocessors, security responsibilities and incident contacts/timing. No model training, research reuse, sale, cross-customer sharing, logo use or publicity is presumed; each requires separate appropriate permission and review.

Agree confidentiality scope, recipient obligations, permitted disclosures and return/deletion handling. Farmer-selected external sharing is manual/opt-in and may create recipient-held copies. Private role controls are prototype safeguards, not audited-security/compliance claims. Do not place passwords or sensitive personal data in pipeline/KPI CSVs.

5. Retention, access and closeout

- Retention period: [RETENTION PERIOD]; requested export window: [EXPORT WINDOW]. Define formats, recipients, identity verification, fees (if any) and delivery responsibility.
- Current UI supports saved HTML reports, browser Print / Save PDF and individual attachment downloads. Bulk export, deletion, account deactivation and backup cleanup are **not promised self-service features**; any manual commitment needs technical confirmation, a named owner and a reviewed schedule.
- Counsel must address legal retention exceptions, backup expiry and copies already shared externally. No immediate complete-erasure promise.
- Verify farm-role access and saved readback before real use; specify customer device/network duties and supplier backup/restore duties for the agreed deployment.

6. Support and operating limits

Channel [CONTACT]; support window [SUPPORT HOURS]; initial response target [RESPONSE TARGET]. Define included hours, severity levels, outages, escalation, exclusions, visit costs and handling outside the support window. Response is not resolution or uptime. Make the manual fallback and reporting path explicit.

The app's public-help PWA and local drafts do not provide offline private-record access; draft save/load does not retain instrument provenance. In-app reminders do not send SMS/background notifications. Testing requests are not transmitted/booked externally and reports are not automatically interpreted. Customer remains responsible for contacting qualified professionals for feed/animal concerns.

7. Payment and taxes

Pilot fee [PILOT PRICE] INR; continuation, if separately agreed, [MONTHLY PRICE] INR/farm/month. Complete [PAYMENT TERMS] for deposit, milestones, invoices, due dates, disputed invoices, late/non-payment, refunds and cancellation. Complete [TAX TREATMENT] with an accountant; no tax registration/rate or gross/net treatment is asserted.

Specify who pays hardware, lab fees, transport and extra visits; include no unquoted pass-throughs. Acceptance, suspension and payment interaction must be explicit. Do not enter payment credentials in this template.

8. Termination and suspension

Counsel to define termination for convenience/breach, notice method and [NOTICE PERIOD], cure opportunity, non-payment suspension and any immediate suspension circumstances. Settle fees for work performed, deposits/refunds, unfinished deliverables and disputed sums in [PAYMENT TERMS].

On exit, follow agreed access, export, retention and confidentiality steps; identify a human owner for manual closeout. Do not delete customer records or promise continued access without reviewed terms and technical confirmation. Stop use of counterpart branding according to separately agreed permissions.

9. Warranties, liability and allocation of risk — counsel to draft

Discuss precisely what delivery commitments are appropriate for a prototype, available remedies, exclusions, mandatory rights and any insurance needs. Proposed liability-cap field: [LIABILITY CAP]. Counsel must determine basis, duration, applicable exclusions/carve-outs and whether any limit is lawful; **no liability limitation is established by this outline**.

Discuss data/confidentiality incidents, third-party claims, device/sample handling, professional reliance and any indemnities without inserting blanket waivers. No disclaimer creates permission to make unsupported safety/performance claims. Neither party should treat scoring as diagnosis, certified testing or a feeding guarantee.

10. Disputes and execution gate

Proposed [GOVERNING LAW] and [DISPUTE PROCESS] require counsel to select jurisdiction, escalation, forum and applicable formalities. Confirm representative authority, final annexes, notices, entire-agreement/change terms and signature method in a separate reviewed execution copy.

Open points: [SUMMARY] · **Next action:** [NEXT ACTION] · **Due:** [DUE DATE]. Customer-facing copy must match the claims matrix; resolve every placeholder and complete the launch checklist before execution.

Pilot Proposal Template

Nutri-Scan pilot proposal

> **DRAFT — legal/accounting review required.** Editable scope proposal, not a binding offer. All examples are planning hypotheses; replace placeholders and attach reviewed terms before kickoff.

Field	Complete before customer approval
Supplier / sponsor	[LEGAL ENTITY] / [FOUNDER NAME] / [CONTACT]
Customer / buyer	[CUSTOMER LEGAL NAME] / [CUSTOMER CONTACT]
Pilot reference	[PILOT ID]
Dates	[PILOT START DATE] through [PILOT END DATE]
Review date	[REVIEW DATE]
Operator / payer / adviser	[ROLE] — record each responsible person and payer separately.

1. Customer problem and baseline

Recent problem and incumbent process: [SUMMARY]

Choose one task, for example retrieving the latest batch report and identifying the pending follow-up. Before training, measure [BASELINE MINUTES] across comparable attempts; document task definition, number of attempts, device and evidence. Do not infer reduced losses or improved animal outcomes from task timing.

2. Proposed scope

Planning example: six weeks, one farm account, up to two operators and ten managed silage batches; four onboarding hours and two follow-up/support hours. Replace with the agreed dates, caps and responsibilities before quoting.

Included activities:

- Set up agreed farm roles and batch identities; train users on observations, private evidence and in-app follow-up.
- Track manually arranged testing requests and store/download trusted customer-supplied reports.
- Demonstrate reviewed serial/manual input and uncalibrated scoring **only with all ten suitable inputs**. Keep synthetic training records separate from real workflow records and exclude them from KPIs.
- Review weekly usage, task timing, follow-up completion, missing inputs and delivery effort.
- Provide a final outcome review and the agreed export/closeout assistance.

Excluded unless separately scoped: hardware supply/validation, sensor calibration, laboratory fees/sample logistics, scientific reference study, multi-farm dashboard, SMS, automatic lab booking, OCR/report interpretation, AI image analysis and professional feeding advice. No lab accuracy, toxin detection, certified calibration, diagnosis or guaranteed safety.

3. Responsibilities and data

Party	Proposed responsibility
[LEGAL ENTITY]	Agreed setup, onboarding, issue tracking, weekly review and manual closeout support; verify saved readback on the pilot setup.
[CUSTOMER LEGAL NAME]	Name buyer/operator/payer, provide appropriate source data, authorize access, maintain devices/connectivity and arrange qualified advice/testing.
Chosen testing provider	Methods, sampling instructions, fees and reports under a separate agreement; no provider is presumed appointed.

Agree [DATA USE PERMISSION], farm-role access, sharing restrictions, [RETENTION PERIOD] and [EXPORT WINDOW]. Training/research, logos and testimonials need separate permission. Current exports are saved HTML reports, browser printing and individual attachments; any broader export/deletion is a manual delivery commitment only if explicitly agreed and technically confirmed.

4. Success and acceptance

Set [ACCEPTANCE TARGET] jointly before kickoff. Illustrative candidates: weekly use in four of six weeks; 80% of agreed follow-ups completed by their due dates; 20% lower median retrieval time across at least five comparable attempts. These are **proposed workflow targets**, not promised improvements or scientific validation.

Measure	Definition / evidence
Weekly active farm	At least one agreed non-demo observation, managed-batch update or follow-up status change in a week.
Follow-up completion	Tasks completed on/before due date / all agreed tasks due in the period; include missed tasks.
Retrieval change	`(baseline median − current median) / baseline median`; same task, comparable attempts, positive baseline; report sample count and exceptions. Current value: [CURRENT MINUTES].
Workflow integrity	Saved identity/readback, Unknown preserved, role access and request/report progression checked on the agreed setup.
Delivery load	Onboarding/support/visit hours and out-of-scope requests logged separately.

Use the KPI log. A pilot with no full screenings can still meet agreed observation/follow-up goals; do not report it as evidence of scoring performance. Final review records accepted items, failures and remedies; silence does not imply acceptance. Payment consequences of unresolved items belong in reviewed [PAYMENT TERMS].

5. Price, support and change control

- Pilot fee: **[PILOT PRICE] INR**, with [TAX TREATMENT] and [PAYMENT TERMS]. Confirm deposit, invoice dates, refund/cancellation handling and travel caps; no model number is an approved quote.
- Support channel [CONTACT], availability [SUPPORT HOURS], initial response target [RESPONSE TARGET]. This is not a resolution or uptime guarantee. Critical feed/animal concerns go to the customer's qualified professional, not the app queue.
- Additions require written scope, effort, data and price approval by both parties before delivery.
- Continuation is a separate offer at [MONTHLY PRICE] per farm/month; no automatic conversion or renewal.

6. Proposed sequence and approval

Kickoff/baseline → setup/readback → weekly reviews → week-five continuation discussion → end-of-pilot acceptance → continuation or export/closeout. Align actual dates with [PILOT START DATE] / [PILOT END DATE].

Next action: [NEXT ACTION] · Owner: [OWNER] · Due: [DUE DATE]. Before execution, complete the agreement discussion template, legal/accounting review and launch checklist. Apply the claims matrix to all attachments.

Pricing And Unit Economics

Pricing and unit economics

> **DRAFT — legal/accounting review required.** Every numeric price, cost, volume and target in this document/CSV is an **illustrative INR hypothesis**, not an approved quote, supplier price, accounting policy or forecast.

Owner: [FOUNDER NAME], [LEGAL ENTITY] · **Review:** [REVIEW DATE]

Unit, scope and assumptions

- Unit = **one farm account per month**; not one animal, sample, scan, cooperative or sensor. A recurring offer is a proposed commercial package, not an in-app billing feature.
- Pilot hypothesis = six weeks, one farm, up to two operators and ten managed batches, with four onboarding hours and two follow-up/support hours. No scientific accuracy acceptance.
- Recurring hypothesis = existing supported software workflow plus 0.75 direct support hours/farm/month at ₹400/hour fully loaded. Extra visits/effort require written scope.
- Price/variable costs exclude GST and other applicable taxes. [TAX TREATMENT] must resolve registration, invoice wording, input credits, withholding and recognition before sale; no tax rate is assumed.
- Collections/payment processing assumption = 2% of revenue. Assume full collection only for this model; add refunds, bad debt and settlement timing when known.
- Lab fees, hardware/BOM, shipping, sensor calibration/replacement, insurance and field validation are **excluded from both price and cost** and must be separately quoted/funded. Never call this an all-inclusive testing package.
- No annual prepayment, subsidies, grant receipts, device margin or referral commission is assumed. No automatic renewal or automatic lab booking.
- Non-labor variable costs: ₹150 hosting/storage + ₹100 routine operations/farm/month. Validate usage and capacity; these are not vendor quotes.

Customer-ready terms use [PILOT PRICE], [MONTHLY PRICE] and [PAYMENT TERMS]; do not copy model numbers into an offer without review.

Six-week pilot example

Illustrative line item	INR
Total pilot fee	9,000
Onboarding: 4 h × ₹400	1,600
Follow-up/support: 2 h × ₹400	800
Local travel allowance	800
Hosting/storage allowance for full pilot	300
Materials/operations allowance	300
Processing: 2% × ₹9,000	180

Illustrative line item	INR
Total direct variable delivery cost	3,980
Contribution before CAC and fixed overhead	5,020
Contribution margin	55.78%

With illustrative acquisition cost (CAC) ₹3,000 per acquired paying farm, pilot contribution after CAC is ₹2,020 before fixed overhead. A pilot fee is one-time, **not MRR**. Additional physical samples and lab comparisons are a separate study, not hidden inside this allowance.

Recurring formulas

```
P = monthly price per farm (INR)
h = direct support hours per farm/month
w = loaded support cost per hour = 400 INR (illustrative)
C = h*w + hosting 150 + operations 100 + 0.02*P
CM = P - C                        # monthly contribution, INR/farm
CM% = 100*CM/P
F = fixed monthly operating cost = 120000 INR (illustrative)
Operating result at N paying farms = N*CM - F
Break-even farms = ceil(F/CM)       # only when CM > 0
CAC = acquisition spend / new paying farms acquired in the same cohort
CAC payback months = CAC/CM         # contribution-based, before churn
MRR = sum of contracted monthly recurring fees for active paying farms
```

If CM ≤ 0, no finite volume break-even exists under these assumptions. Distinguish MRR, invoices issued and cash collected. CAC payback here ignores pilot contribution and assumes unchanged support and zero churn during payback; do not subtract the same pilot contribution twice. Include founder sales time, travel and channel fees in measured CAC.

Scenarios

Same assumptions as the pricing CSV; ₹3,000 CAC and ₹120,000 fixed costs throughout. Stress changes support only.

Illustrative scenario	P INR	Support h	Variable cost INR	CM INR	CM%	Break-even farms	CAC payback months
Low-price hypothesis	999	0.75	569.98	429.02	42.94%	280	6.99
Base hypothesis	1,499	0.75	579.98	919.02	61.31%	131	3.26
Higher-price hypothesis	2,499	0.75	599.98	1,899.02	75.99%	64	1.58
Base with support stress	1,499	1.50	879.98	619.02	41.30%	194	4.85

At base assumptions, 50 paying farms imply ₹74,950 MRR and **−₹74,049/month** after modeled fixed cost; 150 imply ₹224,850 MRR and **₹17,853/month**. These are arithmetic scenarios, not demand/capacity forecasts. At 150 farms, direct support alone is 112.5 h/month; sales, onboarding and travel require additional capacity.

Fixed cost and cash break-even

Illustrative F = ₹80,000 shared team/admin/product time + ₹10,000 baseline tools/cloud + ₹10,000 accounting/admin services + ₹20,000 fixed outreach/travel. The ₹400/hour delivery labor and per-farm hosting above are **incremental**, excluded from these fixed allocations; reconcile payroll, cloud and travel to prevent double counting in a real budget.

The volume break-even above is a steady-state operating contribution calculation, not cash break-even or profit after tax. A working cash model must include:

```
Closing cash = opening cash + customer cash receipts + funding cash received
               - direct cash costs - fixed cash costs - acquisition cash spend
               - capex - validation cash spend - tax/remittances - debt payments
Runway months = unrestricted opening cash / monthly net cash burn
```

Use runway only while burn is positive and relatively stable; model month-by-month otherwise. Upfront pilot cash may be deferred revenue depending on reviewed policy. Refunds/receivables and any purchased devices can delay cash break-even even when CM is positive.

What to measure before fixing prices

1. Ask buyer and payer about actual current spend and who can approve [PILOT PRICE]; test a written offer, not only stated interest.
2. Log support/onboarding/travel minutes and hosting usage by farm; replace every allowance with evidence.
3. Compare conversion, realized price, collections, retention and delivery load by cohort. No LTV claim until retention is observable; a churn-based extrapolation is not observed LTV.
4. Have an accountant review taxes, gross-versus-net pass-through treatment, cost allocations and cash timing; have counsel review exclusions, support limits and [PAYMENT TERMS].

Commercial positioning follows the business plan and claims matrix.

Privacy Notice Template

Customer privacy notice — review template

> **DRAFT — NOT FOR PUBLICATION.** This describes the reviewed MVP and proposed pilot handling; it is not a legal opinion or a compliance certification. The responsible organization must complete its roles, jurisdiction, providers, retention and request procedures with qualified counsel before collecting real customer data.

Responsible organization: [LEGAL ENTITY] · **Privacy contact:** [CONTACT]
Version/effective date: [DATE] · **Next review:** [REVIEW DATE]
Pilot/customer: [PILOT ID] / [CUSTOMER LEGAL NAME]

Why we process information

Nutri-Scan is a prototype for recording feed/silage batches, observations, supplied readings, evidence and follow-up. We process the minimum agreed information to provide the pilot, manage authorized access, support users and protect/recover the service. Scoring is uncalibrated and does not certify feed safety or replace a laboratory.

The final notice must identify the applicable organization roles and lawful grounds for each purpose. A farm owner's authorization does not automatically authorize processing of everyone else's personal information. Obtain required notices and permissions before recording workers, laboratory contacts or other individuals.

What may be recorded and where

Category	Current behavior / customer choice
Account and farm	Username, farm name, role, password hash and session records; the browser receives an HttpOnly session cookie, not a plaintext server token in local storage.
Farm activity	Batch/sample details, observations and notes, reminders, testing-request contacts/status, readings, results and optional reviewed device provenance.
Evidence	Uploaded farm photos and laboratory PDFs/images are private farm records. Normalized photos have metadata stripped, but faces/addresses visible in pixels can remain. Do not upload unrelated or unauthorized material.
Local drafts	User/farm-scoped readings and queue metadata remain in that browser until managed by the user. Drafts do not contain passwords, photos or API responses; instrument provenance is stripped. Logout does not guarantee stored drafts/downloads are erased.
Public offline help	The production PWA caches allowlisted public assets/help, not private API responses, photos, reports or personalized HTML.
Support and recovery	Restricted operator-configured logs/backups and minimized support/incident records; actual providers, location and access must be disclosed below.
Billing	Only the information necessary for reviewed invoices/payments. In-app billing is not implemented; never send passwords, card details or banking credentials in a support message.

Optional microphone, instrument and sharing features

- **Voice:** requires a deliberate browser permission/action. The browser's speech

provider may process audio externally and require a network; Nutri-Scan does not save audio. Identify the selected browser/provider and its policy before offering this feature in the pilot. Manual reading entry remains available.

- **USB serial:** requires port selection and explicit review/confirmation. Device ID,

capture time, selected units and device-reported calibration can be stored with a direct online submission. This is not independently verified calibration.

- **Sharing/export:** review before sending. Downloads, browser PDFs, printouts and

manually shared summaries/photos create copies outside server role controls. Selected recipients/apps apply their own terms. A QR exposes a batch route/identifier, not an access token or public safety certificate.

- **Laboratory requests:** stored in the farm only; creating a request does not send

it to a laboratory. Any manual transfer needs an agreed recipient, purpose and authority.

Access, hosting and other recipients — complete before issue

Required disclosure	Approved entry and evidence
Organization roles and lawful purpose/ground by category	[SUMMARY] / [EVIDENCE LINK]
Actual hosting/backup providers, countries and restricted operator access	[SUMMARY] / [EVIDENCE LINK]
Authorized farm owner/member/viewer access and staff confidentiality	[SUMMARY] / [EVIDENCE LINK]
Payment/support/speech providers actually used and their purposes	[SUMMARY] / [EVIDENCE LINK]
Any international transfers and applicable review/safeguards	[SUMMARY] / [EVIDENCE LINK]
Required disclosures to authorities and applicable process	[SUMMARY] / [EVIDENCE LINK]

No research/model-training reuse, sale, cross-customer disclosure, testimonial, logo or public case study is authorized by this notice alone. Use the separate permission form; do not bundle optional permission into access to the core pilot.

Retention, requests and security limits

Approved pilot retention: [RETENTION PERIOD]; exit/export window: [EXPORT WINDOW]. Before issue, attach a category-specific schedule covering accounts, farm records, attachments, drafts, exports, logs, backups and any separately authorized research. The operations proposal suggests starting periods; these are **not an implemented automatic retention policy**.

Contact [CONTACT] for access/correction/export, permission withdrawal or deletion requests. We must verify identity/authority without asking for passwords or cookies. The final notice must explain applicable rights, acknowledgment/response timing, complaint/escalation route and any legally required retention exceptions. The MVP does not offer complete self-service bulk export, farm erasure or account disablement. Only promise a manual process once its scope, operator and reliability are approved.

Backups and recipient-held exports can remain until approved expiry or disposal; do not promise immediate complete erasure or remote recall of shared files. Any restore must reconcile approved deletions before reopening access. Users should resolve unsynced drafts before clearing local storage and protect shared devices.

Farm roles and private routes are prototype controls, not an audited-security claim. Real-data use requires HTTPS, secure session configuration, restricted administrative access, tested backup/restore and the approved incident process. No system is represented as risk-free or already certified compliant.

Publisher checklist — all unverified

- ☐ Identity, contact, jurisdiction, organization roles and applicable grounds confirmed.
- ☐ Actual providers/locations, optional speech handling and transfers disclosed.
- ☐ Retention, manual rights/closeout process and incident responsibilities tested.
- ☐ Optional permissions remain separate, granular and initially unset.
- ☐ Counsel review, accessible language and version/change notification approved.
- ☐ Final notice issued before collection; evidence of delivery kept privately.

Reviewer/decision: [OWNER] / [DATE] / [STATUS] / [EVIDENCE LINK].

Quote And Order Template

Pilot quotation and order schedule — review template

> **DRAFT — NOT AN OFFER, TAX INVOICE OR READY-TO-SIGN CONTRACT.** Complete scope, costs and accountant/counsel review in a separate customer copy. This template does not collect payment, create an order or authorize data use.

Supplier/contact: [LEGAL ENTITY] / [CONTACT]
Customer/contact: [CUSTOMER LEGAL NAME] / [CUSTOMER CONTACT]
Reference: [PILOT ID] · **Issue:** [DATE] · **Validity end:** [DUE DATE]
Proposed term: [PILOT START DATE] to [PILOT END DATE] · **Review:** [REVIEW DATE]

Order scope

Attach the completed pilot proposal with named farm, user/batch caps, deployment/device/browser, deliverables, included hours, dates and workflow-only [ACCEPTANCE TARGET]. Identify whether any full-screening demonstration uses synthetic inputs or suitable real inputs; do not substitute one for the other.

The prototype does not establish nutrient prediction, lab accuracy, toxin detection, certified calibration, diagnosis or feeding safety. Photos are evidence only; testing requests are local records. No automatic external booking or lab interpretation.

Itemized schedule — complete amounts, do not assume blanks are zero

Item	Quantity/unit and included scope	Amount INR / status
Fixed-fee workflow pilot	Agreed dates, one farm and capped delivery effort	[PILOT PRICE]
Additional approved support or visit	[SUMMARY]	Separately quoted; not included unless approved
Hardware/loan/installation	[SUMMARY]	Excluded unless a reviewed separate schedule is attached
Laboratory work and sample transport	[SUMMARY]	Excluded; method/provider/fees separately approved
Validation/calibration study	[SUMMARY]	Excluded; separate protocol and budget
Applicable taxes/withholding treatment	Accountant-reviewed [TAX TREATMENT]	Complete before issue
Total payable and payment milestones	[PAYMENT TERMS]	Complete after scope/tax review

No modeled price from the pricing scenarios is an approved quotation. State explicitly what is net/gross, refundable and due, who invoices/pays any pass-through, and when work can begin. Use a verified secure payment channel outside this repository; never embed banking/payment credentials here.

Linked terms and readiness prerequisites

- Counsel-approved agreement based on the pilot outline,

including precedence, remedies, cancellation/termination and liability discussion.

- Final privacy notice, permitted core data processing,

approved retention/export process and separate optional permissions where relevant.

- Agreed support [SUPPORT HOURS] / [RESPONSE TARGET]; response is not resolution or uptime.
- Actual technical readiness and

scoped go/no-go approval before real-data use, not merely signature/payment.

Continuation is **not automatic**. Any later recurring offer needs a separate scope, [MONTHLY PRICE], terms and explicit authorization; there is no in-app billing engine.

Review and acceptance evidence — complete privately

Gate	Named reviewer / date / decision / evidence
Scope, effort and exclusion review	[OWNER] / [DATE] / [STATUS] / [EVIDENCE LINK]
Accounting/tax/payment review	[OWNER] / [DATE] / [STATUS] / [EVIDENCE LINK]
Legal/data review	[OWNER] / [DATE] / [STATUS] / [EVIDENCE LINK]
Customer signatory/payer authority	[OWNER] / [DATE] / [STATUS] / [EVIDENCE LINK]
Final approved order/terms/version	[SUMMARY] / [EVIDENCE LINK]

Do not mark acceptance from a verbal expression of interest, an unchecked template or a missing response. Retain the authorized execution record and invoices in restricted commercial storage, not in a public code repository.

Sales Demo Script

Sales demo — today's app and ESP32-compatible emulator

> **DRAFT — legal/accounting review required.** Customer-facing wording must follow the claims matrix. This is a rehearsal script based on current sources, not a record that a live demo was run for this pack.

Presenter: [FOUNDER NAME] · [CONTACT] · **Review:** [REVIEW DATE] · **Duration:** approximately 12 minutes plus optional calibration/offline segment.

Before the meeting

1. Follow the existing same-day runbook for services: frontend `http://localhost:5173`, API `http://localhost:8000`. Use an existing suitable environment; this script needs no added package for the emulator.
2. Use a dedicated demo farm/account, synthetic records and a clearly labelled demo PDF/photo. Prepare a 10+ character password privately; do not put it in slides or CSVs.
3. Rehearse on supported Chromium with Web Serial on localhost/HTTPS. Confirm browser/OS port availability; a PTY may not appear on every setup. Have manual-entry fallback ready and say when it is used.
4. For PWA-only extras, use an already prepared production preview at `http://localhost:5174` per the runbook. Do not promise offline private records.
5. Start the Linux PTY emitter below in a terminal and retain its printed `/dev/pts/...` path. Stop with Ctrl-C after the demo. It emits fixed **synthetic** values with a timestamp generated at emitter startup, not measurements.

```
python3 - <<'PY'
import json, os, pty, time
from datetime import datetime

master, slave = pty.openpty()
print(os.ttyname(slave), flush=True)
frame = {
    "device_id": "host-emulator-01",
    "captured_at": datetime.now().astimezone().isoformat(timespec="seconds"),
    "calibration_status": "unknown",
    "readings": [
        {"field": "moisture_pct", "value": 62.0, "unit": "%"},
        {"field": "ph", "value": 4.2, "unit": "pH"},
        {"field": "temperature_c", "value": 24.0, "unit": "°C"},
        {"field": "humidity_pct", "value": 68.0, "unit": "%"},
    ],
}
line = (json.dumps(frame, ensure_ascii=False, separators=(",", ":")) + "\n").encode()
try:
    while True:
        os.write(master, line)
        time.sleep(1)
except KeyboardInterrupt:
    pass
PY
```

For a physical board, follow the firmware guide: default `DEMO_MODE=true`, calibration unknown, manual capture timestamp refreshed. Default firmware emits demo values, sends no HTTP request and accepts no serial commands. Replacing conversion hooks and validating real sensors is separate work.

Walkthrough and exact talk track

Time	Action	Say / check
0:00 – 1:00	Open app; identify demo farm.	“These are synthetic demo records. Nutri-Scan organizes silage/feed evidence and demonstrates uncalibrated prototype screening. It does not establish feeding safety.” Ask who currently records and reviews batches.
1:00 – 3:00	Open **Farm fieldwork** / **Batches, observations, reminders and testing help**. Create `DEMO-SILAGE-01`, name `Maize silage — West shed`, sample `Silage`, location `West shed`. Save/view a demo photo.	“This batch has a persistent identity. Photos are private evidence, not AI analysis.” Confirm the saved record can be read back.
3:00 – 4:00	**Quick observations**: leave unexamined answers **Unknown**, save.	Show **Observation only — no screening score**. “You can record observations without inventing measurements. Unknown is not No; concerns need qualified review.”
4:00 – 5:00	Add a reminder; complete/reopen it. In **Testing help**, create a farm request with clearly fictional demo contact text. Mark collected, upload/download a trusted demo PDF and mark report received.	“Reminders are in-app tasks. This request is tracked here, not sent or booked with a lab. The uploaded report is downloaded, not automatically interpreted.” A report attachment is required before report-received status.
5:00 – 6:00	Choose batch **Use this batch**, then **Check feed or silage** and reach **Enter readings**.	“Only batch identity is copied. Old readings do not carry over, and all ten screening inputs are required.”
6:00 – 8:00	Open **Read from instrument (USB serial)**; acknowledge protocol/115200 baud; **Choose port and capture**; select printed PTY.	“This is a software emulator of the ESP32-compatible newline JSON transport, not a sensor measurement.” Review `host-emulator-01`, time, units and **Unknown calibration (device report)**. Select the four readings, acknowledge review, press **Confirm selected readings**. Nothing auto-submits.
8:00 – 9:00	Enter the six remaining synthetic fields from the table below. Review all values, confirm and submit directly online.	“These additional values are illustrative. A real screening needs suitable measured nutrient inputs and a defined operator method for visual scores; photos do not populate them.”
9:00 – 10:30	Expand saved result details/provenance, refresh/read back history, download HTML report or use browser **Print / Save PDF**.	“Score, risk, fermentation/spoilage labels and confidence are heuristic outputs from supplied inputs, not validated laboratory findings or probabilities. LOW is not clearance to feed. Selected instrument metadata is retained on this direct online submission.”
10:30– 12:00	Return to useful task, ask for pilot baseline and sponsor.	“Would a six-week pilot of batch retrieval and follow-up be useful? Which task should we measure, who operates it, and who approves payment?” Use proposal; price remains [PILOT PRICE] until reviewed.

Synthetic supplemental inputs — demo only, never field defaults:

Field	Illustrative value	Meaning in demo
Protein	8.8%	Typed synthetic nutrient input, not ESP32 prediction.
Dry matter	38%	Typed synthetic nutrient input.
Crude fiber	28%	Typed synthetic nutrient input; not NDF/ADF.
Mould indicator	0.08	Typed synthetic score on 0–1 scale, not a toxin test.

Field	Illustrative value	Meaning in demo
Colour anomaly	0.05	Typed synthetic visual score.
Surface damage	0.02	Typed synthetic visual score.

Optional two-minute extensions

- **Calibration:** follow the calibration run command, writing only to an appropriate temporary output. Show `synthetic_demo_not_for_field_use`, farm-group splits, MAE/RMSE/bias/R^2 and train-only fitting. Say: “This demonstrates evaluation mechanics. Coefficients are not integrated into the app and do not validate its accuracy.”
- **Offline:** show public multilingual help on the prepared production PWA. In an already-loaded signed-in app, explain deliberate complete-reading draft save/queue and explicit **Sync now**. Drafts discard instrument provenance; do not use that path to demonstrate retained device metadata. Private history, photos, reports and fieldwork saves need reconnection/authentication.
- **Languages/voice:** English/Hindi/Marathi text and optional listening/voice entry depend on browser/device support; translations need qualified review. Voice proposes reviewed readings and never submits them.

Failure handling and close

- If serial is unsupported, unavailable, denied, malformed or times out, disclose the failure and use manual entry of the labelled synthetic values. Do not describe the fallback as successful serial capture. Never fill missing real data with demo numbers.
- If private save/readback fails, show the error and retry after connectivity is restored; do not claim persistence before readback. If printing is blocked, download HTML and use the browser print path.
- Preview any sharing action and use synthetic content only. A QR is an authenticated batch link, not public access or a safety certificate.
- Ask the prospect to explain three limits: prototype output, emulator data, and local-only testing requests. Log misconceptions and next action in the discovery log.

Approved close: “Nutri-Scan demonstrates reviewed ESP32-to-browser acquisition, batch records, observations and follow-up. Real nutrient prediction and contamination-related claims require suitable methods, laboratory-labelled data and separate validation. This demonstration offers no lab accuracy, toxin detection, certified calibration, diagnosis, guaranteed safety or automatic lab booking.”

Student Founder Start Plan

Starting Nutri-Scan as a mechanical engineering student

Start as a college project with one farm—not as a full company yet.
Your first goal is to prove that the workflow is useful before spending heavily.

This plan is tailored to a mechanical engineering student at Sinhgad, Solapur.
No college endorsement, mentor, facility, funding or partnership is assumed.
The steps and 14-day target below are proposed, not completed activities.

Five practical steps

1. **Find one faculty mentor.** Show the working synthetic-data demo and ask for

guidance and farmer introductions. Ask about the college's entrepreneurship or incubation cell **if available**. Check college/institution IP rules before commercializing work or using college resources externally.

1. **Build a small team.** Use your mechanical skills for enclosure design, mounting,

sample-handling design and field usability. Look for one electronics student for sensors and one software student for the app. Do not assume a teammate or module is already available; agree small responsibilities first.

1. **Speak to three dairy farmers.** Seek introductions through family, faculty or

a local veterinarian. Use the discovery guide:

- How do you identify and check stored silage/feed?
- How do you keep photos and laboratory reports?
- Which follow-up task is regularly missed?

Listen for recent examples; do not promise better milk yield or safe feed.
Ask permission for notes and keep identifying information outside Git.

1. **Agree a small trial with one willing farmer.** Start with batch records,

evidence photos, observation-only checks and in-app reminders—not a nutrient or safety test. You do not need expensive sensors to test this workflow. Use a synthetic demo initially; real farm data needs informed permission and the technical readiness gates.
Agree who operates the app, the task being tested, duration and support.

1. **Choose hardware only after the need is clear.** Do not buy an expensive NIR

instrument now. Actual nutrient/safety claims require suitable equipment, laboratory comparisons and expert validation. The ESP32 default/emulator values are synthetic and must be labelled; never use them to fill missing farm readings.

Your next 14-day target

One mentor + two teammates + three farmer interviews + one agreed small pilot.

Proposed window	Practical outcome
Days 1–3	Show the demo to a potential mentor and ask for introductions.
Days 4–7	Recruit teammates and arrange three permissioned farmer conversations.
Days 8–14	Identify one repeated problem and agree a small pilot scope with a willing farm; document outstanding readiness gates before real use.

This target is for discovery and agreement, **not automatic launch approval**.
If access, consent, security or support is unresolved, continue with synthetic demos and planning rather than rushing real-data use to meet the date.

Keep spending and promises small

- Use existing laptops/phones first. Agree a modest travel/materials budget with

your team/mentor before spending; no grant or free college facility is assumed.

- Prove usefulness before major hardware purchases or company-formation spending;

verify applicable requirements before selling or making commercial commitments.

- Consider charging only after scope, actual delivery costs, permissions, taxes and

reviewed terms are clear. A later six-week pilot proposal
can formalize the agreed workflow; it is not required for an initial conversation.

- Measure whether records are easier to retrieve and follow-ups are completed.

Do not claim laboratory accuracy, toxin detection or certified feed safety.

For the wider business and technical plan, return to the
launch guide. Your first action is simply to arrange the mentor demo.

Website And Launch Copy

Website and launch announcement — copy drafts

> **DRAFT — claims/legal review required before publication.** No customer, partner, result, launch date or production availability is asserted. Use the claims matrix; remove unresolved placeholders in approved public copies. This is copy, not a deployed website or a functioning lead form.

Landing-page structure

Hero

Keep feed and silage batch evidence in one guided workflow.

Nutri-Scan is a prototype for recording batch observations, supplied readings, photos, follow-up tasks and laboratory reports. We are exploring a small, scoped workflow pilot with dairy farms and field partners.

Primary action: Discuss a workflow pilot — [CONTACT]
Secondary action: Request a synthetic-data demo — [CONTACT]

Visible beside both actions: Prototype screening is uncalibrated. It is not a laboratory replacement, toxin test or feed-safety certification. Do not use its score as clearance to feed. Availability and scope require individual agreement.

The workflow

1. **Identify the batch:** record feed/silage identity and location.
2. **Record what you observed:** save concerns or Unknown answers without a score;

attach authorized evidence photos.

1. **Follow up:** maintain in-app reminders and local testing-request status; a

named person contacts the chosen laboratory/adviser outside the app.

1. **Review the record:** retrieve saved evidence and downloaded reports with

authorized farm access. Complete screening additionally requires ten suitable supplied inputs and produces uncalibrated prototype outputs.

Who should discuss a pilot?

A dairy farm with repeated batch-record or follow-up work, a named operator and an authorized buyer. Cooperatives, advisers and laboratories may discuss introductions or a separately defined partnership. No partnership, accreditation or cross-farm administration capability is implied.

Proposed pilot

A six-week, one-farm workflow pilot with agreed users, batches, onboarding/support hours, baseline and acceptance measures. We would measure record retrieval, task completion and delivery effort—not feed safety, scientific accuracy or guaranteed financial outcomes. Hardware, laboratory work and research validation are separately scoped. Pricing is supplied only after scope and professional review.

Short FAQ

- **Does a photo test feed?** No. Photos are evidence, not AI diagnosis.
- **Does the app book a laboratory?** No. Requests/status are farm-local records.
- **Can it work offline?** Installed public help can reload offline. Private data

needs connectivity; explicit reading drafts in an already-loaded app have limits.

- **Are the sensors validated?** Default firmware/demo uses synthetic readings;

real sensors need suitable integration and validation.

Footer: [LEGAL ENTITY] · [CONTACT] · final reviewed privacy notice and service terms. Do not link draft legal files as if they were a published policy. A real lead form needs a reviewed notice, minimal fields, safe storage and an operating response owner.

Announcement draft

Nutri-Scan: seeking discussions for a focused feed/silage workflow pilot

We are developing a guided way to organize batch observations, evidence and follow-up. We would like to speak with dairy operators who currently rely on disconnected notes, photos and testing reports. A synthetic-data demo is available by arrangement, subject to rehearsal on the chosen device.

The current prototype does not replace laboratory testing or certify feed safety. Pilot scope, fees, data handling and support would be agreed individually after review. To discuss your workflow or a possible introduction, contact [CONTACT].

Twenty-second spoken introduction

“Nutri-Scan helps organize silage and feed batch records, observations and follow-up. We are looking for one clearly defined farm workflow to test in a bounded pilot. Its screening score is an uncalibrated prototype, not a safety certificate. Who currently keeps your batch evidence and follows up on testing?”

Publication gate

- ☐ Founder verifies actual availability, contact and supported scope.
- ☐ Claims and translated safety wording reviewed; no invented outcomes/logos.
- ☐ Privacy, lead handling, optional marketing permission and terms approved.
- ☐ Approved version, channels, date and accountable owner recorded privately.

README

Nutri-Scan technical and operations launch pack

Prepared against the working tree on **2026-10-15**. This is an operational planning pack, **not a launch approval or evidence that application checks ran**. All checklist boxes start unverified. Assign named owners and attach dated evidence for the exact release, environment and device before changing their status.

Current product boundary

- The API uses an explainable, **uncalibrated prototype scoring engine**. Scores,

risk bands and confidence are screening indicators, not laboratory results, calibrated probabilities or feed-safety certification.

- All ten screening inputs are required. Numeric NIR/vision inputs are adapters;

uploaded photos are private evidence, not automatic image analysis. No validated protein/fiber prediction or contaminant/mycotoxin detection is established.

- Farm accounts use SQLite and HttpOnly sessions with owner/member/viewer roles.

This is an MVP; a broad production multi-tenant release requires further review.

- The ESP32 sketch defaults to **synthetic demonstration readings**, USB serial,

unknown calibration and a manually fixed timestamp. Real sensors require validated integrations. Serial and voice entry propose values for explicit review only.

- Offline drafts work in an already-loaded signed-in app and need explicit sync.

The production PWA's offline reload provides **public help only**, not farm history, private attachments or authenticated offline workflows. Drafts discard instrument provenance. Testing requests and reminders are farm-local records, not lab bookings or SMS services.

Documents and operating order

Document	Use / accountable role
Technical readiness checklist	Establish scope, owners and evidence; release lead
Deployment runbook	Dev, staging, HTTPS deployment, backup, migration and rollback; operations
Hardware integration and wiring	Serial contract, electrical review and device records; hardware lead
QA test plan	Regression, acceptance, target-device and field pilot gates; QA lead
Calibration and model release	Sample/lab provenance, leakage controls and release criteria; calibration lead
Data governance and privacy	Access, retention, deletion, backup and incident operations; data steward
Support and incident runbook	Triage, containment, recovery and escalation; support/on-call lead

Document	Use / accountable role
Risk register	Product, technical, safety and business decisions; release lead
Field operator SOP	Preflight, sample handling, capture, review and handover; field lead
Production launch checklist	Final go/no-go and monitored handover; release authority

Start with readiness, rehearse deployment and QA in staging, resolve the risk register, then complete the final launch checklist. Use the field SOP for every pilot visit and the incident runbook when evidence or safety is in doubt.

CSV templates

These are **manual operational records**, not API import files or the calibration CLI's input schema. Each has a header and one `placeholder_not_for_use` row; remove that row in a controlled operational copy before recording real activity. Blank timestamps/measurements mean **not recorded**, never zero, passed or calibrated.

Template	Row meaning and key
calibration-record.csv	One reference point/check per device sensor; `calibration_record_id` + `point_id`
device-inventory.csv	One physical board/instrument; stable `device_id`
qa-test-results.csv	One execution of one test; `run_id` + `test_case_id`
incident-log.csv	One incident summary, with timeline in restricted evidence; `incident_id`
sample-chain-of-custody.csv	One collection/transfer/receipt/analysis event; `sample_id` + `event_id`

Record conventions

- UTF-8, comma-delimited, one header; quote text containing commas/newlines/quotes.

Preserve headers when copying. Treat spreadsheet formula-like text as text rather than executing it. No passwords, cookies, keys, raw personal contacts or full farm reports in these repository templates.

- `data_status`: `placeholder_not_for_use`, `synthetic_demo_not_for_field_use`, or

`real_field_record`. Real status describes origin, **not validation or approval**. Keep synthetic and real collections in separate controlled files/releases.

- `*_at` and `due_at`: timezone-aware ISO 8601; calendar dates: `YYYY-MM-DD`.

`*_pct`: percent; `temperature_c`: degrees Celsius. Record wet/dry basis explicitly. Numeric cells are finite decimal numbers or blank; do not use `NaN`, `inf` or `0` to fill a missing measurement. SHA-256 fields hold the lowercase 64-digit digest of the referenced artifact when available.

- IDs and `*_ref` fields point to restricted records; use pseudonymous farm/operator

IDs. Evidence references must resolve for reviewers without making files public.

- QA `status`: `not_run|pass|fail|blocked|not_applicable`; a pass requires actual

results/evidence and executor/time. Calibration `result`: `pending|pass|fail`; protocol calibration states remain `unknown|reported_current|reported_expired`.

- Inventory `mode`: `demo|real_pending_validation|validated_for_defined_use`;

inventory `status`: `not_in_service|in_service|quarantined|retired`.
Incident `status`: `open|contained|monitoring|closed`; severity follows the incident runbook. Custody `event_type`: `collection|transfer|receipt|preparation|analysis|disposal`.

Source of truth and reconciliation

Read these existing sources alongside the pack:

1. Project README: current scope, accounts and normal commands.
2. Same-day demo: emulator/ESP32 demonstration and its limits.
3. Fieldwork verification: **previously reported**

checks and remaining real-device gaps, not a result from this documentation task.

1. Fieldwork contract: private routes, attachment

limits, observation/request state machines and current PWA boundary.

1. ESP32 README: serial and sensor hooks.
2. Calibration workflow: synthetic scaffold and

requirements for real reference data.

1. Backend operations, architecture

and farmer usability: supporting context.

Some earlier architecture/usability text predates the fieldwork PWA extension.
For installation/offline behavior use the current fieldwork contract and PWA implementation: public offline help exists; private offline reload does not.
The README's legacy analyze route is authenticated today; the default ESP32 sketch does not POST to it. Deployment records the current runtime configuration, including always-present localhost origins and SameSite=Lax limits.

Evidence and placeholders

`PLACEHOLDER_*`, `.example.invalid` hostnames and sample service/path names are operator substitutions, never existing infrastructure or credentials. Commands are instructions to run in the stated directory with pre-existing dependencies; their inclusion does not mean they were executed. No default account, API key, model registry, automated retention job or production service unit is supplied.

Record a release ID, source revision **and working-tree/artifact digest**, backend and frontend artifact digests, configuration revision (redacted), device/firmware versions, QA results and named approvals in a restricted release evidence location. A Git revision alone cannot identify a build made with uncommitted changes.

Calibration And Model Release

Calibration and model release control

The application currently reports `kind=prototype_screening`, `model_version=prototype-baseline-0.1`, `calibration_status=uncalibrated` in its assessment model. Its engine is a transparent formula, not a trained field model. Device calibration, model calibration and qualification of advisory language are **separate approvals**.

Source procedures: calibration workflow, demo runbook, hardware integration.

1. Lifecycle and accountable owners

Stage	Required record / owner	Promotion gate
Intended use	Sample types, population, outputs/units/basis, exclusions and claims; product/domain lead	Qualified expert agrees the measurement contract
Device qualification	Wiring, firmware/time source, sensor reference checks; hardware lead	Defined-use electrical and repeatability criteria pass
Collection design	Sampling plan, farm/batch/time grouping, lab methods and thresholds; calibration lead	Protocol frozen before examining performance
Collection + reference	Raw instrument data, custody events, lab reports/QC; field/lab leads	Traceable matched sample IDs and acceptable handling/QC
Dataset freeze	Version, manifest/hash, exclusions, split assignment; data steward	Reviewed provenance and leakage audit; synthetic excluded
Fit + validate	Train-only preprocessing/fit, validation decisions and reproducible artifacts; model lead	Predeclared criteria and independent review
Locked test	Held-out farms/batches/time, uncertainty and subgroup results; independent reviewer	Test opened only after design/model frozen; all approved thresholds met
Release	Model card, runtime integration tests, compatibility and rollback; release authority	Signed scope-specific approval and staged QA
Monitor / retire	Drift, reference rechecks, incidents, replacement/withdrawal record; model/calibration leads	Trigger-based revalidation and controlled retirement

No automatic model loader, artifact registry or runtime coefficient import is provided by the current scaffold. Promoting a future model requires an explicitly implemented and tested application change; a JSON coefficient file alone changes no application behavior.

2. Synthetic scaffold — exact demonstration commands

From the root, use an existing private temporary output directory. These commands are instructions, not claimed results from this pack:

```
ls -ld /tmp/omnirush
export CAL_DEMO_DIR="$(mktemp -d /tmp/omnirush/nutriscan-calibration.XXXXXX)"
python3 -m unittest discover -s tools/calibration -p 'test_*.py' -v
python3 tools/calibration/calibrate.py \
  --input tools/calibration/sample_reference.csv \
  --output "$CAL_DEMO_DIR/calibration-demo.json" \
  --fit-calibration
python3 -m json.tool "$CAL_DEMO_DIR/calibration-demo.json"
```

The CLI checks headers/row shape/IDs/finite numbers/duplicates, deterministically splits whole farm groups using SHA-256 ranking, reports baseline MAE/RMSE/signed bias/R^2, and optionally fits one univariate line per target on **train only**.
Fitting needs at least three train pairs and nonconstant input for a target.
Validation/test metrics are separate. The sample CSV has six **synthetic** farm groups; it is not an adequate real evaluation dataset.

Every CLI report is labelled `synthetic_demo_not_for_field_use`, even if another input file is supplied. Do not remove that marker, rebrand the scaffold as field validation or paste coefficients into the backend. Keep real evaluation in a controlled, separately approved workflow with true provenance and review.

The CLI's input header is exactly:

```
sample_id,farm_id,sample_type,moisture_sensor,moisture_lab,protein_nir,protein_lab,fiber_nir,fiber_lab,ph_sensor,ph_lab
```

The new operational calibration record
and custody record are **not**
drop-in inputs to that CLI. Maintain reviewed sample-ID joins and a documented transform if exporting paired measurements to an evaluation dataset.

3. Provenance and laboratory reference

1. Assign immutable sample IDs at collection. Record pseudonymous farm, batch/lot,

sample type/variety, date/time/timezone, operator, storage/handling conditions, device/sensor IDs, firmware, preprocessing and sampling-plan version.

1. Use the qualified sampling plan for representative subsamples, homogenization,

clean containers, preservation, transport time/temperature and disposal. Record whether the field measurement and lab aliquot used the same homogenized material. Do not invent universal holding times: the selected lab method specifies them.

1. Preserve exact raw ADC/spectral/other output before viewing reference results,

with immutable evidence reference and SHA-256. Record repeated readings rather than selecting whichever reading later agrees best with the lab.

1. Log each custody event in sample-chain-of-custody.csv:

transfer parties by ID, seal/container condition, time, storage temperature, handling, received lab ID and related report reference. Do not join by row order.

1. Obtain results from a qualified lab using a documented validated method suitable

for the sample/target. Capture method/version, units and wet/dry basis, receipt, preparation/analysis time, instrument/QC run, replicate results, uncertainty and detection/quantification limits where relevant, analyst/run and report checksum.

1. Predefine exclusions for failed QC, insufficient/contaminated samples, broken

custody, unknown basis or impossible values. Quarantine such records for expert review, retain reasons/corrections and do not silently convert them to ground truth.

1. Resolve unit/basis conversions once in versioned preprocessing. Obtain permission

for collection/use, minimize identifiers and follow governance.
Keep a restricted ID mapping separate from analysis exports.

Sensor calibration records additionally capture reference/buffer lots/certificates, raw counts/voltage, environmental temperature, as-found/as-left values, tolerances, due time and reviewer. Plan pre/post-session checks and assess samples since the last known good check if a sensor drifts or fails.

4. Leakage prevention and locked evaluation

- Freeze dataset hashes and a split manifest **before fitting**. Keep farms together

at minimum; strengthen groups for shared batches/lots, storage stacks, device sessions, near-time sampling and related sources. Use later-time evaluation where intended deployment faces drift.

- Keep duplicate aliquots, repeat instrument scans and lab replicates in one group.

Audit cross-split duplicates/raw hashes and relationships, not just sample IDs.

- Fit scalers, imputation, feature selection, preprocessing and calibration on train

only. Choose hyperparameters/thresholds on validation; do not use test labels to select features, exclusions, acceptable error or model form.

- Lock final model/transform/criteria before opening test results. A test-driven

redesign needs a new independent test set; do not relabel the same test as unseen.

- Preserve collection chronology, access history, exclusions, corrections and all

failed experiments in reviewable controlled records. Synthetic examples, lab-result derived features and post-result manual edits cannot enter the real evaluation.

5. Model card and numeric release gates

Create a controlled card for each scoring/model/advisory release. Required fields:

Section	Required contents
Identity	Release ID/version, artifact SHA-256, source/worktree digest, training/evaluation code and dependency versions, owner/reviewer/date
Intended use	Feed/silage domain, regions/varieties/devices/population, user decision, prohibited uses and human escalation
Inputs/outputs	Required measurements, units/basis/ranges, preparation/preprocessing, missing-input handling, runtime compatibility
Data	Collection period, sample/farm/batch counts, provenance/permission, lab methods/QC/uncertainty, dataset and split hashes, exclusions
Evaluation	Per-target/split/subgroup coverage, missingness, MAE/RMSE/bias/R^2, confidence intervals, repeatability, held-out farm/time performance
Decision behavior	Sensitivity/specificity/false negative/false positive rates at approved thresholds where relevant; uncertainty coverage, abstention and out-of-domain behavior
Limitations	Known failure modes, unsupported contaminants/claims, device/season/language limitations and operator requirements

Section	Required contents
Operations	Monitoring/stop triggers, reference-check cadence, artifact/config compatibility, rollback version and owner, retirement/revalidation plan
Approval	Lab/domain, hardware, data, QA, safety/product and release-authority decisions with evidence refs

Before evaluating the locked test, fill and approve a gate record with **numeric** limits per target/subgroup: minimum independent farms/batches/samples and coverage, maximum MAE/RMSE/absolute bias, acceptable uncertainty, repeatability, and maximum false-negative rate at relevant decision thresholds. These limits are currently **PLACEHOLDER_UNSET** and must be set by qualified domain/lab reviewers, not invented from the synthetic metrics. Unset or unmet required gates mean **no model promotion**. A good aggregate R^2 or higher heuristic confidence is insufficient.

6. Staged release, monitoring and withdrawal

Run shadow evaluation against lab/reference results first, then a bounded authorized pilot with predefined sample coverage/stop rules, then only the approved use. Retain the prior runtime/configuration/artifact and run QA, including old record readback and report/model-version labeling. Historical saved results are not silently recomputed on model change; any reanalysis must be a separately traceable record/workflow.

Monitor device drift, out-of-range/missing inputs, subgroup error, reference failures, operator overrides, safety complaints and unsupported use. Thresholds, sampling cadence and responsible reviewer must be named in the card. On a breach, suspend affected use/claims, identify affected device/time/model/sample ranges, preserve evidence, refer to a qualified lab/adviser and follow the incident runbook. A new sensor, firmware conversion, lab method, season/domain expansion or preprocessing change requires impact review and appropriate recalibration/revalidation, not a version-label-only release.

Data Governance And Privacy

Data governance and privacy operations

This is an **operational policy proposal, not legal advice**. The data steward and release authority must approve scope, retention periods, access and local obligations before real-data launch. No scheduled retention, deletion service or audit-log system is implemented by this document. Current behavior comes from the README, backend README and fieldwork contract.

1. Data map and minimization

Location / category	Current contents and handling	Operational owner
SQLite accounts	Username, farm identity, role, scrypt password hash; session token digests/expiry (not plaintext tokens)	Identity/data steward
SQLite screening/history	Farm/batch/sample IDs, input readings, outputs/advisory, timestamps, request IDs, optional instrument provenance	Farm data steward
SQLite fieldwork	Batch metadata, observation answers/notes, reminders, testing-request contacts/status/dates	Field/data lead
SQLite attachments	Private normalized photo BLOBs and lab report/photo BLOBs; request PDFs are downloaded, not OCR-interpreted	Data/operations lead
Browser local drafts	Complete readings scoped by user/farm, queue/request metadata; no password, token, photo or API response; instrument metadata stripped	Device user/field lead
Browser Cache Storage	Public help/manifest/icons/allowlisted assets only; not private farm data or personalized HTML	Frontend/operations lead
Exports and sharing	Downloaded HTML, browser PDF, QR batch routes, optional previewed summaries/native photo sharing	Exporting user/farm owner
Optional voice	Browser speech service may process microphone input externally/network-dependently; Nutri-Scan does not save audio	Device user/data steward
Host logs/backups/evidence	Operator-configured logs, DB snapshots, calibration/custody/incident/QA records	Operations/data steward

Keep contact details out of batch notes unless needed for the chosen testing provider. Use pseudonymous farm/operator IDs in calibration analysis and operational CSVs; store re-identification mappings in a separate restricted location. Avoid faces, addresses, number plates and unrelated material in photos. Photo metadata stripping does not remove identifying content visible in pixels or document text.

QRs contain a route, not a token or access grant, but may expose a batch identifier. Reports include farm details; preview before sharing. WhatsApp/native share is an explicit user action, not automatic background transmission. Once exported to another device/recipient, files are outside server-side access controls; include them in handling/deletion instructions and avoid uncontrolled support attachments.

2. Access and identity operations

- Owners add member/viewer accounts; owners and members can write scans/evidence/

fieldwork, viewers are read-only. Farm scope comes from the session, not client claims. Verify foreign-resource isolation in QA.

- There is no default account. Give each operator an individual account; prefer

viewer access for review. Verify a farm owner's identity before adding users or assigning legacy data. Document grants, purpose, approver and review/expiry time.

- Proposed review cadence: at pilot onboarding, staff changes and monthly while

active. Host DB/backup operators are privileged across farms; keep that group small, use protected host access and record all administrative data access externally.

- Session cookie: HttpOnly, SameSite=Lax, seven days; Secure requires explicit HTTPS

configuration. Password/session hashes remain sensitive and do not make a stolen DB anonymous. Runtime/backup credentials belong in protected operator-managed secret storage, never Git, frontend `VITE_*`, logs or CSV evidence.

- Member removal, account disable/reset/recovery and farm erasure are not complete

self-service administrative workflows in this MVP. Resolve the operational process for a closed pilot and block broader exposure until approved controls exist. Logging out or deleting a session does not disable future credential-based login.

- Scrub request bodies, `Cookie`, `Set-Cookie`, credentials, attachment bytes and

identifying query strings from logs/support bundles. Minimize proxy URL logging; literal history filters can contain farm/batch information. No audit-log completeness is asserted for the app; maintain a restricted operations/change log separately.

3. Proposed retention schedule — approval required

These are concrete **starting defaults**, not current automatic behavior or legal requirements. Enter approved values, owner and purge job/manual procedure in the release evidence before capture. Document exceptions and expiry; do not silently extend retention because a job was never scheduled.

Category	Proposed operational period / trigger	Disposal / verification
Synthetic staging/demo data	14 days after demonstration/test completion	Destroy isolated DB/profile and test exports; retain minimized QA summary
Real pilot scans, fieldwork, photos/reports	Pilot closure + 90 days unless a documented purpose requires another period	Reviewed scoped erasure/export process; verify parent and attachment removal
Account/member data	Active pilot access; review/remove within 30 days of authorized closure/withdrawal	Apply approved account/farm administrative process; revoke sessions, verify scope
Browser drafts/queue	End-of-shift reconciliation; remove confirmed/unneeded copies within 7 days	User checks history before deleting; resolve queue/conflict first; confirm device storage
Downloaded/shared exports	Task completion, target within 7 days unless explicitly retained	User/recipient handles downloads, print copies, share apps and device backups
Redacted host access/error logs	30 days	Restricted log rotation and verified expiry; incident hold only by approval

Category	Proposed operational period / trigger	Disposal / verification
Incident/change/release/QA summaries	12 months; raw private evidence separately minimized	Keep needed non-identifying outcomes; retire raw evidence by category-specific expiry
Real calibration/custody/raw lab evidence	Model/measurement release lifetime + 12 months	Preserve reproducibility and withdrawals; controlled de-identification/erasure review
Encrypted DB backups	Daily points for 30 days, then expire	Verify off-host expiry/replicas and record deletion-ledger reconciliation on restore

An incident/evaluation hold needs a named authority, reason, restricted scope and review date; track it in the same retention ledger. Do not conceal a pending deletion or promise erasure while a documented exception remains active.

4. Backup, restore and deletion reconciliation

Proposed minimum service objectives for approval: **RPO 24 hours**, **RTO 4 hours**, daily consistent DB backups plus a pre-change backup, encrypted off-host copy, daily backup failure review and monthly restore drill. More frequent capture may require a shorter RPO. These are targets, **not measured service guarantees**.

Use the SQLite backup/restore commands. Record timestamp, checksum, size, integrity/FK check, encryption/storage reference, release pairing, retention expiry and drill result. Restrict directory/file access; HTTPS and mode 0600 do not encrypt a DB at rest. Prove that a restore includes attachments, legacy visibility, roles and lazy fieldwork tables. A backup is not qualified until a restore and authenticated readback have succeeded in isolation.

Keep a restricted **deletion ledger** outside restorable DB snapshots: request ID, authorized scope, opaque farm/user/resource IDs needed to reapply deletion, backups containing the data, hold/expiry, executor/reviewer, completion and verification. This ledger itself requires access limits and retention. On restore, revoke old sessions and apply approved erasures **before reopening access**; otherwise a backup can resurrect deleted records. Restore-time reconciliation is mandatory even when the app rollback succeeds technically.

5. Manual deletion/withdrawal workflow

There is no farm-erasure HTTP endpoint or generic purge command. Do not improvise ad-hoc live SQL or assume deleting one farm row cascades all data.

1. Verify requester identity and authority without asking for their password/cookie.

Log opaque request ID, scope (account, farm, scan, attachment, exports or dataset), requested action and acknowledgment date.

1. Inventory affected accounts/sessions, scans/photos, batches/observations/reminders,

testing contacts/reports, legacy-assigned data, local drafts, logs, exports, datasets/model artifacts and all backup copies. Confirm scope with the owner; identify approved holds and downstream recipients.

1. Data steward approves a reviewed, schema-aware administrative procedure in a

maintenance window. Take the protected pre-operation backup under its approved retention/hold, rehearse on a copy, define row/attachment counts and test foreign keys/remaining farms. If a reliable procedure is unavailable, mark the request pending and escalate; never mark it completed merely by hiding data in the UI.

1. Restrict access/revoke sessions as needed, execute the reviewed scoped procedure,

verify affected bytes/records are gone and other farms unchanged. A logical delete alone does not prove physical secure erasure from SQLite pages, journals, SSDs or filesystem snapshots. Select an approved storage-level erasure/rewrite method and verify it in isolation.

1. Guide the authorized user through draft/download/shared-device cleanup after

resolving unsynced work. Logout clears displayed scope, **not necessarily stored drafts or downloads**. Notify identified recipients through the approved process; record limitations without claiming remote recall.

1. Track encrypted backup expiry or controlled destruction and prevent resurrection

through the deletion ledger. Assess dataset withdrawal and any need to retire or retrain affected models; do not promise removal from an artifact by deleting CSV rows.

1. Record executor/reviewer, actual completion/remaining copies, expiry and user-facing

response. Retain only the minimized evidence necessary to prove the operation.

6. Incident and operational checklist

Suspected exposure, lost device, wrong-recipient share, backup theft or cross-farm access triggers the incident runbook. Contain the affected route/device/account, preserve minimal restricted evidence, identify data categories/time/farms/recipients, and assign the incident/data leads. Determine any external communications or obligations with the responsible organization; this pack does not invent a legal notification deadline.

- ☐ Named data steward, access approvers and recovery/deletion operators recorded.
- ☐ Data map and chosen retention schedule approved; actual purge/rotation duties assigned.
- ☐ Registration/access withdrawal/session revocation gaps resolved for scope.
- ☐ HTTPS/Secure cookie/origin/isolation and private cache checks executed.
- ☐ Device-local storage, exported reports, sharing and speech-provider handling explained.
- ☐ Backup encryption, failure alerting, restore drill and deletion ledger tested.
- ☐ Scoped deletion procedure rehearsed, including attachments/backups/other-farm checks.
- ☐ Incident contacts, redaction procedure and restricted evidence access verified.

Deployment Runbook

Deployment, backup, migration and rollback runbook

Operational recipe; not evidence of a deployment. Use an approved release and pre-existing Python/backend environment and Node/frontend dependencies. Provisioning, service units, TLS certificates, backup encryption and alerting are operator-owned; they are not included in this repository. Existing start/check commands are in the README and backend README.

1. Environment contract

Environment	Network / cookies	Storage / use
Development	localhost 5173 → Vite proxy → 8000; secure cookie false only for local HTTP	Disposable dev DB; no real farm data
Staging	Separate HTTPS hostname, same-origin `/api` proxy, secure cookie true	Separate DB, generated accounts and synthetic fixtures; production-like restore/migration rehearsal
Production / restricted pilot	Approved HTTPS hostname, same-origin `/api` proxy preferred, secure cookie true; backend loopback/private only	Durable absolute DB path, encrypted backups, named operators and approved release scope

Use different databases, accounts, backup locations and configuration for each. Production paths/domains below are **placeholders**. Do not copy farm data into staging without an approved minimized/pseudonymized dataset and retention plan.

Variable	Current meaning
`NUTRI_SCAN_DB`	Backend SQLite file; default is `backend/data/nutri_scan.sqlite3`. Relative overrides resolve from process working directory, so deploy with an absolute path.
`NUTRI_SCAN_ORIGINS`	Comma-separated explicit origins (scheme + host + optional port, no paths/wildcards). Localhost/127.0.0.1 ports 5173/5174 are **always added**; this variable does not replace them. Mutations also accept the API's own origin.
`NUTRI_SCAN_COOKIE_SECURE`	Backend `true` for HTTPS; default false. Session cookie is `nutri_scan_session`, HttpOnly, SameSite=Lax, path `/`, seven-day lifetime.
`VITE_PUBLIC_API_URL`	Public build-time API origin for a separately hosted frontend; use empty for same-origin. Legacy fallback is `VITE_API_URL`. Both must be empty for an unambiguous same-origin build.
`VITE_PROXY_TARGET`	Local Vite dev/preview proxy target; not a production reverse proxy or secret.

Backend variables must be supplied by the shell/supervisor; Uvicorn is not configured here to load a root `.env` automatically. Vite build-time variables are public and embedded into assets. Never place passwords, tokens, TLS keys or backup encryption keys in `VITE_*`, the frontend bundle, firmware or release evidence.

Separate origins: a same-site HTTPS API subdomain may work with credentialed requests and the exact frontend origin configured. Truly cross-site hosting can fail because cookies are SameSite=Lax and browser policy applies. Do not assume CORS changes

that cookie rule. Prove register/login/reload/upload/logout on target browsers or use the same-origin topology. Always-present dev origins need a recorded threat review and appropriate edge/network restrictions before wider exposure.

2. Local development

From `backend/`, using its existing virtual environment (Terminal A):

```
NUTRI_SCAN_COOKIE_SECURE=false .venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8000
```

This uses the default local DB unless explicitly overridden. From `frontend/` (Terminal B), override legacy URL settings to retain the cookie-friendly proxy:

```
VITE_PUBLIC_API_URL= VITE_API_URL= VITE_PROXY_TARGET=http://127.0.0.1:8000 npm run dev -- --host 127.0.0.1
```

Open `http://localhost:5173`. `/api` and `/health` proxy to the backend. Run `docker compose up --build` from the repository root only for the documented development stack: it runs dependency installation and Vite dev at startup, exposes ports and bind-mounts code. It is not the production recipe. No Compose/install command was executed to prepare this pack.

3. Candidate build and staging

Run the QA commands in an isolated checkout/workspace, preserving existing local edits. From `frontend/`, for same-origin hosting:

```
VITE_PUBLIC_API_URL= VITE_API_URL= npm run build
```

Retain the complete `frontend/dist/`, including hashed assets, `sw.js`, manifest, icons and offline help. Record artifact digests and build environment. PWA generation requires Vite's production mode and deployment at **root** `/`; subpath hosting is not supported by the current plugin. Rehearse with a local production preview:

```
VITE_PUBLIC_API_URL= VITE_API_URL= VITE_PROXY_TARGET=http://127.0.0.1:8123 npm run preview -- --host 127.0.0.1 --port 5174 --
strictPort
```

The matching isolated API/DB setup is in QA. For actual staging, serve the candidate static artifact behind its own HTTPS reverse proxy and use a separate absolute `NUTRI_SCAN_DB`; do not use Vite preview as the staging/production service supervisor. Test the same configuration and first authenticated fieldwork request that production will use.

4. HTTPS production topology and startup

Substitute an approved domain, static release path, TLS files and service account. Example backend process command from the deployed `backend/` directory:

```
NUTRI_SCAN_DB=/srv/nutriscan-data/PLACEHOLDER_PRODUCTION.sqlite3 \
NUTRI_SCAN_ORIGINS=https://nutriscan.example.invalid \
NUTRI_SCAN_COOKIE_SECURE=true \
.venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8000 \
  --proxy-headers --forwarded-allow-ips=127.0.0.1
```

An operator-provided supervisor must supply the same environment, working directory, restart policy and restricted user. Use one API process initially; SQLite write contention and process-local login throttling are not solved by adding workers. Restrict registration/exposure at the edge for an approved closed pilot; there is no built-in invitation-only deployment switch. Review recovery/removal/admin gaps before a public multi-tenant launch.

Illustrative Nginx server blocks (configure on the host, not in this docs pack):

```
server {
    listen 80;
    server_name nutriscan.example.invalid;
    return 301 https://$host$request_uri;
}
server {
    listen 443 ssl;
    server_name nutriscan.example.invalid;
    ssl_certificate /etc/PLACEHOLDER_TLS/fullchain.pem;
    ssl_certificate_key /etc/PLACEHOLDER_TLS/private-key.pem;
    root /srv/nutriscan/releases/PLACEHOLDER_RELEASE/frontend/dist;
    client_max_body_size 11m;

    location /api/ {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header Origin $http_origin;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
        proxy_cache off;
    }
    location = /health {
        proxy_pass http://127.0.0.1:8000;
        proxy_set_header Host $host;
        proxy_set_header X-Forwarded-Proto $scheme;
        proxy_cache off;
    }
    location = /sw.js {
        add_header Cache-Control "no-cache" always;
        try_files $uri =404;
    }
    location /assets/ {
        add_header Cache-Control "public, max-age=31536000, immutable" always;
        try_files $uri =404;
    }
    location / {
        add_header Cache-Control "no-cache" always;
        try_files $uri $uri/ /index.html;
    }
}
```

Serve `.webmanifest` as `application/manifest+json` (or `application/json`), JS/CSS with correct types, and `offline.html` as `text/html`. Verify host MIME mappings. Do not rewrite `/api/*` errors to SPA HTML. Do not cache private requests at a CDN; preserve API `Cache-Control: no-store`, `nosniff` and `Set-Cookie`. Keep unknown Hosts rejected by the host's default vhost. Trust forwarded headers only from the actual proxy, and prevent direct public access to port 8000. Add HSTS only after the HTTPS/certificate/subdomain rollout has been rehearsed.

Read-only edge probes after setting the real HTTPS origin:

```
export APP_ORIGIN=https://nutriscan.example.invalid
curl --fail --show-error --silent "$APP_ORIGIN/health"
curl --show-error --silent -D - -o /dev/null "$APP_ORIGIN/api/v1/auth/me"
curl --fail --show-error --silent -I "$APP_ORIGIN/sw.js"
curl --fail --show-error --silent -I "$APP_ORIGIN/manifest.webmanifest"
```

Expected: health 200 with `status: ok`; unauthenticated `auth/me` 401 with `no-store` / `nosniff`; static responses 200 with appropriate types. `/health` is static service status, not a DB readiness probe. Complete authenticated QA in the browser, inspecting cookie flags without exporting cookie values into evidence.

5. Consistent SQLite backup

Back up before startup/migration, legacy assignment, model/app release or deletion. The DB includes account/session hashes and photo/report BLOBs: protect the whole backup as private farm data. Do not copy only a live `.sqlite3` file with ordinary `cp`; use SQLite's backup API to include committed state consistently.

Set **absolute, operator-provisioned** paths. `BACKUP_DIR` must already exist, be owner-only and be on approved encrypted storage outside web roots/repository. Confirm the parent directories, then use a unique output filename:

```
export NUTRI_SCAN_DB=/srv/nutriscan-data/PLACEHOLDER_PRODUCTION.sqlite3
export BACKUP_DIR=/srv/nutriscan-backups/PLACEHOLDER_PRIVATE_DIRECTORY
ls -ld "$BACKUP_DIR" "$(dirname "$NUTRI_SCAN_DB")"
export BACKUP_PATH="$BACKUP_DIR/pre-release-$(date -u +%Y%m%dT%H%M%SZ).sqlite3"
python3 - "$NUTRI_SCAN_DB" "$BACKUP_PATH" <<'PY'
import os, sqlite3, sys
from pathlib import Path
source, target = map(lambda p: Path(p).resolve(), sys.argv[1:])
assert source.is_file() and source != target
assert target.parent.is_dir() and target.parent.stat().st_mode & 0o077 == 0
fd = os.open(target, os.O_CREAT | os.O_EXCL | os.O_WRONLY, 0o600)
os.close(fd)
src = sqlite3.connect(source.as_uri() + '?mode=ro', uri=True)
dst = sqlite3.connect(target)
try:
    src.backup(dst)
    integrity = dst.execute('PRAGMA integrity_check').fetchall()
    foreign_keys = dst.execute('PRAGMA foreign_key_check').fetchall()
    assert integrity == [('ok',)] and not foreign_keys, (integrity, foreign_keys)
finally:
    dst.close()
    src.close()
print('Backup verified:', target)
PY
sha256sum "$BACKUP_PATH"
```

If this fails, mark the artifact failed and investigate; a leftover file is not a verified backup. Record checksum, timestamp, size, release/schema pairing, encryption location, retention expiry and restore evidence in the restricted backup log. The commands set private file permissions but **do not encrypt**; host storage encryption and encrypted off-host transfer must already be configured and tested.

6. Migration and restore rehearsal

There is no versioned migration runner or down-migration command. Backend startup initializes tables and adds missing scan columns; fieldwork tables initialize lazily on first fieldwork access. Rehearse both against a **copy**, then compare counts, attachments, legacy visibility, roles, old scan/report readback and idempotency. Do not infer backward compatibility from “additive”. Record the actual schema and code version together; never hand-edit production schema as an improvised migration.

Restore a verified backup into a **new**, unused file on an existing private test directory. Stop any service that could use the restore destination. Substitute real approved paths; this leaves the source backup and live DB untouched:

```
export RESTORE_DB=/srv/nutriscan-restore/PLACEHOLDER_UNIQUE_RESTORE.sqlite3
ls -ld "$(dirname "$RESTORE_DB")"
python3 - "$BACKUP_PATH" "$RESTORE_DB" <<'PY'
import os, sqlite3, sys
from pathlib import Path
source, target = map(lambda p: Path(p).resolve(), sys.argv[1:])
assert source.is_file() and source != target
assert target.parent.is_dir() and target.parent.stat().st_mode & 0o077 == 0
fd = os.open(target, os.O_CREAT | os.O_EXCL | os.O_WRONLY, 0o600)
os.close(fd)
src = sqlite3.connect(source.as_uri() + '?mode=ro', uri=True)
dst = sqlite3.connect(target)
try:
    src.backup(dst)
    assert dst.execute('PRAGMA integrity_check').fetchall() == [('ok',)]
    assert not dst.execute('PRAGMA foreign_key_check').fetchall()
    # Restored historical sessions must not become valid again.
    dst.execute('DELETE FROM sessions')
    dst.commit()
finally:
    dst.close()
    src.close()
print('Restored with sessions revoked:', target)
PY
```

Reconcile the deletion ledger **before** restored data is exposed. From the selected release's `backend/`, use the restored DB on an unused loopback port:

```
NUTRI_SCAN_DB="$RESTORE_DB" NUTRI_SCAN_COOKIE_SECURE=false \
.venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8123
```

Use the isolated frontend/QA procedure, generate no real outbound shares, and record actual recovery time and recoverable data age. Stop rehearsal processes afterward. For real cutover pause writes at the edge, drain/stop the old API, take a final verified backup, then start the candidate and complete authenticated fieldwork smoke tests before reopening writes.

Legacy assignment is a separate **explicit data-ownership change**, never automatic: after reviewing source legacy farm, destination owner and backup, from `backend/`:

```
NUTRI_SCAN_DB=/srv/nutriscan-data/PLACEHOLDER_PRODUCTION.sqlite3 \
.venv/bin/python -m app.admin assign-legacy \
  --username PLACEHOLDER_EXISTING_OWNER --farm-id PLACEHOLDER_UNOWNED_LEGACY_FARM
```

Verify exact scan IDs and attachments under the intended farm and that other farms cannot read them. Record authorization and before/after counts without raw data.

7. PWA deployment, update and rollback

- Publish the complete static release atomically; retain the previous artifact and

its hashed assets for existing tabs. For a new release, carry forward older hashed assets from the approved artifact store during the transition, or use an equivalent immutable asset route. A new HTML file with missing old chunks breaks open tabs.

- Serve the new `sw.js` with revalidation. Worker cache identity includes public

assets and policy. Updates **wait for explicit user activation**; do not force reload while readings are unsaved. On activation, older namespace caches are removed.

- Inspect Cache Storage: only allowlisted static assets/help/manifest/icons, never

API/auth/history/photos/reports or personalized `index.html`. Public offline help must remain public and carry no account details.

- Test online load, update notification, explicit update, offline reload and all

three help languages on HTTPS target devices. Installation availability depends on browser policy; manual browser installation may be required.

8. Rollback and recovery

Trigger on cross-farm exposure, unsafe claims, corruption/data loss, broken login, failed migration or required acceptance failures. Incident lead owns the decision.

1. Restrict access/pause writes; announce impact and preserve redacted logs plus a

verified post-failure DB snapshot. Record the last known good release and time.

1. If schema/data compatibility was proven in staging, stop the candidate, restore

the prior backend and full frontend artifact/configuration, retain the current DB, then run readback and write smoke tests before reopening.

1. If compatibility is unknown or data changed incompatibly, keep service closed.

Restore the pre-change backup into a new DB as above, invalidate sessions, reconcile deletions and obtain approval for the known recovery-point data loss. Reconcile post-backup operations from preserved evidence; do not blindly replay writes or offline queues. Same farm/request ID with changed content returns 409.

1. Switch the supervisor's DB path/release pointer while stopped, then restart under

the old approved code. Supervisor commands depend on the host's actual unit; no universal `systemctl` unit or deployment script is supplied here.

1. Redeploy the **old worker and its assets together**, revalidate `sw.js`, and guide

users through explicit update after saving/handling drafts. Do not indiscriminately clear all site storage: it can destroy unsynced drafts. Verify cache/privacy again.

1. Re-run scoped QA, verify counts/attachment downloads, monitor, communicate recovery

and complete the incident review. Document any unrecoverable scans, permissions changes or sample provenance gaps.

Use governance for backup/deletion controls and production launch checklist for go/no-go evidence.

Field Operator Sop

Field operator standard operating procedure

Use for an **authorized pilot or clearly labelled synthetic demonstration**.
The field lead supplies the approved sampling method, qualified adviser/lab contacts and support route before travel. Nutri-Scan is an uncalibrated screening prototype; neither a low-risk score nor a photo establishes that feed/silage is safe.

Source workflows: same-day demo,
fieldwork contract,
farmer usability and
hardware instructions.

1. Preflight and start-of-shift record

- Confirm scope (demo versus real pilot), site permission, operator training and

support/qualified adviser availability. Record date/time/timezone, release, operator ID and weather/storage conditions as needed by the sampling plan.

- Confirm phone/browser/USB support, charged equipment, reachable HTTPS app and

correct individual account/farm. Owner/member can submit; viewer can only read. Do not share accounts or store passwords in notes/templates.

- Inspect electronics, leads, enclosure and clean sample-contact surfaces. Do not

modify powered wiring, attach raw pH probes or put a 5 V output on ESP32 ADC.

- Check device ID, firmware/mode, real time source and calibration/reference records

against inventory. Perform the approved reference check and record it in calibration-record.csv.

- If reference checks fail, calibration is unverified/expired, equipment is damaged

or capture time is stale, stop using that instrument for real measurements. Record the issue and use a qualified alternative or observation-only workflow.

- For a demo, label devices, batch IDs and every associated file **synthetic demo —

not field/lab measurements**. Default ESP32/emulator numbers cannot populate real farm records. The serial wire has no demo flag, so the external label is essential.

2. Identify the batch and take a representative sample

1. In **Farm fieldwork**, create/select the exact feed/silage batch. Record human

batch ID, name/location and known preparation/opening dates/quantity. Do not guess dates, invent expiry or convert quantity units silently. Batch identity (ID + sample type) is immutable after creation; fix errors through the field lead.

1. Assign immutable sample ID(s), including aliquot relationships, **before**

measurement. Follow the qualified plan for representative subsamples, mixing, clean containers, preservation/transport limits and protective equipment. Do not taste or deliberately inhale suspect material.

1. Record collection and each handover/receipt/preparation/analysis event in

sample-chain-of-custody.csv.
Include timezone-aware time, sample/batch/device/operator IDs, seal/condition, storage temperature/handling and restricted evidence reference. Keep the actual ID mapping private. Unknown conditions remain explicitly unknown/blank.

1. Clean equipment between samples per method to prevent carryover. If handling,

contamination or custody is uncertain, quarantine the record/sample for qualified review; never “correct” readings after seeing a lab report.

3. Choose observation or full screening

Observation only: choose **Quick observations**, answer visible mould, unusual heat, unusual smell and cover damage as Yes/No/Unknown. Leave unknown as Unknown, save and confirm **Observation only — no screening score**. Add private evidence if authorized. Concerning/unknown observations require qualified follow-up, not an invented numeric diagnosis.

Full screening: Use this batch copies identity only, not previous readings.
The four steps are sample → batch → readings → review/submit. All ten inputs are required; incomplete readings cannot be submitted or saved as complete-reading drafts:

Input group	What to enter
Storage/sensor	Moisture %, pH, temperature °C, humidity % from suitable measured methods
Nutrient	Protein %, dry matter %, crude fiber % from a suitable report/instrument with known basis
Visual prototype inputs	`mold_score`, `color_anomaly`, `surface_damage`, each 0–1 from the trained operator's approved scoring method; a photo does not generate these automatically

Use **How to measure…** and language/listening help, but do not replace the approved method with a guessed value. Missing is not zero. If any required actual value or validated scoring method is unavailable, use observation-only tracking and arrange qualified measurement/testing instead of filling demo values.

4. Optional instrument/voice capture

1. For a validated serial device, close the serial monitor, connect safely and open

Read from instrument (USB serial) on reading step 3. Supported Chromium needs HTTPS/localhost, user permission, the documented JSON protocol and 115200 baud.

1. Explicitly choose the port, review device/time/calibration/reference/units/values,

select only trustworthy readings, acknowledge review and confirm. Calibration status is reported by the device, not verified by the app.

1. On timeout, denied permission, unplug or invalid output, record the failure and

continue with a measured manual alternative. The app neither commands the device nor submits a scan automatically. A new capture replaces prior provenance; edits to captured values remove their provenance.

1. Voice is optional and requires microphone consent. Review every proposed value;

browser speech may use a network/provider service. Nutri-Scan does not save audio. Typed entry remains available. Listening/recognition on real devices can differ from software mocks and draft translated safety guidance needs expert review.

5. Final review, save and interpretation

- Compare every value/unit/basis to the actual measurement record, plus batch/sample,

capture time/source and any device provenance. Correct mistakes **before** final confirmation; edits invalidate the prior confirmation.

- Submit once when online. Wait for the saved result/scan ID, then refresh history

and confirm readback. A spinner/error is not proof that the scan was saved; inspect history before repeating an uncertain submission. Record sample-to-scan linkage in the controlled field record.

- Read saved advisories and prototype status. Never certify safety or change feeding

decisions solely because of score/confidence. If spoilage, unusual heat/smell, suspect material or conflicting lab results are reported, stop using the affected screening as assurance and contact the qualified adviser/lab under the site plan.

- Photos are optional private evidence, not diagnosis: up to five JPEG/PNG/WebP,

each <=5 MiB. Avoid identifying bystanders/locations and upload only with authority.

6. Offline, drafts and retry

- In an **already-open signed-in app**, complete actual readings, choose **Save

draft**, then** Queue** if an upload is intended. Unsaved edits are not saved. Reconnection alone does not upload; press **Sync now** explicitly while signed in with permission for the same farm.

- Check draft/queue counts, last confirmed queued upload and saved history. Failed

uploads keep actionable state; a 409 needs history/support review before changes. Do not create a new request ID just to force an ambiguous queued item through.

- **Drafts retain readings but discard instrument device/time/calibration provenance.**

Keep an independent protected measurement/custody log. If provenance is required, use direct online capture/submission under the approved plan or escalate; never claim draft sync retained it. Incomplete readings need a protected field record, not invented values to satisfy draft validation.

- An offline reload of the installed PWA shows **public help only**. Farm history,

fieldwork writes, reports/photos and authenticated reload need reconnection. Do not clear browser/site storage or accept disruptive updates with unresolved local work. Follow the field lead's approved backup/reconciliation procedure.

7. Follow-up, reports and sharing

- Create an in-app reminder and check it manually; it sends no SMS or background

alert. Complete/reopen it as actual work progresses.

- For testing help, contact the chosen qualified provider **outside the app**,

confirm method/handling/time/price yourself and record only necessary contacts. The saved request is local tracking, not a booked lab service. Mark collected with the actual date; upload a trusted report (PDF <=10 MiB or supported photo, max five) before marking report received. Do not treat basic PDF validation as malware scanning or an uploaded report as automatically interpreted by Nutri-Scan.

- Download saved-result HTML; use browser Print/Save PDF if needed. Reports include

farm identifiers and reproduce saved readings, not later form edits. Review before sharing. Use the preview-first summary and verify recipient/identity/photo choices; identity/photos are off by default. Text actions do not send private photo URLs.

- Batch QR routes require sign-in and grant no access. A localhost QR cannot serve

another phone; use the approved reachable HTTPS deployment.

8. End-of-shift handover and stop conditions

Reconcile samples, custody events, scans, device checks, drafts/queues, reminders, testing requests and lab handovers with the field lead. Confirm every purported saved record by readback, note unresolved items and retain only approved protected evidence. Perform any required post-session reference check; a failure triggers review of samples since the last known good check. Clean/store equipment per method.

Resolve/retain unsynced work before deleting drafts. Log out, secure the device, protect downloads/print copies and follow retention/privacy; logout alone does not erase local drafts or exported files. Record device returns, remaining battery/damage and next calibration/referral tasks.

Stop and escalate unsafe wiring, failed reference checks, unknown sample identity, privacy exposure, data loss or potentially harmful interpretation through support/incidents. Record time, scope and observed facts; do not ask the farmer to treat a prototype score as confirmation of safety.

Hardware Integration And Wiring

Hardware integration and wiring

Current source: ESP32 README,
sketch,
same-day demo, and
instrument validator.
This document does not certify a wiring design or a sensor's accuracy.

1. Demo mode versus real measurement

Item	Current implementation	Required before real use
`DEMO_MODE`	`true`; illustrative 62% moisture, pH 4.2, 24°C, 68% humidity	Label the entire session synthetic; replace/validate hooks before disabling demo mode
Moisture	ADC hook with placeholder dry/wet counts	Material-specific reference method, measured transfer function and repeatability
pH	ADC hook with placeholder voltage/slope	Conditioned probe interface, appropriate sample preparation, buffer calibration and temperature handling
Temperature/humidity	Non-demo hooks return `NAN`	Actual validated DHT/BME-like library/module integration and error handling
Clock	Fixed `MANUAL_CAPTURED_AT`; no RTC/network sync	Validated real capture-time source for real sessions, including timezone and drift check
Nutrient values	Not measured by this sketch	Suitable validated report/NIR instrument; never derive protein/fiber from demo readings
Network	No automatic Wi-Fi, credentials, POST or serial commands	A separately reviewed integration if networking is introduced later

Non-finite/out-of-range hook results are omitted. The browser requires at least one valid reading and the screening still requires all ten inputs. A plausible emitted number does not make the placeholder conversion calibrated. Even after updating a manual timestamp before a demo build, repeated frames carry the same timestamp; that is not continuous real measurement time.

2. Electrical preflight — hardware lead sign-off required

Disconnect power before changing wiring. Identify the **exact board variant**, module supply/output limits, ADC configuration and interface circuit from datasheets. The following pins are editable placeholders, not universal ESP32 pin assignments:

Signal	Sketch default	Integration caution
Conditioned pH analog output	GPIO 35	**Never connect a raw pH probe or a 5 V output directly to ADC.** Use the probe manufacturer's high-input-impedance signal-conditioning/interface circuit and verify its output range.

Signal	Sketch default	Integration caution
Moisture analog output	GPIO 34	Verify material suitability, common ground and worst-case signal voltage; supply voltage is not the same as safe output voltage.
DHT-like data	GPIO 4	Placeholder only; review pull-ups/logic voltage and board boot-strapping constraints; no measurement library is present.
BME-like I2C SDA / SCL	GPIO 21 / 22	Placeholder alternative; verify module supply, pull-ups to safe logic voltage, address and wiring length.
Power / ground	Board/module specific	Use a suitable protected supply, common reference as required by the interface, strain relief and dry enclosure; avoid unintended USB/external supply back-powering.

- ESP32 ADC inputs are **not 5 V tolerant**. “0–3.3 V conditioned output” is not a

promise of linear/full-scale ADC accuracy; confirm the actual board's permitted input and usable ADC range at the selected attenuation. Measure output limits before attaching the ADC. Add appropriate conditioning/protection under qualified hardware review; do not choose a resistor divider without considering probe loading.

- GPIO 34/35 on classic ESP32 boards are input-only and lack internal pull-ups;

other variants may differ. Avoid assuming the same pin map on every ESP32 family.

- The sketch sets 12-bit reads but does not validate reference voltage, attenuation,

ADC nonlinearity, grounding/noise, sensor excitation or temperature compensation. Do not treat `3.3/4095`, raw-count endpoints or pH slope constants as calibration.

- Keep electronics/connectors dry and separate from sample containers. Clean sensor

contact surfaces between samples using the manufacturer's procedure. Do not taste samples or deliberately inhale suspect material. Follow the qualified sampling plan for protective equipment, biological hazards and disposal.

Record the reviewed wiring revision, voltage measurements, board/module datasheets, sensor IDs and reviewer in the device inventory.
Hold a device out of service for damaged cables, unknown voltage limits, failed reference checks, contamination or expired calibration under the approved method.

3. ESP32-to-browser serial contract

- **115200 baud, 8 data bits, no parity, one stop bit, no flow control.**
- UTF-8 JSON: one object followed by LF; the browser accepts CRLF too. No boot

banners, debug text or pretty-printed multi-line objects in the capture stream.

- Template emits approximately once per second. The app sends **no commands** and

reads one frame per explicit capture. Port selection is included in the **15-second** window; maximum frame size **16,384 bytes**.

- Required keys: `device_id`, `captured_at`, `calibration_status`, `readings`.

Optional: `calibration_reference`. Unknown keys, duplicate object keys and duplicate reading fields are rejected; do not add a `demo` or `protocol_version` wire key.

- `device_id`: 1–120 characters; optional reference: 1–200; neither accepts surrounding

whitespace/control characters. Stable eFuse-derived `nutriscan-esp32-<hex>` is an identifier, **not authentication**.

- Timestamp: `YYYY-MM-DDTHH:mm:ss.fraction` (fraction up to six digits).

It must be an actual valid timezone-aware timestamp; the parser does not establish that a clock is fresh or trustworthy.

- Calibration: `unknown|reported_current|reported_expired`, a device claim rather

than app verification. Check the underlying calibration record independently.

Allowed `readings` array: **1–7 unique fields**, finite JSON numbers (not strings), exact units and bounds:

Fields	Unit	Inclusive range
`moisture_pct`, `humidity_pct`, `protein_pct`, `dry_matter_pct`, `fiber_pct`	`%`	0–100
`ph`	`pH`	0–14
`temperature_c`	`°C`	-20–80

The three visual scores cannot be captured over this protocol. An example frame below is **synthetic demo data**, with a fixed illustrative timestamp, not a field measurement. Keep the demo label in session records because the wire has no demo flag:

```
{"device_id":"host-emulator-01","captured_at":"2026-10-15T12:00:00+05:30","calibration_status":"unknown","readings":
[{"field":"moisture_pct","value":62.0,"unit":"%"},{"field":"ph","value":4.2,"unit":"pH"},
{"field":"temperature_c","value":24.0,"unit":"°C"},{"field":"humidity_pct","value":68.0,"unit":"%"}]}
```

4. Bring-up and capture procedure

1. Log device/mode/firmware digest; inspect wiring with power removed. For a demo,

keep `DEMO_MODE=true` and update the manual timestamp before compiling. For real operation, use a separately validated sensor/time implementation and reference checks.

1. Open the sketch in an existing Arduino IDE/ESP32 core environment, choose the

correct board and port, compile/upload and retain the actual build/upload result.
No successful firmware compile or physical wiring check is asserted by this pack.

1. Check serial output at 115200. Close the serial monitor before the browser opens

the port; only one process should own it. The README's host smoke only checks JSON and keys; it is not full protocol, calibration or electrical validation.

1. On Linux, an optional host transport check (substitute the actual port) is:

```
stty -F /dev/ttyUSB0 115200 cs8 -cstopb -parenb -ixon -ixoff raw -echo
timeout 10s python3 -c 'import json,sys; [print(json.dumps(json.loads(line))) for line in sys.stdin]' < /dev/ttyUSB0
```

A timeout exit status of 124 is expected when bounded streaming ends; inspect output/errors and valid frames, rather than calling that exit code a QA pass.
For no-hardware transport use the exact PTY emulator.

1. In a supported Chromium browser on HTTPS or localhost, sign in as owner/member,

enter screening step 3, open instrument capture and explicitly select the port. Review ID, actual time, calibration reference/state, values and units. Select readings and acknowledge review before confirming. No automatic scan is submitted.

1. Fill remaining actual measurements and perform the normal final review/submit.

Direct online saves retain selected-field provenance. Editing a captured value removes that field's provenance; changing source/device/time removes the capture. A new capture replaces previous capture provenance, not all previous form values.

1. Refresh history and confirm saved values/provenance. On timeout, unplug, malformed

frame or unsupported browser, record the failure and use measured manual entry; never fill absent values from the demo constants.

5. Device and sensor calibration records

Use inventory for board ID, asset tag, sensor/module serials, firmware and wiring revision, mode, owner, clock source, validated voltage limits and calibration references. Store actual sensor serials and multiple calibration IDs in the controlled `sensor_inventory_ref` record.

Use calibration-record.csv for **each reference point**: sensor, method/version, buffer/reference lot and certificate, sample/basis, environmental temperature, raw ADC count/voltage, as-found and as-left value, reference value/unit, tolerance, error (`as_left - reference`), result, due time, reviewer and raw evidence hash. Multiple points share a calibration record ID but have distinct point IDs. A failed as-found check can affect earlier samples even if adjustment makes as-left pass; assess that interval explicitly.

Choose calibration frequency and acceptance tolerance with the module supplier and qualified measurement lead; there is no app-enforced due-date or device registry. Keep sample-linked checks in the custody record and follow calibration/model release before using sensor results to substantiate accuracy claims.

Production Launch Checklist

Production launch and monitored handover checklist

Initial decision: not assessed / no approval recorded. This checklist does not convert the MVP into a production multi-tenant service. Distinguish a synthetic demo, a restricted pilot and broader production; approval applies only to the recorded scope. Every checked item needs named owner/date/evidence in the release record.

Release record — fill before review

Field	Required value
Release ID / candidate	PLACEHOLDER_RELEASE_ID; source revision plus working-tree/artifact digest
Scope / environment / URL	PLACEHOLDER_SCOPE; authorized farms/users/devices/sample types/languages; HTTPS origin
Backend / frontend / config	PLACEHOLDER_ARTIFACT_DIGESTS; redacted config revision; DB/schema pairing
Firmware / sensors / model	PLACEHOLDER_VERSIONS; demo/real status, calibration refs; current model is uncalibrated prototype
Cutover window / rollback artifact	PLACEHOLDER_TIME_WITH_TIMEZONE; previous known-good artifact/DB/backup refs
Named accountable people	PLACEHOLDER_RELEASE_AUTHORITY; operations, QA, hardware/calibration, data, domain, field, support and backups
Evidence location / support hours	PLACEHOLDER_RESTRICTED_EVIDENCE_REF; staffed hours and out-of-band escalation

Before the cutover window

Product, safety and field ownership — product/domain/field leads

- ☐ Readiness completed for the exact scope;

unresolved items and risk register have named dispositions.

- ☐ Prototype/lab-replacement limits verified in UI, exports, sharing, training and

launch script. No unsupported feed-safety, nutrient accuracy or contaminant claims.

- ☐ If releasing a calibrated model, model/data/lab gates

have approved numeric limits, locked real evaluation evidence and signed card.
Otherwise retain explicit uncalibrated status; synthetic metrics authorize no claims.

- ☐ Target hardware wiring, actual sensor hooks/time source, inventory and reference

checks approved. Demo units/data unmistakably separated from real pilot records.

- ☐ Operator SOP rehearsed; qualified adviser/provider contacts

and actual sampling/handling limits available; translation safety review completed.

- ☐ Local-only requests/reminders and public-only offline reload explained; no

promised lab booking, SMS, private offline history or draft provenance retention.

Deployment, identity and data — operations/data leads

- ☐ Candidate artifacts/configuration identified and staging rehearsal completed

under the deployment runbook; actual services/owners known.

- ☐ HTTPS/certificate renewal, DNS, root-path static assets and proxy forwarding

verified. API not directly public; `NUTRI_SCAN_COOKIE_SECURE=true` and exact origins set; deployed cookie login/reload/logout exercised on target browsers.

- ☐ Always-present dev origins, SameSite=Lax and server-own-origin behavior reviewed.

Any separate-origin topology proven; no wildcard CORS/private CDN cache.

- ☐ Registration exposure, access withdrawal, account recovery/reset and session

revocation procedures resolved for scope. Incomplete controls block broader launch or require explicitly approved restricted-pilot containment.

- ☐ Absolute durable SQLite path/permissions, disk/attachment headroom and measured

expected load acceptable. No assumption that more workers fix SQLite/throttling.

- ☐ Latest consistent backup verified, encrypted off-host copy available, restore

drill meets approved numeric RPO/RTO, sessions/deletion reconciliation rehearsed.

- ☐ Startup + first fieldwork lazy migration and any separate legacy assignment

verified on a copy; prior artifact/data compatibility or restore fallback proven.

- ☐ Governance retention/access/deletion schedule

approved with actual operators/jobs; local-device/export handling and backup expiry owned.

- ☐ Runtime secrets/TLS/backup credentials restricted; no secrets in frontend,

firmware, templates, logs or evidence; upload/PDF trust procedure confirmed.

QA and recovery — QA/support leads

- ☐ Required regression commands and acceptance cases executed

on the final candidate; actual results/outputs recorded, no historical counts reused.

- ☐ Cross-farm/role/Origin/private attachment/no-store gates pass. Viewer remains

read-only. All required input and idempotent retry paths pass.

- ☐ Saved-report/download/print, batch/history, observation-only, request/reminder,

explicit draft sync/provenance limits and field pilot smoke have evidence.

- ☐ Real-device serial permission/unplug/time/calibration and voice/manual fallback

tested for advertised devices; unavailable tests are blocked, not passed.

- ☐ PWA install/update/explicit activation/offline help/cache privacy and rollback

tested on the actual HTTPS host/browser matrix; coherent old hashed assets retained.

- ☐ Incident contacts/coverage, alert routing, rollback decision authority, restricted

evidence access and incident exercise confirmed.

Cutover — operations lead with release authority

1. Confirm go/no-go authorization and operator/support presence. Notify pilot users

of maintenance, explicit sync/update behavior and approved stop/manual fallback.

1. Pause writes at the edge, drain/stop the previous API, take the final verified

backup and record counts/checksum/time. Preserve the prior code/config/static artifact; do not overwrite the only rollback copy.

1. Deploy the complete candidate backend/static/worker/config pairing. Start under

the approved supervisor, initialize core tables, then exercise authenticated fieldwork to cover lazy schema creation. Do not auto-run legacy ownership assignment.

1. Run edge health/unauthenticated privacy probes and authorized scope-specific

read/write/attachment/report smoke. Verify DB readback, session flags, cache policy and sample/scan/provenance linkage. Delete/dispose of launch fixtures by policy.

1. Reopen only the authorized scope if checks pass. For unsafe claims, privacy/data

failures, corruption, broken core flows or unknown migration compatibility, keep service restricted and execute the rollback procedure.

Observation and support handover

Proposed initial observation: a **staffed first two hours** plus daily review for the first seven pilot days; approve coverage and actual numeric alert/stop thresholds in the release record before opening. These are planning defaults, not an SLA.

- ☐ Watch availability, request failures/latency, 401/403/429 patterns, disk/DB

growth/locks, backup age/failures, missing scans/queue conflicts, attachment errors, PWA update failures, sensor/reference drift and safety/privacy complaints.

- ☐ Confirm at least one real backup completes on schedule and assign the next

restore/access/retention review. A green `/health` is not sufficient evidence.

- ☐ Log deviations/incidents and next update times. Stop affected workflows for

cross-farm exposure, failed sensor checks, data loss or harmful interpretation.

- ☐ Hand over current release/DB/backup/rollback refs, device roster, sample/custody

gaps, unresolved queues, accepted exceptions, support contacts and next shift owner.

Final decision and approval

Record **go / restricted_go / no_go** with actual time, permitted scope, conditions, accepted risk IDs, expiry/next review, and named release-authority, operations, QA, data and domain approvals. Hardware/calibration approval is required for real-device claims. Missing critical safety/privacy/restore evidence means no real-data go-live. Publish only the approved scope; retain the decision and QA results in restricted evidence.

Qa Test Plan

QA test plan and acceptance criteria

Execution status: not run by this documentation pack. Prior results in fieldwork verification describe an earlier run; they do not certify this candidate, production HTTPS or physical sensors. QA lead records release/configuration/device versions and every execution in qa-test-results.csv.

1. Test policy and fixtures

- Test generated accounts and **clearly labelled synthetic demo data** against a

disposable SQLite DB, not the normal backend DB or real farm accounts. Use at least two farms, an owner/member/viewer, feed and silage, and repeated batch IDs across farms/sample types. No real external sharing, lab booking or SMS is needed.

- Run commands from the stated directories using existing dependencies. Browser

scripts additionally need an existing Playwright module, Chromium and runtime libraries; these are not installed by `npm test`. Record `blocked` if unavailable.

- Preserve stdout/stderr and exit status in restricted evidence. Include expected

versus actual behavior, issue reference and operator/time. Scrub passwords/cookies.

- `pass`: all assertions/expected outcomes met with evidence; `fail`: observed mismatch;

`blocked`: prerequisite missing; `not_run`: no execution. `not_applicable` needs a scope rationale and approver. Do not convert missing device testing into a pass.

2. Exact regression commands

From `backend/`:

```
.venv/bin/python -m unittest discover -s tests -p "test_*.py"
```

From `frontend/`:

```
npm test
VITE_PUBLIC_API_URL= VITE_API_URL= npm run build
```

From the repository root, for the independent synthetic calibration scaffold:

```
python3 -m unittest discover -s tools/calibration -p 'test_*.py' -v
```

Pass criteria: commands exit 0, no unexpected test failures/skips, build emits the production PWA files and no unresolved error/warning affecting scope. Record actual counts; do not expect historical 48/309 counts to remain fixed. Existing docs note a

FastAPI/Python deprecation warning; review the actual output. There is no configured separate lint/typecheck script. `git diff --check` from the root can check tracked whitespace; it does not check new untracked documentation by itself.

3. Isolated live browser setup

Ensure ports 8123/5174 are free; **do not stop existing user services to free them**. If occupied, arrange a separate test window/workspace. Live farmer/fieldwork scripts assert frontend port 5174; pointing them at arbitrary existing preview servers is unsafe because the scripts create test accounts/data.

From the root, create a fresh private disposable test location outside the repository:

```
ls -ld /tmp/omnirush
export SMOKE_DIR="$(mktemp -d /tmp/omnirush/nutriscan-launch-qa.XXXXXX)"
python3 -c 'import os; print(os.environ["SMOKE_DIR"])'
```

Use that **same printed path** as `SMOKE_DIR` in each terminal. Terminal A, from `backend/`, starts the disposable API:

```
NUTRI_SCAN_DB="$SMOKE_DIR/smoke.sqlite3" NUTRI_SCAN_COOKIE_SECURE=false \
.venv/bin/python -m uvicorn app.main:app --host 127.0.0.1 --port 8123
```

Terminal B, from `frontend/`, builds the app with no absolute API override and proxies to that API. A build replaces `dist/`; use a disposable test workspace if an existing build must be retained:

```
VITE_PUBLIC_API_URL= VITE_API_URL= npm run build
VITE_PUBLIC_API_URL= VITE_API_URL= VITE_PROXY_TARGET=http://127.0.0.1:8123 \
npm run preview -- --host 127.0.0.1 --port 5174 --strictPort
```

Terminal C, from `frontend/`, set an existing module path (omit that variable only if `playwright` already resolves), then run:

```
export NUTRISCAN_PLAYWRIGHT_MODULE=/PLACEHOLDER_EXISTING_INSTALL/node_modules/playwright
NUTRISCAN_SMOKE_URL=http://127.0.0.1:5174 NUTRISCAN_ALLOW_TEST_ACCOUNTS=1 \
node tests/farmer-flow.browser.mjs
NUTRISCAN_SMOKE_URL=http://127.0.0.1:5174 NUTRISCAN_ALLOW_TEST_ACCOUNTS=1 \
node tests/fieldwork-live.browser.mjs
NUTRISCAN_PREVIEW_URL=http://127.0.0.1:5174 node tests/pwa-live.browser.mjs
```

Optional isolated/mock browser scripts from `frontend/`, with the same Playwright prerequisite; they manage their own fixtures/servers:

```
node tests/fieldwork.browser.mjs
node tests/reading-assist.browser.mjs
node tests/pwa.browser.mjs
```

Pass: script exit 0 and expected assertions fulfilled; inspect actual output. Mocked serial/speech/share APIs do not prove permission prompts, physical transport, audio or external delivery. Stop only the test processes you started, retain needed redacted evidence and dispose of the isolated DB/profile under the retention policy.

4. Acceptance cases

Execute these against the candidate. Rows identify **required behavior**, not results. For API writes use the correct trusted `Origin` and authenticated session unless the test deliberately checks missing/foreign identity. Unknown/foreign object lookups should not leak another farm's existence.

ID	Exercise	Pass criteria
AUTH-01	Register, login, reload, logout; bad login	New farm/owner, password length >=10; HttpOnly session, no token/password in localStorage; logout revokes session; bad login 401; HTTPS cookie has Secure and SameSite=Lax
AUTH-02	Owner creates member/viewer; roles attempt writes	Owner manages members; member can scan/upload; viewer reads/downloads but scan/fieldwork/upload writes return 403; non-owner member creation denied
AUTH-03	Second farm, forged `farm_id`, guessed IDs, foreign attachments	No cross-farm read/write; explicit mismatched farm filter/body 403 where supported; missing/foreign private resources 404; no-session routes 401
AUTH-04	Missing/untrusted Origin on mutation, allowed Origin, repeated bad logins	Missing/untrusted Origin 403; legitimate origin succeeds; throttle returns 429 under its documented limit; all private responses/errors no-store/nosniff
SCAN-01	Blank, malformed, nonfinite and out-of-range inputs; actual zero	UI blocks missing/invalid values and requires all ten; zero remains zero; visual inputs 0–1; percent inputs 0–100, pH 0–14, UI/instrument temperature -20–80; no fabricated defaults
SCAN-02	Four steps, Back, edit after confirmation, submit/refresh	Back retains values; edits invalidate review; persisted result/IDs match reviewed request; prototype and uncalibrated metadata visible
SCAN-03	Repeat same farm/request ID with same/changed payload after restart	Equivalent request returns same saved scan, no duplicate; changed payload 409; scope remains farm-local
HIST-01	Search text/exact batch, UTC date bounds, filters and pagination	Literal matching; inclusive UTC dates; all expected farm records reachable; feed/silage not merged; missing trend values remain missing
REPORT-01	Export then edit unsaved form; refresh/export history; popup blocked	HTML represents saved record/ID only; print hides controls but keeps measurements/advisory/disclaimer; PDF saved via browser dialog; blocked popup explains HTML fallback
PHOTO-01	Valid JPEG/PNG/WebP; metadata-bearing, bad type, too large, sixth photo	Private evidence saved/readable by farm; normalized and metadata stripped; <=5 MiB and max five/parent; invalid payload/type/size/count rejected; no AI-image claim
SHARE-01	QR on another device; preview summary and selected evidence	QR contains batch route, no token or access grant; sign-in required; farm identity/photos off by default; text share has no private URLs/tokens; photos only through supported explicit native file sharing
DRAFT-01	Save/queue offline in open signed-in app; reconnect/retry; switch user	No implicit upload on reconnect; explicit sync idempotent; failures remain actionable; no other user's drafts visible; no photos/tokens/API responses stored
DRAFT-02	Save/load instrument-origin draft	Readings can persist but device/time/calibration provenance is stripped; warning visible; resulting record is not presented as retaining instrument provenance
FIELD-01	Create/edit batch; duplicate identity; Use this batch	Unique farm/batch/sample, immutable batch/sample identity; valid dates/quantity; reuse copies identity only, never old readings
FIELD-02	Save Yes/No/Unknown observations, including untouched answers	Unknown stays unknown; `observation_only`; no engine call, score, confidence or safety verdict
FIELD-03	Reminder complete/reopen; local date due groups	Correct persistence and calendar grouping; no promise of SMS/background delivery
FIELD-04	Request → collected → report_received, invalid backward/date paths	Forward transitions/date rules enforced; report required before received; tracker never says booked/sent automatically
FIELD-05	Request report upload/download; photo/report limits	Max five attachments; photos <=5 MiB, PDF <=10 MiB; basic PDF signature/EOF validation, attachment download/no inline execution; no OCR/malware-scan claim

ID	Exercise	Pass criteria
SERIAL-01	Valid 1–7 reading frame, select subset, cancel/edit/recapture	Explicit user port/review/application only, no commands/submit; unselected readings unchanged; provenance matches selected values; edits drop affected provenance
SERIAL-02	Unknown/duplicate keys/fields, numeric strings, wrong units/time, oversize, timeout/disconnect	Invalid frames not applied; no invented readings or scan; clear failure/manual fallback; 16,384-byte/15-second boundaries exercised
VOICE-01	Consent denied, stop/unmount, reviewed value, language unavailable	No automatic microphone use or submission; typed fallback always available; browser/provider/network dependence clear; app stores no audio
PWA-01	HTTPS production install, inspect caches, offline reload in three languages	Correct manifest/icons/root scope; only public assets/help cached; API/auth/photos/reports network-only; offline page exposes no farm data/login form
PWA-02	Candidate update, explicit activation and old-release rollback	No forced activation while unsaved; coherent worker/assets; old namespaces removed on activation; private cache boundary survives both versions
OPS-01	Backup, restore, startup + first fieldwork call, role/attachment readback	Integrity/FK checks pass, accounts/scans/attachments recover, sessions revoked in restore, deletion ledger reconciled; approved RPO/RTO met
CAL-01	Synthetic scaffold tests/report and label review	CLI report is explicitly `synthetic_demo_not_for_field_use`; coefficients not connected to app; no lab-accuracy claims
UX-01	390px and target devices; English/Hindi/Marathi; listening and keyboard	No hidden required control/overflow blocker; readable text/icons and focus; actual voices/translation approval recorded; manual fallbacks usable

The raw manual API `SensorData.temperature_c` currently accepts finite numbers without the UI/instrument -20–80 bound. Do not assert API parity for that field; record the difference and decide whether a broader-input API is acceptable for the release scope. Client validation is not a substitute for server validation.

5. Field pilot smoke — field lead + QA witness

Use authorized pilot accounts on the pilot environment and actual approved devices; label any demo fixtures synthetic. Preserve real and synthetic records separately.

1. Record site/date, target phone/browser/OS, network, release, device/sensor IDs,

calibration/time checks and operator training. Confirm support can be reached.

1. Sign in; create/select the correct batch/sample; capture one observation with

Unknown preserved and confirm it has no score.

1. Run reference checks, then an actual instrument capture (or measured manual

fallback), review all ten inputs and submit **only if complete**. Refresh saved history/provenance; compare readings against the operator record.

1. Download a saved HTML report and review the print/PDF flow. Test authenticated

QR access from another device. Review a share preview without sending externally.

1. On a disposable complete-reading draft, disconnect/reconnect and demonstrate

explicit sync, count/history confirmation and the provenance-loss warning. Test offline public help separately; do not expect private history after offline reload.

1. Demonstrate reminder complete/reopen and local testing-request/report tracking

with a clearly synthetic trusted report; never claim a lab was contacted.

1. Record serial unplug/permission denial, real speech/manual fallback, logout/shared

device handling, discrepancy/referral and end-of-shift reconciliation.

Pass: every in-scope step meets acceptance criteria, no unaccounted sample/scan/queue item, no misleading safety claim and no unresolved critical/high failure. Stop real-data capture for failed reference checks, unsafe hardware or farm isolation failures. Record unavailable conditions as blocked and schedule a witnessed rerun.

6. Exit and regression policy

Required regression commands and all scope-required acceptance cases must pass for the exact candidate. Critical safety/privacy/data-loss failures are no-go; high functional failures block affected workflows. Lower-priority exceptions need a named owner, bounded scope, workaround, expiry and release-authority approval. Device-specific features cannot be advertised as verified while their real-device checks are blocked.

On a fix, rerun the failing case and affected neighboring cases; rerun the broader required suites when the candidate changes. Attach QA summary to production launch, and log incidents/discrepancies under the support runbook.

Risk Register

Technical, product, safety and business risk register

Initial assessment on **2026-10-15**; all entries are **open / mitigation unverified**. Likelihood/impact are qualitative planning estimates (`L|M|H`), not measured rates. Named individuals, numeric thresholds, review dates and residual-risk acceptance belong in the restricted release record. Role titles below are unassigned owners until a person accepts them. No mitigation is considered completed by documenting it.

ID / category	Failure and consequence	Likelihood / impact	Mitigation and required evidence	Accountable role / review trigger
R01 safety/product	Prototype risk/confidence mistaken for lab accuracy or safe-to-feed approval	H / H	Prominent uncalibrated disclaimer in results/reports/shares/training; qualified referral; no certification/contaminant claims; wording review and field comprehension evidence	Domain/product lead; every claims/score/translation change
R02 hardware/safety	Raw pH probe, 5 V ADC signal, unsuitable supply or wet damaged wiring harms board/operator or corrupts readings	M / H	Exact-board datasheets, conditioned interface, powered-off inspection, measured voltage limits, protected enclosure and qualified wiring sign-off	Hardware lead; every module/wiring change or damage
R03 validity	Demo constants, placeholder ADC conversions, missing temperature/humidity hooks or stale manual time appear to be real observations	H / H	Label demo units/sessions; separate synthetic records; implement validated hooks/clock; inventory/calibration evidence and actual reference tests	Hardware/calibration lead; each device/firmware/session
R04 data/model	Unrepresentative samples, lab basis mismatch, contaminated material or split leakage produces optimistic accuracy	H / H	Custody and lab QC, immutable raw hashes, train-only transforms, farm/batch/time grouping, locked test and subgroup thresholds; independent release review	Calibration/data leads; dataset/model/method/domain change
R05 security/privacy	Cross-farm reads/writes or trusted-Origin/cookie misconfiguration exposes farm data	M / H	Farm-scoped auth/role/attachment tests, HTTPS/Secure cookie and exact origins; review always-on localhost/own origins and edge restrictions	Security/operations lead; deployment/auth/proxy change
R06 identity/business	Public account creation abuse, incomplete reset/removal workflow or process-local throttle enables unauthorized use/support overload	H / H	Restricted-pilot edge access, named account/admin process and incident contacts; resolve public registration/withdrawal/recovery controls before broad launch; abuse/load evidence	Product/security lead; exposure expansion or staff change
R07 operations/data	SQLite write contention, attachment growth, disk exhaustion or corruption loses scans/availability	M / H	Single-process starting topology, real load/capacity measurement, storage alerts, bounded uploads, consistent encrypted backups and restore drill; capacity-based DB migration decision	Operations lead; load/storage trend or backup failure
R08 release/data	Additive/lazy schema changes or legacy assignment break old code, misassign ownership or make rollback unsafe	M / H	Copy-based startup + first-fieldwork migration rehearsal, ownership review, before/after IDs/counts, compatibility pairing, pre-change backup, rollback drill	Release/operations lead; migration/assignment/release

ID / category	Failure and consequence	Likelihood / impact	Mitigation and required evidence	Accountable role / review trigger
R09 offline/data	User assumes private offline reload, auto-sync or preserved provenance; loses drafts/traceability or duplicates a scan	H / H	Public-help-only messaging, explicit sync/idempotency tests, draft provenance warning, protected independent capture log, end-of-shift reconciliation and no blind storage clearing	Field/frontend lead; queue/PWA/storage change
R10 PWA/privacy	Cache policy change/CDN caches private data, or mixed worker/assets strand old tabs	M / H	Allowlist cache inspection, private no-store/network-only tests, atomic artifact/worker update, retained old hashed assets, explicit activation and rollback test	Frontend/operations lead; build/cache/CDN update
R11 browser/product	Web Serial, microphone, speech language, native sharing, printing or install unsupported on target devices	H / M	Actual HTTPS phone/browser matrix, permission/unplug tests and usable measured manual/readable/download fallback; do not equate mocks with device evidence	QA/field lead; new device/browser/OS
R12 attachments/privacy	Trusted-looking PDF is malicious; photos/reports/exports/shared devices reveal identity	M / H	Basic PDF validation explicitly not scanning; trusted-provider handling, isolated review, photo metadata stripping, preview-first sharing, local/export deletion guidance and privacy incident drills	Data/support lead; new file flow or exposure
R13 UX/safety	Draft translations, unclear units/visual scores or inaccessible controls change an operator's decision	M / H	Expert English/Hindi/Marathi review, units/basis training, unknown stays unknown, no fabricated values, 390px/actual-device operator comprehension checks	Domain/field lead; copy/language/workflow change
R14 product/safety	Local testing request/reminder implies real lab booking, certified provider, report interpretation or SMS	H / H	Explicit local-tracker labels, manual provider/contact confirmation, report-received gating, qualified interpretation and SOP comprehension test	Product/field lead; testing/reminder integration change
R15 governance/business	Unexecuted retention/deletion, resurrected backup data, lost device or unowned support creates privacy/reputation cost	M / H	Approved schedule/owners, deletion ledger, restore reconciliation, access reviews, local-device cleanup and incident/on-call exercise	Data/operations lead; restore, withdrawal or lost device
R16 business/scale	Sensor/lab/logistics costs, support load, poor repeatability or low farmer adoption defeats viability	H / H	Bounded pilot records actual cost per usable sample, lab turnaround, repeat visits, support tickets and farmer comprehension/value; pre-agreed budget/service thresholds before expansion	Business/product lead; pilot review or budget breach
R17 contract/validity	Raw manual API accepts finite temperatures outside UI/serial -20–80 range; integrations assume identical validation	M / M	Document current contract, inventory clients, test UI/serial bounds separately and decide server-range policy before third-party API exposure; record exception/scope	Backend/product lead; external integration/schema change

Review and acceptance procedure

1. At readiness review, assign a named owner, target date, evidence reference,

current/residual rating and proposed disposition (`mitigate|avoid|accept|block`).

1. Review weekly during a pilot, before each candidate release and immediately

after any trigger/incident. Track actual likelihood evidence rather than reducing ratings because a checklist exists.

1. Safety/privacy/data-loss risks with missing critical controls block real-data

operation. A synthetic-only demo can proceed only under that explicit scope; it does not accept the same risks for farmers or production data.

1. Release authority and the relevant domain/data owner approve bounded residual

risk with a reason, scope, expiry and stop trigger. No numeric validation gate may be waived by citing a synthetic scaffold result.

1. Escalate materialized risks through support/incidents

and feed corrective actions into QA,
readiness and
final launch.

Sources for current limitations: README,
fieldwork contract,
real-device gaps,
ESP32,
calibration workflow.

Support And Incident Runbook

Support and incident runbook

Before launch: replace `PLACEHOLDER_SUPPORT_ROUTE`, `PLACEHOLDER_ON_CALL_CONTACT`, `PLACEHOLDER_DATA_STEWARD` and `PLACEHOLDER_QUALIFIED_ADVISER` in the restricted contact sheet. Name primary/backup owners, actual coverage hours and escalation route. These placeholders are not working contact details or a 24/7 support promise.

Use incident-log.csv for incident summaries and a restricted linked timeline for updates. Do not place passwords, cookies, full private reports, raw farm contacts or identifiable screenshots in public tickets.

1. Intake and ordinary support

Collect: opaque ticket ID, user role/farm alias, time/timezone, release, page/action, browser/OS, online/offline state, error/status, reproduction steps and affected record IDs only where needed. Ask permission for minimized screenshots. Never ask the farmer to send a password/session token or run an unreviewed database command.

Symptom	First checks / resolution
Login returns 401/403	Check correct account, HTTPS/Secure cookie, origin/proxy configuration and system time; 403 may be Origin/role policy, not a bad password. No in-app password reset is supplied.
Login returns 429	Respect Retry-After; review throttling/proxy client-IP behavior. Do not repeatedly retry or disable controls as a quick fix.
Scan cannot submit	Confirm all ten actual inputs, valid units/bounds and final review; member/owner role. If measurements are unavailable, use observation-only tracking rather than demo fillers.
Queue appears stuck	Confirm correct user/farm, connectivity and explicit Sync now. Check saved history/request ID before retry; unchanged requests are idempotent. A 409 needs review, not a new ID to force a duplicate.
Device details missing after draft load	Expected allowlist limitation: readings retained, instrument provenance stripped. Use independent field records; do not claim the original device timestamp survived.
Serial/voice unavailable	HTTPS/localhost, supported browser, permission, correct port/115200, close serial monitor, inspect timeout/frame. Keep manual measured entry available. Speech may depend on an external browser service/network.
Cannot upload evidence	Check supported type, <=5 MiB photo/<=10 MiB PDF, max five per parent, valid file and writer role. PDF checks are basic; do not open untrusted reports.
Print/install fails	Permit report popup or download HTML then print. Install depends on browser; test built root-path PWA over HTTPS. Offline reload is public help only.
Missing testing/reminder notification	Requests are farm-local tracking, not lab bookings. Reminders are in-app, not SMS/background notifications. Confirm manually with the selected provider.
Conflicting score/lab result or suspect feed	Stop relying on the score for that batch, preserve IDs/measurements, refer to a qualified lab/adviser. Scores/confidence are uncalibrated and never authorize feeding.

2. Severity and response targets

Targets below apply **during the explicitly staffed coverage window** and must be approved before launch. Record actual acknowledgment times; never backfill a target as an achieved response.

Severity	Examples	Target acknowledgment / coordination
SEV1 critical	Cross-farm exposure, active credential/backup compromise, suspected harmful safety claim/use, widespread corruption/data loss	15 minutes; incident lead + operations + data/domain lead immediately; stop affected real-data use
SEV2 high	Core login/scan unavailable, failed restores/backups threatening RPO, major wrong result/provenance, failed calibration affecting collected samples	1 hour; incident lead coordinates same-window containment and qualified review
SEV3 medium	One optional feature/device failing with safe manual fallback; isolated report/UI defect	1 business day; support/feature owner tracks workaround and fix
SEV4 low	Cosmetic/documentation issue without behavior or safety impact	3 business days; backlog with owner and next review

Outside coverage, document the actual escalation channel and safe stop/manual fallback instructions. Upgrade severity if scope is unknown and safety/privacy could be affected; incident lead records the reasoning and later reclassification.

3. Response sequence

1. **Declare / assign.** Create incident ID, severity, detection time, affected

environment/version, reporter alias, incident lead, operations owner, data/domain reviewer and next update time. Start a timezone-aware timeline.

1. **Contain.** Pause affected capture/writes/exports or restrict the edge; quarantine

affected hardware/sample batches. For private cache exposure, prevent further distribution and correct worker/static policy under controlled recovery. For potential feed harm, tell operators to obtain qualified advice, not to use a higher/lower prototype score as a safety test.

1. **Preserve evidence.** Keep a protected consistent DB snapshot, artifact/config

digests, redacted logs, request/scan/sample/device IDs, clock/calibration status, error counts and minimal screenshots. Follow custody/access controls. Do not delete or alter the only record of what happened.

1. **Assess.** Determine start/end time, farms/users/records, app/firmware/model

version, data categories, potential recipients, affected calibration interval and whether backup/deletion obligations are involved. Separate observed facts from hypotheses and list unknowns.

1. **Recover.** Fix/redeploy or use the rollback runbook.

Do not change DB schema, replay queues or downgrade code against an unknown schema ad hoc. A DB restore needs session revocation and deletion reconciliation.

1. **Verify.** Reproduce the failed path in isolation; rerun scoped

QA acceptance, isolation/attachment/readback checks and the affected real-device/reference checks. Observe service/queue/error metrics for an incident-lead-approved period before reopening.

1. **Communicate / close.** Report impact, containment, user action, remaining uncertainty and next update. Data steward coordinates privacy communications; domain lead approves safety wording. Record actual recovery/data loss and obtain closure approval. Schedule corrective actions, owners and due dates.

4. Operator diagnostics

Use actual environment paths/hostnames. These read-only probes do not authenticate or prove DB health. Expected unauthenticated private status is 401, not 200:

```
export APP_ORIGIN=https://nutriscan.example.invalid
curl --fail --show-error --silent "$APP_ORIGIN/health"
curl --show-error --silent -D - -o /dev/null "$APP_ORIGIN/api/v1/auth/me"
curl --fail --show-error --silent -I "$APP_ORIGIN/sw.js"
```

Authorized host operators can check a protected DB **read-only**; results may be resource-intensive on a large DB, so schedule appropriately. This does not migrate or create tables. From any directory, with the correct absolute path:

```
export NUTRI_SCAN_DB=/srv/nutriscan-data/PLACEHOLDER_PRODUCTION.sqlite3
python3 - <<'PY'
import os, sqlite3
from pathlib import Path
path = Path(os.environ['NUTRI_SCAN_DB']).resolve()
assert path.is_file()
db = sqlite3.connect(path.as_uri() + '?mode=ro', uri=True)
try:
    print('integrity:', db.execute('PRAGMA integrity_check').fetchall())
    print('foreign_key_violations:', len(db.execute('PRAGMA foreign_key_check').fetchall()))
finally:
    db.close()
PY
```

Record output in restricted evidence. A green health endpoint with a failed DB check is not recovery. Inspect disk/backup status through the host's existing monitoring, and confirm an authenticated scan/fieldwork read on an authorized test scope without logging response bodies or secrets.

Emergency session revocation

This is a **privileged, global write**: it signs out every user, not just the reporter. Incident lead approval, edge restriction, stopped/drained API, verified backup and confirmed environment are prerequisites. With the correct `NUTRI_SCAN_DB`:

```
python3 - <<'PY'
import os, sqlite3
from pathlib import Path
path = Path(os.environ['NUTRI_SCAN_DB']).resolve()
assert path.is_file()
db = sqlite3.connect(path.as_uri() + '?mode=rw', uri=True)
try:
    with db:
        revoked = db.execute('DELETE FROM sessions').rowcount
    print('Sessions revoked:', revoked)
finally:
    db.close()
PY
```

This does **not** disable accounts, reset passwords or prevent a compromised credential from logging in again. Keep affected access restricted until an approved identity recovery/removal process resolves the compromise. Rotate compromised host, TLS/backup or future integration credentials using the responsible secret store; there is no shared application session-signing secret to rotate in this MVP.

5. Communication, post-incident review and handover

Use this redacted update structure:

> Incident PLACEHOLDER_ID, detected PLACEHOLDER_TIME: observed impact and affected workflow/scope. Containment in place: PLACEHOLDER_ACTION. Operator action: PLACEHOLDER_ACTION. Known unknowns: PLACEHOLDER_UNKNOWNS. Next update: PLACEHOLDER_TIME via the approved support route. Prototype outputs remain screening indicators; seek qualified advice for feed/silage safety concerns.

For SEV1/SEV2 propose a review within two business days after stabilization. Capture timeline, cause/contributing conditions, detection/response gaps, actual data loss, sample/model implications, backup/restore results, deletion reconciliation, affected user follow-up, failed controls and concrete prevention actions with owners/dates. Update the risk register, SOP/training and relevant QA cases.
Close only after evidence is reviewed, access is safe, affected samples are dispositioned by the domain lead and residual work is owned. Hand over open incidents explicitly at every shift change; silence is not a completed handover.

Technical Readiness Checklist

Technical readiness checklist

Status: unassessed. Every item below needs an owner, date, evidence reference and decision in the release record. Use QA results for executed checks. Missing evidence is `not_run` or `blocked`, never a pass.

1. Scope and ownership — release lead

- ☐ Select scope: synthetic demo, restricted real-data pilot, or broader production.

State farms, users, languages, sample types, devices and permitted claims.

- ☐ Assign release, operations, QA, hardware, calibration, data, field and support

leads plus backups and an out-of-band contact route. Fill actual support hours.

- ☐ Record the exact build/source/configuration and approved rollback artifacts.

Preserve unrelated working-tree work; do not deploy an unidentified dirty tree.

- ☐ Confirm the uncalibrated-prototype statement is visible in onboarding, saved

results, reports, share previews, demo scripts and operator training.

- ☐ Explicitly exclude lab equivalence, certified contaminants and automated photo

diagnosis until the calibration gates are met.

- ☐ Review risks; document residual risk owners and expiry dates.

2. Hosting and persistence — operations lead

- ☐ Rehearse deployment in an isolated staging database.

Dev Compose and Vite preview are not production process management.

- ☐ Use HTTPS with a valid certificate, root-path PWA hosting, restricted backend

network access and explicit trusted reverse-proxy settings.

- ☐ Verify `NUTRI_SCAN_COOKIE_SECURE=true`, allowed frontend origins and actual

cookie behavior in deployed browsers. No wildcard credentialed CORS.

- ☐ Review always-added localhost origins, server-own-origin acceptance and

SameSite=Lax constraints; edge controls and topology must match the approved scope.

- ☐ Use an absolute `NUTRI_SCAN_DB` on durable local storage with restricted access;

confirm free space, attachment growth, SQLite writer load and restart persistence.

- ☐ Complete consistent encrypted backups, checksum/integrity checks, restore

drill and migration rehearsal including first fieldwork access (lazy table creation).

- ☐ Approve numeric RPO/RTO and backup schedule. Proposed defaults in

governance require actual operational evidence.

- ☐ Provide an owned service supervisor, TLS renewal, monitoring and alert routing.

`/health` alone does not prove DB readability, capacity or authenticated writes.

3. Identity and privacy — data/security lead

- ☐ Verify owner/member writes, viewer read-only, cross-farm isolation, trusted

Origin checks and private `no-store` responses, including errors/attachments.

- ☐ Confirm there is no default account or embedded credential. Keep runtime

configuration and protected TLS/backup credentials out of builds and logs.

- ☐ Resolve public registration, account withdrawal, password recovery and session

revocation for the chosen scope. These administrative workflows are incomplete
in the MVP; use a documented restricted-pilot procedure or block broader launch.

- ☐ Approve the retention/access/deletion policy, local draft/shared-device handling,

sharing education and restore-time deletion reconciliation.

- ☐ Assess trusted PDF handling (basic validation is not malware scanning), logging

redaction and optional browser speech-provider processing.

4. Hardware and data validity — hardware/calibration leads

- ☐ Complete wiring review against the exact

ESP32 board/module datasheets; no raw pH probe or 5 V ADC connection.

- ☐ Record inventory, firmware digest,

pins, voltage limits, time source and sensor IDs for each device.

- ☐ Mark demo units and synthetic data unmistakably. Do not treat `DEMO_MODE=false`

as validated; moisture/pH conversions are placeholders and temperature/humidity
hooks need implementation before real measurement.

- ☐ Record reference checks, calibration intervals/tolerances and signed results

using calibration records.

- ☐ Test real port permission, unplug/reconnect, invalid frames, timeout, partial

readings and reviewed-only application on actual target devices.

- ☐ Establish sample/lab chain of custody and immutable raw records for any

accuracy evaluation. Synthetic scaffold metrics cannot approve field use.

5. Product and field operation — QA/field leads

- ☐ Execute regression and acceptance tests for the candidate;

retain outputs and discrepancies without importing historical pass counts.

- ☐ Verify all ten inputs, saved-record reports, batch/sample separation, photos, QR authentication and explicit idempotent offline sync.
- ☐ Verify observation-only records have no score; requests do not imply lab booking; reminders have no SMS/background delivery promise.
- ☐ Verify built PWA update/rollback, public-only Cache Storage and offline help in English/Hindi/Marathi. Private responses must remain network-only.
- ☐ Demonstrate the loss of device/time/calibration provenance through draft save/load and define a traceable manual fallback for field sessions.
- ☐ Obtain qualified review of translated safety guidance and target-device readability/voice behavior; mocked browser speech does not prove a real microphone.
- ☐ Train operators with the SOP, escalation script and sample handling plan. Unknown observations and absent measurements stay unknown.

6. Decision

The release authority records `go|restricted_go|no_go`, permitted scope, blocked items and next review date. Safety, farm isolation, durable data/restore and required QA failures are **no-go** for real-data operation. An approved synthetic demo does not approve a field pilot. A pilot does not approve laboratory or safety claims. Complete the production checklist for the actual cutover and handover.

Source baseline: README, fieldwork contract, verification/gaps, ESP32.

Validation Report

Launch pack document validation

Review date: 2026-10-15

Status: document checks passed. This records checks run for these saved planning materials, not product-release approval, legal/accounting advice, a security audit or scientific validation. Application behavior was not changed for this documentation task.

Saved scope

- 46 launch artifacts: **35 Markdown documents and 11 CSV templates**.
- One master index, business/customer/commercial drafts, technical/operations pack,

reusable records, worked financial scenarios and a proposed 90-day action tracker.

- A student-founder start plan covering mentor/team formation, three farmer

interviews, a proposed 14-day target and a low-spend, observation-first approach.

- Four generated convenience PDFs are saved under `docs/launch/pdf/`: the complete

launch pack, business pack, technical pack and student start plan. They contain source documentation only; regenerate them after Markdown/CSV edits.

PDF generation used the locally available Playwright Chromium renderer. The outputs were opened as valid PDF 1.4 files and ended with `%%EOF`: launch pack (8 pages), business pack (8 pages), technical pack (8 pages), and student plan (4 pages). Page counts were independently checked with the system `file` utility; generated PDFs are convenience exports, not the editable source of truth.

- Root `README.md` linked to the pack and clarified to distinguish current input

adapters/prototype scoring from the presentation's future architecture.

- No company registration, public publication/deployment, outreach, partner execution,

payment collection or actual launch approval performed.

Consolidated checks actually run

The session-local read-only validator at `/tmp/omnirush/validate_nutriscan_launch.py` was run from the repository root with:

```
python3 /tmp/omnirush/validate_nutriscan_launch.py
```

This temporary validator is diagnostic tooling for the current workspace, not an application dependency or a portable file shipped in the launch pack. Its checks:

- Resolve local Markdown destinations and heading anchors in the launch pack plus

root README navigation; ignore illustrative timestamp syntax inside inline code.

- Confirm every launch artifact is reachable from the master index; check Markdown

table widths/separators, balanced fenced blocks and trailing whitespace, including new/untracked files that Git's whitespace check alone would miss.

- Parse all CSVs with strict quoting and consistent row widths/unique headers;

reject obvious formula-leading and nonfinite cells. Values stay synthetic, illustrative or placeholder-labelled rather than fabricated customer records.

- Check square-bracket business placeholders against the 46-token canonical register;

verify action dates/dependencies/statuses and that no checklist is pre-approved.

- Recalculate all four pricing scenarios with decimal arithmetic: variable cost,

contribution, margin, ceiling break-even count and CAC payback.

- Recalculate all 12 cash-flow rows and roll-forwards: active/new/churned farms,

receipts, delivery/acquisition costs, net cash flow, closing cash and cumulative gap. The worked scenario first goes negative in month 5 and ends at −₹841,382.46; this is not a forecast or a verified financing need.

- Parse shell examples with `bash -n` without running deployment, deletion, migration

or service commands; parse embedded Python and JSON examples without operating them.

- Extract the synthetic sales-demo frame (using a fixed aware timestamp for this

check) and pass it through the current frontend `parseInstrumentFrame` in Node; verify four readings, unknown calibration and humidity unit `%`.

- Scan launch files for common private-key/token/credential-URL patterns. This is a

bounded heuristic check, not proof that a repository or a deployment has no secrets.

- Run `git diff --check`; it passed. All launch files are separately whitespace-checked.

The first run identified five Markdown hard-break trailing-space lines in the outreach draft; these were converted to blank-line breaks. A timestamp-format expression was initially misread as a link by the diagnostic parser; inline-code handling was corrected. The consolidated rerun passed after those changes.

Claims and review boundaries

Customer-facing one-pager, business plan, pricing, pilot terms, sales script, FAQ, outreach and website copy were reviewed against the pack's claims matrix. Drafts explicitly distinguish uncalibrated scoring, synthetic firmware, evidence-only photos, local laboratory requests, in-app reminders, public-help offline reload and private data needing connection. No lab accuracy, toxin detection, certified calibration, diagnosis, feeding guarantee or automatic laboratory booking is represented as available.

Prices/costs/volumes remain illustrative. Legal/privacy documents are drafting aids, not executed agreements or an implemented retention/deletion service. Completed forms, real customer contacts, financial records and release evidence belong outside Git.

Not rerun or established by this task

- Backend/frontend/calibration application suites, production build and live browser

flows were **not rerun** solely for these documentation edits. Prior results are recorded in `docs/fieldwork-verification.md` and `docs/calibration-workflow.md`.

- Shell/Python syntax checks do not prove deployment, restore, Nginx configuration, firmware compilation, real USB/microphone behavior or production compatibility.
- External URLs, provider identity/qualification, customer demand, tax/regulatory applicability, insurance, actual costs and available cash were not independently verified.
- Founder details, approved prices/dates, customer scope, professional review and real-device/security/restore evidence remain required before customer-ready use.

12 Month Cash Flow

month	row_status	currency	new_farms	churned_farms	active_farms	monthly_price_inr	direct_cost_per_farm_inr	cac_inr	opening_cash_inr	customer_receipts_inr	funding_received_inr	direct_cash_cost_inr	fixed_cash_cost_inr	acquisition_spend_inr	other_reviewed_cash_costs_inr	net_cash_flow_inr	closing_cash_before_gap_inr	cumulative_funding_gap_inr
1	ILLUSTRATIVE_NOT_A_FORECAST	INR	0	0	0	1499.00	579.98	3000.00	600000.00	0.00	0.00	0.00	120000.00	0.00	0.00	-120000.00	480000.00	0.00
2	ILLUSTRATIVE_NOT_A_FORECAST	INR	0	0	0	1499.00	579.98	3000.00	480000.00	0.00	0.00	0.00	120000.00	0.00	0.00	-120000.00	360000.00	0.00
3	ILLUSTRATIVE_NOT_A_FORECAST	INR	1	0	1	1499.00	579.98	3000.00	360000.00	1499.00	0.00	579.98	120000.00	3000.00	0.00	-122080.98	237919.02	0.00
4	ILLUSTRATIVE_NOT_A_FORECAST	INR	2	0	3	1499.00	579.98	3000.00	237919.02	4497.00	0.00	1739.94	120000.00	6000.00	0.00	-123242.94	114676.08	0.00
5	ILLUSTRATIVE_NOT_A_FORECAST	INR	2	0	5	1499.00	579.98	3000.00	114676.08	7495.00	0.00	2899.90	120000.00	6000.00	0.00	-121404.90	-6728.82	6728.82
6	ILLUSTRATIVE_NOT_A_FORECAST	INR	3	0	8	1499.00	579.98	3000.00	-6728.82	11992.00	0.00	4639.84	120000.00	9000.00	0.00	-121647.84	-128376.66	128376.66
7	ILLUSTRATIVE_NOT_A_FORECAST	INR	4	0	12	1499.00	579.98	3000.00	-128376.66	17988.00	0.00	6959.76	120000.00	12000.00	0.00	-120971.76	-249348.42	249348.42
8	ILLUSTRATIVE_NOT_A_FORECAST	INR	6	0	18	1499.00	579.98	3000.00	-249348.42	26982.00	0.00	10439.64	120000.00	18000.00	0.00	-121457.64	-370806.06	370806.06
9	ILLUSTRATIVE_NOT_A_FORECAST	INR	7	0	25	1499.00	579.98	3000.00	-370806.06	37475.00	0.00	14499.50	120000.00	21000.00	0.00	-118024.50	-488830.56	488830.56
10	ILLUSTRATIVE_NOT_A_FORECAST	INR	10	0	35	1499.00	579.98	3000.00	-488830.56	52465.00	0.00	20299.30	120000.00	30000.00	0.00	-117834.30	-606664.86	606664.86
11	ILLUSTRATIVE_NOT_A_FORECAST	INR	15	0	50	1499.00	579.98	3000.00	-606664.86	74950.00	0.00	28999.00	120000.00	45000.00	0.00	-119049.00	-725713.86	725713.86
12	ILLUSTRATIVE_NOT_A_FORECAST	INR	20	0	70	1499.00	579.98	3000.00	-725713.86	104930.00	0.00	40598.60	120000.00	60000.00	0.00	-115668.60	-841382.46	841382.46

90 Day Action Plan

action_id	plan_status	phase	earliest_day	target_day	owner	action	dependency_ids	required_evidence	status	completed_at	evidence_ref
A01	PROPOSED_NOT_APPROVED	discovery	1	3	[FOUNDER NAME]	Confirm one farm workflow and buyer user payer roles		Written ICP and intended use	not_started		
A02	PROPOSED_NOT_APPROVED	finance	1	7	[OWNER]	Replace illustrative cost and cash assumptions with reviewed budget	A01	Approved spending cap and restricted finance evidence	not_started		
A03	PROPOSED_NOT_APPROVED	discovery	4	14	[FOUNDER NAME]	Conduct permissioned discovery interviews and record objections	A01	Recent problem examples and buyer authority evidence	not_started		
A04	PROPOSED_NOT_APPROVED	claims	10	17	[OWNER]	Review one-pager website copy and target-device synthetic demo	A01;A03	Approved claims and actual rehearsal record	not_started		
A05	PROPOSED_NOT_APPROVED	legal_data	8	21	[OWNER]	Resolve entity IP data roles privacy notice and optional permissions	A01	Qualified review and reliable authorized closeout process	not_started		
A06	PROPOSED_NOT_APPROVED	partners	8	28	[OWNER]	Verify relevant adviser lab introduction and separate study needs	A01	Verified contact purpose and no implied partnership	not_started		
A07	PROPOSED_NOT_APPROVED	commercial	15	28	[FOUNDER NAME]	Qualify scoped pilot and review quotation proposal taxes and payment	A02;A03;A05	Named sponsor operator payer and approved terms	not_started		
A08	PROPOSED_NOT_APPROVED	readiness	15	35	[OWNER]	Rehearse secure deployment access isolation backups restore and QA	A01;A05	Actual scope-specific evidence and unresolved risk dispositions	not_started		
A09	PROPOSED_NOT_APPROVED	hardware_optional	15	35	[OWNER]	If real device is included validate wiring hooks time and reference checks	A01;A02	Device-specific evidence or explicit demo-only exclusion	not_started		
A10	PROPOSED_NOT_APPROVED	pilot_gate	29	35	[OWNER]	Approve or pause pilot after terms payment prerequisites and all gates	A04;A07;A08;A09	Named restricted-go or no-go decision and baseline metric method	not_started		
A11	PROPOSED_NOT_APPROVED	pilot_onboarding	36	42	[OWNER]	Begin permitted pilot scope and verify saved readback and operator comprehension	A10	Onboarding records and staffed first-use checks	not_started		
A12	PROPOSED_NOT_APPROVED	pilot_delivery	36	77	[OWNER]	Review weekly workflow KPIs missing data support costs and incidents	A10	Weekly restricted evidence excluding demo rows	not_started		
A13	PROPOSED_NOT_APPROVED	pilot_closeout	77	77	[OWNER]	Review six-week workflow acceptance issues and remedies	A11;A12	Customer-reviewed outcome and actual costs	not_started		

action_id	plan_status	phase	earliest_day	target_day	owner	action	dependency_ids	required_evidence	status	completed_at	evidence_ref
A14	PROPOSED_NOT_APPROVED	continuation	78	90	[FOUNDER NAME]	Obtain explicit paid continuation or complete approved exit process	A13	Separate reviewed order plus actual collection or closeout record	not_started		
A15	PROPOSED_NOT_APPROVED	research_planning_optional	78	90	[OWNER]	If funded scope a separate lab reference validation program	A02;A06	Reviewed protocol budget and rights not an accuracy release	not_started		

Customer Discovery Log

row_status	interview_id	interview_date	segment	respondent_role	consent_status	problem_and_incumbent_summary	buyer_user_payer_summary	measurement_availability_summary	current_spend_and_price_evidence	baseline_minutes	evidence_link	qualification_status	next_action	owner	due_date	review_status
PLACEHOLDER_NOT_AN_INTERVIEW	[INTERVIEW ID]	[DATE]	[SEGMENT]	[ROLE]	[CONSENT STATUS]	[SUMMARY]	[SUMMARY]	[SUMMARY]	[SUMMARY]	[BASELINE MINUTES]	[EVIDENCE LINK]	[STATUS]	[NEXT ACTION]	[OWNER]	[DUE DATE]	LEGAL_ACCOUNTING_REVIEW_REQUIRED

Partner Pipeline

row_status	partner_id	partner_type	contact_reference	permission_status	current_stage	scope_and_methods_summary	fees_or_referral_terms_summary	last_contact_date	next_action	owner	due_date	evidence_link	review_status
PLACEHOLDER_NOT_A_PARTNER	[PARTNER ID]	[SEGMENT]	RESTRICTED_CONTACT_REFERENCE_ONLY	[CONSENT STATUS]	[STATUS]	[SUMMARY]	[SUMMARY]	[DATE]	[NEXT ACTION]	[OWNER]	[DUE DATE]	[EVIDENCE LINK]	LEGAL_ACCOUNTING_REVIEW_REQUIRED

Pilot Kpi Log

row_status	pilot_id	week_start_date	kpi_name	kpi_value	unit	numerator	denominator	baseline_minutes	current_minutes	support_minutes	record_type	evidence_link	owner	notes	review_status
PLACEHOLDER_NOT_MEASURED	[PILOT ID]	[DATE]	[KPI NAME]	[KPI VALUE]	DEFINE_UNIT_BEFORE_USE	[KPI VALUE]	[KPI VALUE]	[BASELINE MINUTES]	[CURRENT MINUTES]	[KPI VALUE]	SET_REAL_OR_DEMO	[EVIDENCE LINK]	[OWNER]	[SUMMARY]	LEGAL_ACCOUNTING_REVIEW_REQUIRED

Pricing Scenarios

scenario	row_status	currency	unit	price_inr	support_hours_per_month	support_rate_inr_per_hour	hosting_inr_per_month	operations_inr_per_month	processing_rate	cac_inr	fixed_cost_inr_per_month	variable_cost_inr_per_month	contribution_inr_per_month	contribution_margin_pct	break_even_farms	cac_payback_months	review_status	scope_note
low_price	ILLUSTRATIVE_NOT_A_QUOTE	INR	farm_per_month	999	0.75	400	150	100	0.02	3000	120000	569.98	429.02	42.94	280	6.99	LEGAL_ACCOUNTING_REVIEW_REQUIRED	Excludes taxes hardware lab shipping and validation; no demand or retention evidence
base	ILLUSTRATIVE_NOT_A_QUOTE	INR	farm_per_month	1499	0.75	400	150	100	0.02	3000	120000	579.98	919.02	61.31	131	3.26	LEGAL_ACCOUNTING_REVIEW_REQUIRED	Excludes taxes hardware lab shipping and validation; no demand or retention evidence
higher_price	ILLUSTRATIVE_NOT_A_QUOTE	INR	farm_per_month	2499	0.75	400	150	100	0.02	3000	120000	599.98	1899.02	75.99	64	1.58	LEGAL_ACCOUNTING_REVIEW_REQUIRED	Excludes taxes hardware lab shipping and validation; no demand or retention evidence
base_support_stress	ILLUSTRATIVE_NOT_A_QUOTE	INR	farm_per_month	1499	1.50	400	150	100	0.02	3000	120000	879.98	619.02	41.30	194	4.85	LEGAL_ACCOUNTING_REVIEW_REQUIRED	Support-only stress; excludes taxes hardware lab shipping and validation; no demand evidence

Calibration Record

data_status	calibration_record_id	point_id	device_id	sensor_id	sensor_type	firmware_version	wiring_revision	calibrated_at	operator_id	method_ref	reference_type	reference_lot	reference_certificate_ref	sample_id	measurement_basis	temperature_c	raw_adc_counts	raw_voltage_v	as_found_value	reference_value	as_left_value	unit	tolerance_abs	error_as_left_minus_reference	result	due_at	reviewer_id	reviewed_at	raw_evidence_ref	raw_evidence_sha256	notes
placeholder_not_for_use	PLACEHOLDER_CALIBRATION_ID	PLACEHOLDER_POINT_ID	PLACEHOLDER_DEVICE_ID	PLACEHOLDER_SENSOR_ID	PLACEHOLDER_SENSOR_TYPE	PLACEHOLDER_FIRMWARE_VERSION	PLACEHOLDER_WIRING_REVISION		PLACEHOLDER_OPERATOR_ID	PLACEHOLDER_METHOD_REF	PLACEHOLDER_REFERENCE_TYPE	PLACEHOLDER_REFERENCE_LOT	PLACEHOLDER_CERTIFICATE_REF	PLACEHOLDER_SAMPLE_ID	PLACEHOLDER_WET_OR_DRY_BASIS							PLACEHOLDER_UNIT			pending		PLACEHOLDER_REVIEWER_ID		PLACEHOLDER_RESTRICTED_EVIDENCE_REF		Template row only - remove before operational use; blank values are not measurements

Device Inventory

data_status	device_id	asset_tag	board_model	board_serial	sensor_inventory_ref	firmware_version	firmware_sha256	wiring_revision	mode	ph_adc_pin	moisture_adc_pin	dht_data_pin	bme_sda_pin	bme_scl_pin	validated_adc_max_v	clock_source	calibration_record_refs	calibration_due_at	custodian_id	assigned_farm_id	status	last_checked_at	wiring_evidence_ref	notes
placeholder_not_for_use	PLACEHOLDER_DEVICE_ID	PLACEHOLDER_ASSET_TAG	PLACEHOLDER_BOARD_MODEL	PLACEHOLDER_BOARD_SERIAL	PLACEHOLDER_SENSOR_INVENTORY_REF	PLACEHOLDER_FIRMWARE_VERSION		PLACEHOLDER_WIRING_REVISION	demo							PLACEHOLDER_TIME_SOURCE	PLACEHOLDER_CALIBRATION_REFS		PLACEHOLDER_CUSTODIAN_ID	PLACEHOLDER_FARM_ID	not_in_service		PLACEHOLDER_WIRING_EVIDENCE_REF	Template row only - no device or voltage limit has been validated

Incident Log

data_status	incident_id	detected_at	reported_by_id	environment	release_id	severity	status	summary	affected_scope_ref	incident_lead_id	containment_at	containment_summary	restored_at	data_loss_assessment_ref	evidence_ref	evidence_sha256	next_update_at	closed_at	reviewer_id	follow_up_ref	notes
placeholder_not_for_use	PLACEHOLDER_INCIDENT_ID		PLACEHOLDER_REPORTER_ID	PLACEHOLDER_ENVIRONMENT	PLACEHOLDER_RELEASE_ID		open	PLACEHOLDER_FACTUAL_SUMMARY	PLACEHOLDER_AFFECTED_SCOPE_REF	PLACEHOLDER_INCIDENT_LEAD_ID				PLACEHOLDER_DATA_LOSS_ASSESSMENT_REF	PLACEHOLDER_RESTRICTED_EVIDENCE_REF				PLACEHOLDER_REVIEWER_ID	PLACEHOLDER_FOLLOW_UP_REF	Template row only - no actual incident or response time is asserted

Qa Test Results

data_status	run_id	test_case_id	release_id	environment	executed_at	executor_id	device_browser_ref	status	expected_result	actual_result	evidence_ref	evidence_sha256	issue_ref	reviewer_id	notes
placeholder_not_for_use	PLACEHOLDER_RUN_ID	PLACEHOLDER_TEST_CASE_ID	PLACEHOLDER_RELEASE_ID	PLACEHOLDER_ENVIRONMENT		PLACEHOLDER_EXECUTOR_ID	PLACEHOLDER_DEVICE_BROWSER_REF	not_run	PLACEHOLDER_ACCEPTANCE_CRITERION		PLACEHOLDER_RESTRICTED_EVIDENCE_REF		PLACEHOLDER_ISSUE_REF	PLACEHOLDER_REVIEWER_ID	Template row only - no test execution or pass is asserted

Sample Chain Of Custody

data_status	sample_id	aliquot_id	event_id	event_type	event_at	farm_id	batch_id	sample_type	sampling_plan_ref	operator_id	from_custodian_id	to_custodian_id	device_id	calibration_record_id	container_id	seal_condition	sample_condition	temperature_c	handling_notes	lab_sample_id	lab_method_ref	lab_report_ref	raw_evidence_ref	raw_evidence_sha256	deviation_ref	notes
placeholder_not_for_use	PLACEHOLDER_SAMPLE_ID	PLACEHOLDER_ALIQUOT_ID	PLACEHOLDER_EVENT_ID	collection		PLACEHOLDER_FARM_ID	PLACEHOLDER_BATCH_ID		PLACEHOLDER_SAMPLING_PLAN_REF	PLACEHOLDER_OPERATOR_ID	PLACEHOLDER_FROM_CUSTODIAN_ID	PLACEHOLDER_TO_CUSTODIAN_ID	PLACEHOLDER_DEVICE_ID	PLACEHOLDER_CALIBRATION_ID	PLACEHOLDER_CONTAINER_ID	PLACEHOLDER_SEAL_CONDITION	PLACEHOLDER_SAMPLE_CONDITION		PLACEHOLDER_HANDLING_NOTES	PLACEHOLDER_LAB_SAMPLE_ID	PLACEHOLDER_LAB_METHOD_REF	PLACEHOLDER_LAB_REPORT_REF	PLACEHOLDER_RESTRICTED_EVIDENCE_REF		PLACEHOLDER_DEVIATION_REF	Template row only - no real sample or custody event is asserted